

SURVEY

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Artificial Intelligence-driven privacy preservation in the internet of vehicles: a comprehensive systematic literature review

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Abstract

The rapid growth of the Internet of Vehicles (IoV) and the increasing use of Connected and Autonomous Vehicles (CAVs) are causing many privacy and security problems that traditional protection methods can not handle. IoV networks have a lot of different types of data flows, distributed infrastructure, and real-time communication needs. This makes it very hard and important to protect privacy. The constant sharing of data between automobiles, roadside infrastructure, and cloud platforms makes IoV ecosystems vulnerable to attacks that try to manipulate the data, steal it, or make guesses about it. This paper seeks to deliver a systematic and thorough assessment of Artificial Intelligence (AI)-based privacy-preserving strategies in the IoV, tackling the absence of a cohesive framework in existing literature. We employ a Systematic Literature Review (SLR) to examine and classify essential methodologies, such as Federated Learning (FL), Differential Privacy (DP), Homomorphic Encryption (HE), Blockchain-enhanced AI, Adversarial Machine Learning (AML), and anomaly detection, into a six-domain taxonomy. This taxonomy provides a systematic framework for evaluating the advantages, compromises, and scaling constraints of different methodologies. The findings indicate that although AI-driven methodologies have considerable promise in protecting vehicle data, substantial obstacles remain regarding computational overhead, real-time adaptation, and scalability in high-mobility environments. Our work systematically maps existing approaches, revealing performance constraints and unresolved research gaps while delineating specific future research trajectories, including lightweight model design, hybrid privacy frameworks, and integration with emerging technologies such as edge intelligence and 6G networks. The examination of Quality of Service (QoS) and Quality of Experience (QoE) factors indicates that computational overhead (25%), scalability challenges (20%), latency issues (15%), accuracy trade-offs (15%), optimization requirements (15%), and deep learning model complexity (10%) persist as the primary impediments to the real-time, large-scale implementation of privacy-preserving AI in vehicular networks.

Keywords Artificial intelligence-driven privacy, Internet of vehicles, Federated learning, Differential privacy, Blockchain security, Adversarial machine learning

Introduction

Significant privacy issues have been created by the Internet of Vehicles (IoVs) explosive expansion and the growing acceptance of connected and autonomous cars [1]. The privacy issues raised by large data exchange between infrastructure, vehicles, and cloud services are not being satisfactorily addressed by conventional security mechanisms. IoV networks' complexity results from heterogeneous data sources, real-time communication needs, and different service expectations. Edge computing, Federated Learning (FL), and blockchain are among the newly developing technologies under investigation to improve IoV's privacy preservation [2, 3]. Ensuring data confidentiality while preserving the effectiveness of vehicle communication still presents a difficult task, though. Modern privacy-preserving technologies are needed to protect user information, stop illegal access, and minimize inference attacks in IoV systems. By using real-time data analysis, predictive modeling, and automated decision-making, Artificial Intelligence (AI) is increasingly helping smart privacy solutions to be realized. Utilizing AI-based technologies, IoV systems can strengthen data security, guarantee adherence to privacy rules, and raise user confidence [4].

Especially in Machine Learning (ML)-driven security systems, recent developments in AI have produced strong privacy-preserving methods [5–7]. By allowing distributed training of models across linked vehicles without direct data sharing, FL lowers privacy risks [8–10]. Differential privacy (DP) adds statistical noise to datasets, stopping illegal entities from accessing private data [11]. Computations on encrypted data are made possible by Homomorphic Encryption (HE), guaranteeing confidentiality during handling. Blockchain AI is also starting to show promise as a distributed method for protecting vehicle communications while preserving data integrity [12]. By helping IoV systems identify and stop inference attacks, Adversarial Machine Learning (AML) methods improve network robustness. Real-time identification of privacy violations made possible by AI-driven anomaly detection systems helps to reduce possible vulnerabilities [13]. Notwithstanding these developments, several problems still exist, including the computational overhead of privacy-preserving AI models, scalability problems, and real-time adaptation to dynamic vehicle environments [14]. Dealing with these issues calls for an all-encompassing assessment of privacy-preserving AI-driven solutions specifically for IoV systems [15].

The incorporation of AI into the IoV has facilitated significant progress in automation, communication, and security; yet, it has concurrently heightened privacy issues stemming from extensive data production and real-time connectivity. Even though there are more and more studies looking into privacy-preserving technologies like FL, DP, HE, Blockchain-based AI, and AML, the research that is already out there is still scattered, not comparable, and only applies to certain fields. Each study only looks at one method or a small number of use cases. As a result, no unified and thorough methodology consistently ranks, categorizes, and evaluates different methodologies based on computation, scalability, latency, accuracy, and energy efficiency. Despite the growing interest in AI-driven privacy protection in the IoV, the existing literature lacks a cohesive and systematic analysis that can consolidate various approaches within a singular evaluative framework. This fragmentation has led to a dispersion of knowledge, making it more difficult to evaluate how well AI-driven privacy-preserving solutions work in IoV and how flexible they are. To fill this gap, the current study performs a Systematic Literature

Review (SLR) to find, sort, and critically evaluate current AI-based privacy-preserving techniques. After looking at a lot of previous research, we suggest a six-domain taxonomy that includes: (1) FL for secure data sharing and distributed training; (2) DP for protecting user data with statistical noise; (3) HE for secure computation over encrypted vehicular data; (4) blockchain-integrated AI for decentralized privacy and trust management; (5) AML for defense against inference and poisoning attacks; and (6) anomaly detection for finding privacy violations and cyber intrusions in vehicular networks. This classification system gives us an organized and comparative way to look into AI-based privacy-preserving methods, pointing out their pros, cons, and how they may be used in real-world next-generation automotive systems. Our study enhances the comprehension of AI's involvement in vehicular privacy, elucidates unresolved difficulties, and pinpoints emerging research prospects through a methodical methodology. The main contributions of this work are: (i) the creation of a six-domain taxonomy that brings together scattered research; (ii) a comparison and cross-layer analysis of computational and operational trade-offs; (iii) the discovery of integrative synergies like FL + DP and FL + Blockchain across edge, fog, and cloud layers; (iv) the application of privacy-preserving methods to larger IoT ecosystems, such as smart cities and healthcare; and (v) the combination of open research problems related to non-IID data, adaptive learning, model interpretability, and regulatory compliance. Overall, this analysis goes beyond just describing things by providing an analytical framework that shows how AI-driven privacy-preserving solutions work together, build on each other, and change over time in real-world vehicle contexts that are complicated. The main contributions of this work are as follows:

- Suggesting a new taxonomy for privacy-aware AI-IoV that includes FL, DP, HE, blockchain-based AI, AML, and anomaly detection;
- Providing a side-by-side look at compute cost, scalability, latency, accuracy, and energy use that meets the needs of vehicles in real time;
- Finding ways that edge, fog, and cloud can work together and describing hybrid architectures that can protect vehicle privacy in a way that is both scalable and flexible;
- Generalizing across domains to healthcare, smart cities, and broader IoT, suggesting potential transferability and flexibility;
- Combining problems like non-IID data, adaptivity, interpretability, following the rules, and making clear approaches toward real-time, scalable, trust-aware IoV privacy.

This work is organized mostly as follows: "[Related work](#)" goes over related works; "[Methodology](#)" offers our methodical literature review approach; "[Privacy challenges in IoV](#)" addresses current privacy issues in IoV; "[AI techniques for privacy preservation](#)" introduces our suggested taxonomy and classifies AI-driven privacy preservation techniques; "[Results and analysis](#)" assesses and contrasts several approaches; "[Future work and research challenges](#)" emphasizes open research challenges; "[Conclusion and limitations](#)" ends the study with important results and future directions. Also, Table 1 shows an Abbreviation Table.

Table 1 Abbreviation table

Whole term	Abbreviation	Whole term	Abbreviation
Additive Secret Sharing	ASS	Homomorphic Encryption	HE
Adversarial Machine Learning	AML	Homomorphic, Federated Learning-Enabled Pedestrian Detection System	HMFLS
AI-based Trust-aware and Privacy-preserving System	ATPS	Intelligent Transportation Systems	ITS
Artificial Intelligence	AI	Internet of Things	IoT
Artificial Intelligence of Things	AIoT	Internet of Vehicles	IoV
Autonomous Vehicle	AV	k-Nearest Neighbors	KNN
Base Station	BS	Local Differential Privacy	LDP
Blockchain-based Privacy-preserving Federated Learning	BPFL	Long Short-Term Memory	LSTM
Blockchain-enabled Federated Reinforcement Learning	BFRL	Machine Learning	ML
Blockchain-enabled Internet of Vehicles	BloV	Multi-access Edge Computing	MEC
California Consumer Privacy Act	CCPA	Multi-Agent Reinforcement Learning	MARL
Confidentiality, Integrity, and Authenticity	CIA	Multilayer Perceptron	MLP
Connected and Autonomous Vehicles	CAV	On-Board Unit	OBU
Context-Aware Privacy-Preserving for Autonomous Driving	CAPPAD	Optimized Unary Encoding	OUE
Controller Area Network	CAN	Partial Ordering-based Trust Management	POTM
Convolutional Autoencoder Aided Anomaly Detection	CAaDet	Physical Unclonable Function	PUF
Convolutional Neural Network	CNN	Public Key Infrastructure	PKI
Decentralized Privacy-Preserving Deep Learning	DPDL	Quality of Experience	QoE
Decentralized Vehicular Ad Hoc Networks	DVANETs	Quality of Service	QoS
Deep Learning Engine	DLE	Reinforcement Learning	RL
Deep Q-Network	DQN	Rivest–Shamir–Adleman	RSA
Differential Privacy	DP	Roadside Unit	RSU
Differential Privacy-preserving Deep Learning Caching Framework	DP-DLCF	Secure Hierarchical Decomposition	SHD
Distributed Proof of Work	dPOW	Self-Sovereign Identity	SSI
Edge Computing	EC	Shamir's Secret Sharing	SSS
Electric and Flying Vehicles	EnFVs	Social Internet of Vehicles	SIoV
Electric Vehicle	EV	Software-Defined Networking	SDN
Federated Learning	FL	Systematic Literature Review	SLR
Fog-assisted Internet of Vehicles	Fa-IoV	Trajectory Privacy Preserving	TPP
Fully Homomorphic Encryption	FHE	Two-Layer Privacy-Preserving	TLPP
Gated Recurrent Unit	GRU	Vehicle-to-Cloud	V2C
Gaussian Mixture Model	GMM	Vehicle-to-Everything	V2X
General Data Protection Regulation	GDPR	Vehicle-to-Grid	V2G
Generative Adversarial Network	GAN	Vehicle-to-Infrastructure	V2I
Visual Internet of Vehicles	V-IoV	Vehicle-to-Roadside	V2R
Wasserstein Generative Adversarial Network	WGAN	Vehicular Ad Hoc Network	VANET
You Only Look Once	YOLO	Vehicular Social Network	VSN
Zero-Knowledge Proof	ZKP	Vertical Heterogeneous Networks	VHetNets

Related work

Significant developments in the IoV, IoT security, autonomous vehicles, and next-generation transportation networks follow from the fast spread of AI applications in security, privacy, and optimization. But there are also new challenges with cybersecurity, privacy, and ethics that need AI-driven solutions like blockchain, FL, anomaly detection, and reinforcement learning models. Key related studies on AI-driven security and privacy preservation methods across autonomous vehicles, IoT security, intelligent biometrics, blockchain integration, and AI-driven cyber threat detection are examined in this part. Rising acceptance of Connected and Autonomous Vehicles (CAVs) poses serious privacy and security concerns. Blockchain-enabled Federated Reinforcement Learning (BFRL) for data security and privacy protection in CAVs was investigated by Alam [16]. Their paper focused on risks, including remote hacking, sensor manipulation, and inference attacks on vehicle networks. Also, data analytics, system optimization, and security systems have all been much enhanced by combining AI with the IoT. Examining AI-driven privacy mechanisms in IoT, an SLR by Marengo [17] found real-time ML models and edge computing as enablers of improved data security. Their study underlined, nevertheless, the necessity of fresh laws to control AI-driven data protection in IoT environments. Comparably, Aouedi et al. [18] looked at Intelligent IoT (IIoT) and the junction of AI, security, and privacy issues in uses including smart healthcare, transportation, and industrial automation. Although their research classified current IIoT security systems, it lacked an adaptive security mechanism, including AI, to dynamically handle real-time cyber threats. Although seamless communication in modern vehicles depends on Controller Area Networks (CANs), their vulnerability to hacking and illegal data modification results from the absence of cryptographic privacy standards. Leveraging AI-driven intrusion detection and threat mitigating strategies, a paper by Ahmad et al. [19] offered a thorough review of anomaly detection systems in CANs. Their research classified AI-based anomaly detection models and underlined issues with scalability, data security, and real-time AI implementation in vehicle networks. Their research emphasized the need for privacy-preserving anomaly detection systems, though it lacked an experimental validation framework for real-world CAN security applications. Also, Alqahtani and Kumar [20] investigated AI-driven solutions to strengthen Electric and Flying Vehicles (EnFVs) cybersecurity and looked at obstacles. Emphasizing the growing complexity of threats in digital transportation systems, their research starts with an overview of EnFV cybersecurity problems. Moreover, new problems in privacy preservation, security enforcement, and distributed data protection systems have been presented by the IoV. Examining the function of blockchain and AI in IoV networks, Wang et al. [21] proposed blockchain intelligence as a single framework for improving trustworthiness and securing automotive communication. Their paper explored the difficulties of real-time integration and computational efficiency and classified blockchain-driven security systems. Comparably, Shen et al. [22] underlined how distributed consensus systems enhance AI security models in IoT environments and investigated blockchain used in AI-powered IoT. Both studies, however, underlined the need for scalable, low-latency AI solutions for IoV implementations concerning privacy. Also, Awad et al. [23] offered a thorough review of AI-powered biometrics. Their study did find significant difficulties with biometric data privacy, adversarial attacks, and regulatory compliance. Nonetheless, security flaws remain a major issue since IoT systems run and greater risk of data leaks,

privacy invasions, and cyberattacks. Using a comprehensive survey of IoT security issues, Abed and Anupam [24] compiled real-world case studies of the most recent IoT security events. Emphasizing AI-driven intrusion detection, anomaly detection, and adaptive security mechanisms, their study examined how AI might identify IoT-related cyber threats. Though it offered a thorough classification of security architectures, the study lacked thorough experimental validation of AI-based solutions for real-time IoT threat mitigation. Beyond cyberattacks, IoT systems present physical security issues that call for thorough AI-driven security models. Examining IoT security from a more general standpoint, Humayun et al. [25] classified issues into cybersecurity threats, physical security risks, and AI-powered countermeasures. Emphasizing AI-based adaptive security models and offering solutions with intrusion detection, AI-enhanced encryption, and real-time threat monitoring, their study offered a disciplined framework, but it lacked real-world deployment validation and the suggested AI security models were not tested for scalability in vast IoT environments. Also, improving security and privacy in IoV networks has mostly turned to blockchain solutions. Emphasizing secure communication, consensus mechanisms, and fraud detection in automotive networks, Surapaneni et al. [26] examined Blockchain-enabled IoV (BIOV) architectures. Their paper underlined how blockchain technology guarantees tamper-proof data exchange, fights against cyberattacks, and vehicle identity security. Their paper also highlights major scalability and latency issues when using blockchain in high-speed vehicle networks, though. In this regard, ML models have been made possible by AI development, enhancing IoV network security. Proposing Reinforcement Learning (RL)-driven task offloading systems, Ali et al. [27] investigated ML-based approaches for secure IoV edge computing and Vehicle-to-Everything (V2X) communication. Their study found, however, difficulties with real-time processing limits and energy-efficient ML models. Likewise, Reddy and Muralidhar [28] concentrated on FL for vehicle safety and road traffic analysis, stressing FL's capacity to protect data privacy by employing ML model training over scattered IoV nodes. The study did note, though, limits in FL model convergence and inter-vehicle synchronization. By including social interactions among infrastructure, drivers, and vehicles, the Social Internet of Vehicles (SIOV) extends IoV. Examining SIOV privacy issues, Butt et al. [29] classified them as either communication privacy, location privacy, or identity protection. Though it lacked actual confirmation of blockchain integration in SIOV environments, the study also covered blockchain-based privacy solutions. Comparably, Xing et al. [30] investigated how FL might protect SIOV data and suggested cooperative ML models for privacy protection and threat detection. Still unresolved, though, were questions about multi-agent trustworthiness in SIOV. Also, in-vehicle edge computing, blockchain and AI together improve traffic control, safe data exchange, and anomaly detection. Inspired by Zamanirafe et al. [31], a scalable IoV combining distributed ML models with blockchain-based consensus systems was investigated. The work revealed safe data aggregation using Roadside Units (RSUs), but it also found difficulties with consensus efficiency and AI model adaptation to dynamic vehicle networks. Comparably, Amiri-Zarandi et al. [32] examined ML-based privacy preservation methods in IoT and classified them into privacy-aware AI architectures, adversarial defense, and anomaly detection. Still, a major constraint, though, real-time ML scalability in IoT and vehicle environments stayed limited. As well as Liu et al. [33] offered a thorough analysis of privacy concerns in AI-driven systems, classifying AI privacy contacts into private ML,

AI-assisted privacy protection, and AI-driven privacy attacks. Because of their dynamic inter-vehicle communication and real-time content sharing, Vehicular Social Networks (VSNs) present special privacy concerns. Examining privacy-preserving methods for content distribution in VSNs, Wang et al. [34] classified access-control, anonymity-preserving, and encryption-based technologies. The study did, however, find issues in trade-offs between usability and privacy in VSN-based data sharing. In Vehicle-to-Grid (V2G) networks, Electric Vehicles (EVs) depend on safe authentication mechanisms to stop unwanted access. Investigating privacy-preserving authentication methods in V2G, Chavali et al. [35] looked at blockchain-driven and encryption-based authentication systems. Still unresolved, though, were difficulties with lightweight cryptographic solutions for V2G. In addition, for optimizing vehicle network security, RL has become rather popular. Reviewing Multi-Agent Reinforcement Learning (MARL) methods for resource management, trust enforcement, and privacy preservation in IoV have been investigated by Mianji et al. [36]. Though scalability remained a question, their study classified RL models into centralized, distributed, and hybrid MARL architectures. Also, examining Intrusion Detection Systems (IDSs) in IoV, Taslimasa et al. [37] classified IDS mechanisms into hybrid, anomaly-based, and signature-based models. Although the paper offered a thorough taxonomy of IDS methods, the actual application of AI-powered IDS in vast-scale IoV networks remained difficult.

Driven by the fast acceptance of CAVs and their interaction with cloud services, edge computing, and AI, the IoV has experienced hitherto unheard-of expansion. However, because of the massive sharing of private information among infrastructure, vehicles, and outside networks, this growth has presented serious privacy issues. Conventional security systems are vulnerable to data breaches, inference attacks, and illegal access since they find it difficult to handle the dynamic character, heterogeneous data sources, and real-time communication needs of IoV systems. Though FL, blockchain, and AML have been investigated as possible fixes, ensuring data confidentiality while preserving efficiency still presents a major obstacle. Current privacy-preserving techniques limit their efficacy in large-scale vehicle networks due to computational overhead, scalability constraints, and real-time adaptability problems. As IoV systems must not only protect data transmissions but also comply with worldwide privacy rules and preserve high degrees of user trust, the need for strong, AI-driven privacy-preserving solutions has grown more urgent than ever. Particularly those using real-time analytics, predictive modeling, and automated decision-making, AI-based solutions have shown encouraging promise in tackling these problems. Several cutting-edge privacy-preserving methods driven by ML have been presented: HE for safe computation, DP for noise injection, FL for distributed model training, and Blockchain AI for distributed security. Without sharing raw data, FL lets several vehicles cooperatively train AI models, so lowering privacy concerns related to centralized processing. DP introduces statistical noise to datasets, so it challenges adversarial entities to deduce sensitive information. HE makes sure that computations on encrypted data stay private through processing, stopping data leaks during model training. Blockchain-integrated AI solutions have meanwhile become a promising way to improve transparency, security, and confidence in vehicle communications. While AI-powered anomaly detection systems offer real-time privacy breach identification, lowering possible vulnerabilities in IoV networks, AML techniques strengthen IoV security by identifying and mitigating inference attacks. Notwithstanding

these developments, several important research gaps still exist, especially with relation to computational efficiency, real-time adaptability, and scalability of privacy-preserving AI models in fast, data-intensive vehicle networks. Although several studies have suggested AI-based privacy preservation methods, the body of current research has not been methodically evaluated. Without offering a consistent framework to evaluate their efficacy in IoV environments, most previous work has concentrated on isolated AI-driven privacy models, including FL, blockchain security, or adversarial defense mechanisms. The absence of thorough, ordered research makes it challenging to ascertain the practical relevance and efficiency of these approaches in actual vehicle networks. Utilizing a SLR, which analyzes, classifies, and assesses AI-driven privacy-preserving technologies in IoV systems, our work fills in this void. FL for secure distributed training, DP for anonymized data protection, HE for privacy-preserving computation, Blockchain AI for decentralized security, AML for inference attack mitigating, and Anomaly Detection for real-time privacy breach identification are six main domains into which we propose a new taxonomy for AI privacy techniques. This ordered taxonomy offers a methodical framework for analyzing how AI-driven solutions improve privacy, their benefits, disadvantages, and pragmatic relevance in IoV systems. Employing this extensive classification and evaluation, our research reveals important voids in AI-driven privacy mechanisms, including high computational costs, challenges in real-time adaptation, and the necessity of hybrid AI models balancing efficiency and security. Our work attempts to close the gap between theoretical developments and practical implementation in next-generation vehicle networks by methodically comparing several AI privacy-preserving technologies. Our work provides a methodical road map for industry players, legislators, and researchers, enabling the creation of scalable, effective, privacy-conscious AI solutions for IoV security. Utilizing this methodical approach, we progress the field of AI-driven privacy preservation by employing a well-defined research direction, addressing unresolved issues, and opening the path for safe, intelligent transportation systems that protect user data while guaranteeing effective vehicle communication. Also, Table 2 shows the related reviews in the relevant studies.

Methodology

We investigated related literature on privacy-preserving methods in vehicle networks in the past sections. An SLR is presented in this part to give a comprehensive understanding of how AI-driven approaches are implemented to improve security, privacy, and efficiency in the IoV. The main goal of this review is to categorize and assess how well these approaches address important problems, including data security, privacy leakage, computational efficiency, real-time adaptability, and resilience against cyber threats. We want to close the knowledge gap and offer a disciplined framework for evaluating six different AI-driven approaches by grouping state-of-the-art research into a taxonomy: FL, DP, HE, blockchain AI models, AML, and AI-based anomaly detection. We investigate the implementation and performance of these techniques over IoV applications, including secure data sharing, location privacy, vehicle communication, and real-time traffic monitoring. We also go over issues, including computational overhead, scalability restrictions, security flaws, and future directions of research that would affect intelligent transportation systems.

Table 2 Related reviews

Author(s)	Main Idea	Scope	Advantage	Disadvantage
Alam [16]	Investigating AI-driven privacy and security in CAVs using BFRL.	CAVs, Data Security	Clearly defined scope with systematic literature review methodology; categorized AI-based privacy solutions for CAVs.	Limited discussion on real-time blockchain implementation challenges in CAVs; lacks experimental validation.
Marengo [17]	AI in IoT privacy: a systematic review of real-time AI-based privacy solutions.	IoT, Privacy, AI Security	A comprehensive evaluation of AI-based IoT security frameworks, providing real-world case studies.	Lack of evaluation of AI-based security performance across various IoT scenarios; missing future research directions.
Aouedi et al. [18]	Security challenges in IIoT with AI-based privacy-preserving techniques.	IIoT, smart transportation, healthcare	An extensive review of AI-driven IoT security discussed privacy preservation strategies.	Did not include an adaptive security framework to counter real-time cyber threats in IIoT networks.
Ahmad et al. [19]	AI-based anomaly detection for CANs.	Automotive security, AI-driven IDS	A detailed taxonomy of AI-driven anomaly detection in CANs highlighted privacy risks.	Did not provide experimental testing for AI-driven anomaly detection in CANs; lacked benchmarking comparisons.
Alqahtani and Kumar [20]	AI-driven security solutions for EnFVs.	Smart transportation, EnFV security	Explored blockchain-driven vehicular security; provided classification of blockchain-based privacy models.	Lacks empirical validation of blockchain-driven vehicular security models; missing performance benchmarks.
Wang et al. [21]	Blockchain intelligence for secure data privacy in IoV.	IoV, blockchain-driven security	A comprehensive review of blockchain-based privacy preservation in IoV discussed implementation challenges.	Lack of scalability testing for blockchain privacy-preserving solutions in large-scale IoV networks.
Shen et al. [22]	Blockchain for AI-powered IoT security and efficiency.	AIoT, blockchain security	Surveyed AI-driven security for AIoT; identified key scalability and efficiency concerns.	Did not examine real-world AIoT security deployments; lacks a discussion on regulatory compliance.
Awad et al. [23]	AI-powered biometric authentication for secure IoT environments.	Biometrics, AI-driven authentication	Reviewed AI-powered biometric authentication in IoT; analyzed privacy and security risks.	Lack of practical implementation guidelines for AI-based biometric authentication in IoT.
Abed and Anupam [24]	AI-based IoT security frameworks with real-world case studies.	IoT security, AI-enhanced cybersecurity	Summarized AI-based intrusion detection systems in IoT, including real-world IoT cyber attack analysis.	Limited real-world deployment validation of AI-driven intrusion detection models in IoT environments.
Humayun et al. [25]	AI-integrated security frameworks for IoT and physical security.	AI in IoT, privacy-aware security	Classified AI-driven IoT security models based on physical and cybersecurity challenges; proposed AI-enhanced encryption.	Did not include large-scale deployment testing of AI-enhanced encryption models for IoT security.
Surapaneni et al. [26]	Blockchain-enabled IoV for secure vehicle communication.	IoV, blockchain, security	Explored security enhancements in BloV; provided security level classification.	Limited real-world experimentation for blockchain-enabled security in high-speed vehicular environments.
Ali et al. [27]	ML-based security and V2X optimization in IoV.	IoV, ML, V2X security	Investigated ML-based V2X security solutions; proposed reinforcement learning approaches for privacy protection.	Did not evaluate energy efficiency and real-time adaptability of ML-driven V2X security models.

Table 2 (continued)

Author(s)	Main Idea	Scope	Advantage	Disadvantage
Reddy and Muralidhar [28]	FL for road traffic analysis and safety.	IoV, traffic safety, FL	Reviewed FL privacy models in IoV; discussed synchronization and convergence challenges.	Did not analyze inter-vehicle synchronization in FL models for IoV.
Butt et al. [29]	Privacy challenges and blockchain-based SloV solutions.	SloV, blockchain, privacy	Surveyed blockchain privacy-preserving methods for SloV; analyzed security protocols.	Lacks experimental validation for blockchain-based privacy techniques in SloV environments.
Xing et al. [30]	FL for SloV data security.	SloV, FL, data security	A comprehensive evaluation of AI-based IoT security frameworks provided real-world case studies.	Lack of evaluation of AI-based security performance across various IoT scenarios; missing future research directions.
Zamanirafe et al. [31]	Distributed ML and blockchain for scalable IoV.	IoV, edge computing, blockchain	Clearly defined scope with systematic literature review methodology; categorized AI-based privacy solutions for CAVs.	Limited discussion on real-time blockchain implementation challenges in CAVs; lacks experimental validation.
Amiri-Zarandi et al. [32]	ML-based privacy preservation techniques in IoT.	IoT, privacy, AI security	A detailed taxonomy of AI-driven anomaly detection in CANs highlighted privacy risks.	Did not provide experimental testing for AI-driven anomaly detection in CANs; lacked benchmarking comparisons.
Liu et al. [33]	Privacy-preserving AI in IoV and ML-based security risks.	AI, privacy, security	Explored blockchain-driven vehicular security; provided classification of blockchain-based privacy models.	Lacks empirical validation of blockchain-driven vehicular security models; missing performance benchmarks.
Wang et al. [34]	Privacy-preserving content dissemination in VSNs.	VSNs, privacy-preserving content sharing	Summarized AI-based intrusion detection systems in IoT, including real-world IoT cyber-attack analysis.	Limited real-world deployment validation of AI-driven intrusion detection models in IoT environments.
Chavali et al. [35]	Authentication mechanisms and privacy in V2G networks.	V2G, authentication, privacy	Reviewed AI-powered biometric authentication in IoT; analyzed privacy and security risks.	Lack of practical implementation guidelines for AI-based biometric authentication.
Mianji et al. [36]	MARL for IoV security and privacy.	IoV, MARL	Classified AI-driven IoT security models based on physical and cybersecurity challenges; proposed AI-enhanced encryption.	Did not include large-scale deployment testing of AI-enhanced encryption models for IoT security.
Taslimasa et al. [37]	AI-powered IDS in IoV networks.	IoV, AI-based IDS	A comprehensive review of blockchain-based privacy preservation in IoV discussed implementation challenges.	Lack of scalability testing for blockchain privacy-preserving solutions in large-scale IoV networks.
Our Work	SLR on AI-driven privacy preservation in IoV, proposing a structured taxonomy.	IoV, AI-based privacy, FL, blockchain, AML, anomaly detection	Surveyed AI-driven security for AIoT; identified key scalability and efficiency concerns. Developed a comprehensive AI-driven taxonomy for IoV privacy; systematically compared AI techniques; and provided detailed future research directions.	-

Formalization of questions

The primary objective of this SLR is to investigate and evaluate significant AI-driven contributions to privacy preservation in IoV. The key Research Questions (RQs) guiding this review are:

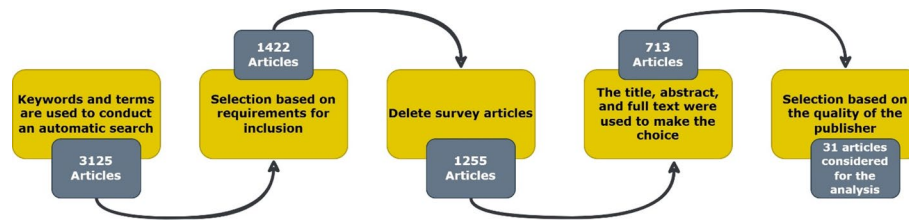


Fig. 1 PRISMA flow diagram of the study identification, screening, eligibility, and inclusion

Table 3 Search terms and keywords

S#	Keywords	Related research question(s)
1	"Federated Learning" and "IoV Security"	RQ1, RQ2
2	"Privacy-preserving AI" and "Homomorphic Encryption"	RQ1, RQ4
3	"Blockchain AI" and "Vehicular Networks"	RQ1, RQ2, RQ3
4	"Adversarial Machine Learning" and "Privacy Risks"	RQ1, RQ3
5	"Differential Privacy" and "Intrusion Detection"	RQ1, RQ3, RQ4
6	"Anomaly Detection" and "Vehicular Cybersecurity"	RQ1, RQ3
7	"AI-driven Trust Management" and "IoV Security"	RQ1, RQ2
8	"Deep Learning" or "Privacy Preservation" and "VANET"	RQ1, RQ4
9	"Secure AI Training" or "Federated Learning" and "IoV Applications"	RQ1, RQ2, RQ4
10	"Privacy-aware AI" or "Security-enhanced AI" and "Smart Transportation"	RQ1, RQ3, RQ4

- *RQ1: Which AI-based privacy-preserving techniques are most commonly applied in IoV, and how do they enhance security in areas such as vehicular data sharing, location privacy, and anomaly detection? Discussed in Sect. "Privacy challenges in IoV".*
- *RQ2: How do FL and blockchain technologies contribute to decentralized security in vehicular networks, and what are their advantages over traditional security models? Discussed in Sect. "Privacy challenges in IoV".*
- *RQ3: What are the existing gaps in scalability, computational efficiency, and real-world implementation of AI-driven privacy-preserving techniques in IoV? Located in Sect. "AI techniques for privacy preservation".*
- *RQ4: How can AI techniques be optimized to improve privacy, security, and computational efficiency in large-scale IoV networks? Addressed in Sect. "Results and analysis".*

Selection and evaluation of literature

Figure 1 shows how the study used a four-step systematic selection procedure to find relevant papers. We searched for keywords in some of the best academic databases, such as IEEE Xplore, ACM Digital Library, Elsevier (ScienceDirect), and SpringerLink. The search used the terms in Table 3 to look through conference proceedings, peer-reviewed journal publications, and specialized research. The first search found 3,125 articles. This amount was brought down to 1,422 publications based on set inclusion criteria that focused on relevance to IoV, AI-driven privacy preservation, and technical rigor. The next step was to get rid of survey papers and other non-research contributions, which left 1,255 articles.

After that, we looked at the title, abstract, and full text to see how technically deep and practically useful the articles were. This led to 713 papers that were eligible. Ultimately, after assessing publication quality, topic relevance, and methodological rigor, 31 studies

were selected for comprehensive examination. The selected studies encompass a diverse array of methodologies, including FL for decentralized privacy preservation, DP for secure location and identity management, and HE for encrypted data analytics. Additionally, they include blockchain for secure vehicle data exchange, AML for resilience against poisoning and inference attacks, and anomaly detection for immediate intrusion detection. These six categories form the basis of our proposed taxonomy, which will be analyzed in detail in the subsequent sections to evaluate their relevance and potential for improving IoV security systems.

Four important scholarly databases were chosen for the literature review to make sure it was both broad and thorough: IEEE Xplore, ACM Digital Library, Elsevier, and SpringerLink. We chose these repositories based on three main things. IEEE and ACM are the best places to publish research in computer science, artificial intelligence, and vehicular technology. They host the most important publications and conferences for IoV and privacy-preserving research. Second, Elsevier and Springer provide multidisciplinary coverage across engineering, data privacy, and intelligent transportation systems, ensuring that cross-domain perspectives are incorporated. Finally, all four publishers have strict peer-review criteria and are indexed in Scopus, Web of Science, and DBLP. This guarantees that only high-quality, extensively referenced papers are included. By concentrating on these digital libraries, we guaranteed that the examined works embody authoritative and representative contributions to the domain of AI-driven privacy protection in the IoV.

The methodology was specifically created to directly tackle the study target, which is to create a thorough taxonomy of AI-driven privacy-preserving techniques in the IoV and to pinpoint research deficiencies for subsequent advancement. Three RQs were created to help with this, and each step of the approach was linked to them. We used systematic database searches (IEEE, ACM, Elsevier, Springer) and the multi-stage filtering method shown in Fig. 1 to answer RQ1 (What AI-driven solutions are currently applied for privacy preservation in IoV?). We answered RQ2 (How can these techniques be categorized into a cohesive taxonomy?) by putting the 31 papers we chose into six unique domains: FL, DP, HE, blockchain, AML, and anomaly detection. Table 3 shows a summary of these areas. RQ3 (What are the open problems and potential research directions?) was formulated by a thorough examination of limits, cross-domain applicability, and performance trade-offs, as discussed in "[Future work and research challenges](#)". This organized linkage guarantees that the approach consistently tackles the research objective while preserving clarity in the construction of the final taxonomy and research directions.

Privacy challenges in IoV

Vehicles, infrastructure, and cloud-based services constantly exchange real-time data in the dynamically linked ecosystem that is the IoV. However, this linked character raises serious privacy issues that cannot be satisfactorily addressed by conventional security systems. The enormous volume of sensitive vehicle data produced, handled, and shared across heterogeneous networks exposes vehicles and users to risks, including data leakage, cyber threats, and trust-related authentication vulnerabilities, posing privacy challenges. Covering data leakage risks, sophisticated cyber threats, authentication, and trust problems requiring immediate mitigating strategies, this part offers a thorough study of privacy challenges in IoV.

Data leakage risks in vehicle networks

Large-scale automotive data produced and exchanged by IoV networks includes location tracking, driver behavior analytics, V2X interactions, in-car sensor telemetry, and traffic pattern recognition. Although intelligent transportation systems depend on constant data exchange, if intercepted or used improperly, this raises serious privacy issues. Unauthorized tracking, sensitive data exposure, vehicle sensor vulnerabilities, and third-party data misuse are the most important privacy hazards in IoV systems [38]. Unauthorized location tracking, where IoV-enabled vehicles continuously broadcast GPS data to fleet management systems, ride-sharing platforms, and navigation services, is among the most urgent problems. Should they be intercepted, malevolent actors can track vehicle movements, deduce driving patterns, and even project future paths, enabling possible stalking, focused crimes, and invasions of privacy. Sensitive personal data exposure is also a growing issue since connected and autonomous cars save driver-specific information, including biometric authentication records, voice command history, and financial transaction data (e.g., toll payments and fuel purchases). A data breach could compromise user identity, allow financial fraud, and enable an adversary to perform unauthorized profiling utilizing this data [39].

Beyond personal user hazards, vehicle sensor data vulnerability presents an operational hazard to IoV systems. For real-time decision-making, vehicles rely on lidar, radar, camera feeds, and in-vehicle telematics; if this sensor data is leaked or altered, attackers could guide vehicle control systems, causing operational failures, road accidents, or traffic disturbances. Third-party data abuse is still a growing problem as many cloud-based analytics, infotainment apps, and AI-driven driver assistance systems gather enormous volumes of driver and vehicle data. Third-party service providers may extract, analyze, and re-use user data without express permission and strong privacy enforcement systems, breaching GDPR and CCPA, and generating questions on data ownership and responsibility [40]. Since IoV environments need real-time data processing across the distributed edge and cloud architectures, conventional encryption systems and static access control policies fall short of offering sufficient protection against these inadvertent data leaks. Advanced AI-based privacy-preserving methods have become feasible answers to these problems. DP adds statistical noise to vehicle datasets, preventing adversaries from obtaining individual-level insights and so preserving data utility for analysis. By enabling distributed AI model training through FL, vehicles can cooperate in learning patterns without exchanging raw data, so lowering privacy risks. HE guarantees safe computation on encrypted data, enabling cloud services to process private vehicle data without decryption, thus reducing unwanted access risks. Combining secure multi-party computation and Zero-Knowledge Proofs (ZKPs) with these AI-driven systems provides scalable privacy-preserving solutions for protecting IoV networks while preserving effective vehicle communication and real-time processing capability [41].

Cyber threats in iov: inference attacks, man-in-the-middle, and advanced privacy violations

As V2X networks grow, the IoV has evolved into a high-value target for cybercriminals using privacy weaknesses to execute advanced cyberattacks. Critical security issues introduced by the integration of distributed vehicle ecosystems, wireless communications, and ML-based decision-making challenge conventional defense systems. Among the most urgent threats are inference attacks, Man-in-the-Middle (MITM) exploits,

spoofing, AI-powered data breaches, and vulnerabilities in Over-the-Air (OTA) updates. By inferring sensitive vehicle and driver information from ML model vulnerabilities, inference attacks seriously compromise vehicle privacy [42]. Unlike direct breaches, inference attacks combine telemetry data, traffic movement, and behavioral patterns to forecast vehicle locations, driver habits, and future travel paths. By letting attackers reverse-engineer AI models and extract private data, adversarial ML methods increase these hazards even more. To stop inadvertent data exposure, privacy-aware ML models, adversarial defense strategies, and safe FL architectures, all of which are urgently needed, are highlighted here [43].

Adversaries intercept and use IoV communications, including wireless networks, Vehicle-to-Cloud (V2C) connections, and Vehicle-to-Roadside (V2R) unit interactions in MITM Attacks. Attackers can inject false commands, change real-time vehicular communication streams, and carry out data exfiltration, manipulating traffic, illegal vehicle control, or major service interruptions [44]. Blockchain-backed security systems and AI-powered IDS are indispensable to maintain message integrity and identify abnormalities in real-time, given the high-speed character of vehicle data exchange. Since autonomous cars depend on digital signatures and cryptographic authentication keys to validate vehicle and infrastructure identities, spoofing and identity theft aggravate security issues even more. Attackers can pose as emergency vehicles, pretend to be RSUs, and fool IoV systems into carrying out illegal actions, thus posing significant traffic safety risks. Maintaining seamless interoperability in multi-vendor IoV systems depends on the adoption of privacy-preserving authentication models, such as ZKPs and blockchain-based decentralized identity verification, reducing identity spoofing risks [45].

The increasing reliance on AI-driven vehicle intelligence raises fresh concerns about privacy breaches driven by AI. Attackers can use ML weaknesses to control neural networks, inject adversarial noise, and fool autonomous car AI models into producing false security decisions or incorrect object detection classifications. Underlining the need for AI model robustness testing, adversarial ML defenses, and privacy-aware federated intelligence, these adversarial learning techniques can cause false security alerts, traffic flow manipulation, or disruption of autonomous navigation. Privacy breaches in OTA updates also cause great concern since IoV networks usually use OTA updates to improve vehicle functionalities, fix vulnerabilities, and maximize AI-driven decision-making [46]. Insecure update channels, however, let enemies inject harmful firmware, retrieve private vehicle logs, or change system settings to generate long-term security flaws. Ensuring tamper-proof, safe software updates in IoV environments depends on end-to-end encryption in OTA update channels together with blockchain-based software integrity validation. The degree and complexity of these cyber threats highlight how urgently real-time anomaly detection in IoV, AI-enhanced intrusion prevention, and ongoing privacy monitoring are needed. Although privacy-aware ML architectures, distributed blockchain security solutions, and AI-based adversarial defense models offer interesting countermeasures, their scalability, computational efficiency, and real-time adaptability in large-scale, high-mobility vehicle networks remain open challenges that demand more study and optimization [47].

Trust and authentication issues

Dynamic vehicle mobility, distributed network architectures, and multi-party communication needs all help to make trustworthy, privacy-preserving authentication in IoV networks intrinsically difficult. Without a centralized trust authority, Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) authentication becomes even more difficult; thus, IoV systems become open to illegal access, identity theft, and malicious impersonation attacks [48]. Decentralized identity management risks, cryptographic key management restrictions, interoperability issues in multi-vendor IoV systems, and privacy leaking in Vehicular Social Networks (VSNs) generate these authentication challenges. Since no uniform identity verification system exists across heterogeneous vehicle networks, distributed identity management risks present a major obstacle in IoV. Without appropriate authentication systems, attackers can use identity spoofing methods to impersonate legitimate vehicles, RSUs, or cloud service providers, enabling MITM attacks and illegal data interceptions. By removing third-party control over vehicle credentials and guaranteeing tamper-proof authentication through distributed ledger technology, blockchain-based decentralized Identity verification systems using Self-Sovereign Identity (SSI) frameworks offer a possible solution [49].

Another important problem is Key Management and Authentication Failures, since IoV networks rely on safe authentication based on cryptographic key exchanges. In high-density vehicle environments, where thousands of vehicles constantly authenticate with infrastructure nodes, traditional Public Key Infrastructure (PKI)-based authentication models find scalability problems. While guaranteeing effective, seamless authentication in real-time vehicle environments, quantum-resistant cryptographic techniques, lightweight cryptographic algorithms, and multi-factor AI-driven authentication models can improve privacy protection. Operability problems in multi-vendor IoV systems aggravate confidence-related problems even more. Modern IoV systems combine IoT sensors, cloud service providers, heterogeneous vehicle models, and V2X infrastructures created by several manufacturers [50]. Trust fragmentation between systems, protocol mismatches, and authentication inconsistencies resulting from this variety lead to Context-aware authentication can be facilitated by AI-powered trust evaluation models, which examine anomaly detection systems, vehicle activity logs, and past behavior patterns, while preserving user privacy. Furthermore, federated trust models using blockchain-based reputation management can produce flexible, distributed trust systems that change depending on car activity and interactions [51].

Privacy leakage in VSNs is a growing issue whereby vehicles actively exchange data in socially conscious IoV systems for crowd-sourced navigation intelligence, real-time traffic event detection, and cooperative route optimization. Lack of privacy-preserving trust systems exposes vehicles to location tracking, behavioral profiling, and targeted cyberattacks, thus raising the possibility of identity correlation and re-identification hazards. Blockchain-driven reputation systems and FL-enabled trust models can help sustain participant anonymity while supporting privacy-aware intelligence sharing [48]. In large-scale VSN systems, however, maintaining effective data aggregation, real-time processing, and strong privacy guarantees remains a difficult task. Dealing with these trust and authentication issues in IoV calls for scalable cryptographic key management systems, distributed identity verification methods, and flexible, AI-driven privacy-preserving authentication tools. Although suggested promising solutions include blockchain

authentication, AI-enhanced behavioral analysis, and multi-agent trust frameworks, their real-time feasibility, computational efficiency, and large-scale deployment capacity still depend on much study and improvement [52].

AI techniques for privacy preservation

By letting vehicles, infrastructure, and cloud services exchange data in real-time, the IoV is changing contemporary transportation. While this connectivity improves efficiency and safety, it also raises serious privacy concerns. Vehicles continuously transmit sensitive information, including location history, driving behavior, and even biometric data, across complex networks. Unfortunately, traditional privacy measures like encryption and access control often fall short due to the fast-moving, high-demand nature of IoV environments. Consequently, privacy-preserving methods driven by AI have become a promising approach to safeguarding data while preserving functionality. ML, combined with advanced cryptographic methods and decentralized security models, is now playing a key role in securing vehicular data. Among these techniques, FL keeps raw data on local devices and lowers exposure to possible breaches while enabling distributed AI training. By including controlled noise in data and stopping attackers from reconstructing sensitive information, DP adds a layer of security. HE allows computations on encrypted data, ensuring privacy without requiring decryption. Additionally, blockchain AI models provide decentralized trust, securing transactions and enabling privacy-focused data sharing across IoV networks. IoV systems are still susceptible to adversarial attacks, in which hostile actors use AI models to access private information or interfere with smart decisions, even with these developments. AML strengthens AI defenses, making models more resistant to attacks such as inference-based data extraction and model poisoning. AI-driven anomaly detection is another vital field since it allows real-time monitoring to find and stop privacy violations, so preventing cyber threats before they become more severe. Despite these innovations, AI-driven privacy solutions still face challenges, including computational overhead, communication delays, scalability limitations, and the need for real-time adaptation in large-scale IoV networks. Developing a thorough security strategy for connected cars is challenging because of the lack of a consistent framework to classify and assess current privacy-preserving methods. To fill this gap, we introduce a new taxonomy for AI-driven privacy preservation in IoV, organizing existing methods into six key categories: (1) FL for decentralized AI training without exposing raw data, (2) DP for safeguarding vehicular metadata through statistical noise, (3) HE for conducting AI computations on encrypted data without compromising confidentiality, (4) Blockchain AI for decentralized trust and secure data sharing, (5) AML to strengthen AI defenses against data extraction and poisoning attacks, and (6) AI-based anomaly detection to monitor and respond to privacy breaches in real-time. This taxonomy offers a methodical approach for assessing and advancing IoV network AI-driven privacy models. Our work prepares the ground for next-generation automotive privacy solutions that provide a mix of security, efficiency, and scalability in intelligent transportation systems by addressing important challenges, including optimizing computational efficiency, lowering latency, and improving real-time adaptability. Figure 2 displays the proposed taxonomy for IoV domain AI techniques for privacy preservation.

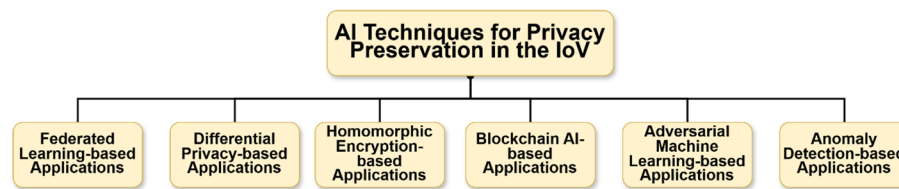


Fig. 2 Proposed Taxonomy for AI methods for privacy preservation in the IoV domain

Federated learning-based applications

Emerging as a great answer for guaranteeing efficiency, security, and privacy in the IoV is FL. FL reduces the data exposure risks by allowing distributed model training without raw data, thereby optimizing learning over several vehicle nodes. FL is frequently coupled in IoV systems with other security-enhancing technologies, including HE for privacy-preserving computations, blockchain for safe data sharing, and DP to guard against inference attacks. These integrations enable cars to jointly train AI models while maintaining local data confidentiality, thus enhancing autonomous driving, traffic prediction, and real-time decision-making. To solve important issues, including location privacy, safe data aggregation, and effective resource management, several models have been proposed to improve FL's performance in IoV. Furthermore introduced to lower communication overhead and enhance the scalability of FL-based models are methods including distributed consensus mechanisms and group leader choice. The adoption of these techniques has improved security, communication efficiency, and real-time adaptability inside large-scale vehicle networks. In this regard, to address privacy, security, and efficiency issues in the IoV, Ullah et al. [53] proposed a framework that combined blockchain technology with FL. Their approach guaranteed that data stayed on local devices, safeguarding user privacy while using a consortium blockchain to improve security and openness. Their method also included HE and a Dynamic Scaling Factor to enable privacy-preserving computations, best use of available resources, and enhanced computational efficiency. Their model also extracted meaningful insights from heterogeneous vehicle data streams using Gated Recurrent Units (GRU) inside a Convolutional Neural Network (CNN). Their framework underwent extensive testing, showing better security, communication efficiency, model accuracy, and convergence speed than more traditional approaches. Their privacy and security study verified that the framework resisted several cyberattacks. Their method gave vehicle data sharing a safe and effective means, enabling creative uses and improving traffic safety in IoV networks. Also, to lower location tracking risks while preserving high Quality of Service (QoS) and security efficiency, Adnan et al. [54] proposed a privacy-preserving framework that enhanced existing shadowing schemes in Vehicular Adhoc Networks (VANETs) and the IoV. Their approach presented a computationally effective group leader selection mechanism based on centeredness, rule adherence, and On-Board Unit (OBU) resources to lower overhead by 20%. By combining FL with DP, their system safeguarded private location data without sacrificing data utility, so improving location accuracy under privacy restrictions. By removing the single point of failure inherent in conventional shadowing methods, their method optimized group leader-based privacy schemes, enhancing scalability and resilience. Their long-running simulations involving up to 300 vehicles confirmed the scalability across vast vehicle networks and resilience against adversarial attacks of their framework. While improving security and performance in large-scale IoV systems, their

framework effectively lowered higher tracing ratios, reliance on single group leaders, and reduced utility in distance-based privacy techniques. Moreover, aimed to solve privacy leakage concerns and increase data aggregation efficiency, Xu et al. [55] proposed a privacy-preserving data aggregation protocol for the IoV with FL. Their approach secured model parameters and stopped leakage in semi-trusted RSUs employing Shamir's Secret Sharing scheme, Paillier Homomorphic Encryption, and blinding factor protection. By including dynamic training time windows, which let vehicles with varying computing resources engage effectively without greatly postponing the worldwide model updates, their model improved data aggregation efficiency. Their approach also maximized fault-tolerance systems, lowering the regular interaction of Trusted Authorities (TA), minimizing communication overhead, and enhancing scalability. While their performance evaluation showed great computational and communication efficiency in real-world vehicle environments, their security analysis verified that the protocol was resilient against adversarial model inference attacks and collusion threats. In addition, aimed to solve privacy, security, and communication overhead issues, Mun et al. [56] proposed a privacy-enhanced data aggregation scheme for FL in IoV. Their approach secured model aggregation and stopped privacy leaks from vehicle-generated datasets by combining Additive Secret Sharing (ASS) with HE and Asymmetric Encryption (RSA). Their method combined vehicles and chose only qualifying vehicles for involvement to lower communication overhead and increase scalability using an RSU-based group leader mechanism. Originally, their model included all RSUs and chosen vehicles in training, but later global model updates were carried out by randomly chosen groups to prevent malicious data injection attacks while preserving high accuracy. Their findings showed lowered communication costs, computational efficiency, and good preservation of privacy. Their method balanced security, efficiency, and predictive accuracy for traffic flow prediction and vehicle object detection applications, guaranteeing FL scalability in big IoV networks. Also, to improve data privacy and security in VANETs, Ali et al. [57] proposed a privacy-preserving vehicular prediction model that combined FL with blockchain technology. Their approach used FL to guarantee sensitive location data stayed on local devices by enabling distributed training of vehicle trajectory prediction models at the edge of the network. Their method also included a distributed database built on a blockchain to guard vehicle-generated data against illegal access and manipulation. Their model enabled quick response in crucial events by using vast-scale vehicle data to estimate proximity to emergency services and predict future vehicle locations. Their optimization system tracked vehicle speed and developed an attack model including countermeasures, enhancing the security and resilience of VANET communications. Their findings showed better emergency response efficiency, great predictive accuracy, and vehicle data security improvement. While guaranteeing strong privacy protection, their suggested approach greatly helped with traffic management, emergency response, and safety-critical vehicle applications. Finally, By integrating FL with distributed consensus mechanisms for vehicle-fog and inter-fog collaborative learning, Liang et al. [58] proposed a hierarchical decentralized learning framework that sought to improve security and efficiency in IoV networks. Their method addressed shortcomings in current distributed learning systems, including inadequate topological adaptability and accuracy degradation. Their system introduced a network-level masking mechanism to remove the need for masking pairing during inter-fog handover, preserving data privacy

by combining masking techniques with local training. They also tuned the consensus matrix to improve IoV network signaling efficiency. Their distributed consensus method guaranteed efficient cancellation of masking, enhancing learning accuracy and preserving strong privacy protection. Validating their superiority in safe and adaptive IoV learning environments, their experimental evaluation on two benchmark datasets showed that their framework outperformed current distributed learning methods in terms of learning accuracy, privacy protection, and signaling efficiency.

FL-based privacy-preserving models have certain difficulties, even if they provide great benefits for IoV security. The main advantage of FL is its capacity to distribute sensitive vehicle data, lowering the risks associated with centralized data storage. Blockchain integration guarantees tamper-proof and open data exchanges, so strengthening security and confidence. Furthermore, enhancing privacy is HE and DP's ability to enable computations on encrypted data and stop inference attacks. These methods, meanwhile, bring computational overhead and communication delays that might affect real-time IoV operations. Dependency on consensus-based learning systems and semi-trusted RSUs might also cause congestion and call for effective optimization techniques to strike a compromise between security and performance. Scalability is still a major issue, especially in big IoV networks where energy consumption and computational complexity have to be carefully controlled. Notwithstanding these difficulties, FL-based methods keep developing and provide interesting solutions for distributed, safe intelligence in connected cars. Table 4 shows federated learning-based applications in the IoV privacy preservation settings.

Differential privacy-based applications

Protecting private data in the IoV has become a major challenge for which DP has become indispensable. Preserving privacy while keeping data utility remains a major difficulty given the enormous volumes of vehicle data being exchanged across connected infrastructures. DP is a necessary mechanism for reducing inference attacks and stopping unwanted access since conventional encryption methods sometimes find it difficult to balance security with real-time communication needs. DP guarantees that, even in aggregated form for analysis, individual data points remain indistinguishable by adding controlled statistical noise to datasets. To enable distributed, privacy-preserving AI training while reducing data leakage, several uses have combined DP with FL. DP has also been applied in charging session projections for EVs, safe data-sharing systems, and vehicle intrusion detection. These techniques show the promise of DP in protecting IoV environments since they solve security flaws while maximizing network efficiency. Aiming to improve intrusion detection in the IoV, Chen et al. [59] proposed FDL-IDM guaranteed privacy protection. To avoid direct data sharing and lower privacy risks, their model used FL for distributed training. They also incorporated DP by minimizing communication overhead through noise perturbation to model updates before aggregation, so maintain privacy. Their method obtained spatiotemporal driving behavior data and segmented it depending on sender identity and temporal sequences to generate state matrices that changed with time. Their model reduced information loss by including an attention mechanism to combine outputs across hidden layers and time steps, thus enhancing the time series representation. In FL training, they also used a random scheduling technique, choosing random clients and dynamically changing the learning

Table 4 Federated learning-based privacy preservation methods in the IoV settings

Authors	Main idea	Advantage	Disadvantage	Method	Simulation platform	Dataset	Size of dataset	Application domain	Environments?
Ullah et al. [53]	Introducing a framework for privacy-preserving data sharing and security enhancement.	Ensured data privacy, High security, and optimized resource allocation	High overhead	FL + Blockchain, and HE with GRU-enhanced CNN	Python	Public dataset	Small	IoV security and intelligent transportation systems	IoV networks, V2X communication
Adnan et al. [54]	Introducing a framework for location privacy preservation in IoV using group leader selection.	Low overhead, improved accuracy	Potential performance bottlenecks and computational overhead in large-scale deployments	Optimized shadowing schemes with FL and DP	Python	Public dataset	Up to 300 vehicles	Vehicular networks, location privacy	IoV, VANETs, V2X communication
Xu et al. [55]	Proposing an FL-based data aggregation protocol for securing model parameters.	Improved privacy protection and reduced TA dependency	High overhead, dependency on semi-trusted RSUs for model aggregation	Shamir Secret Sharing, Paillier HE, and blinding factor protection	Python	Public dataset	Small	IoV, FL security, and privacy-preserving data aggregation	IoV networks, V2X communication
Mun et al. [56]	Proposing an FL-based data aggregation scheme using HE and ASS for IoV.	Low communication overhead, High scalability	High overhead	HE, ASS, and asymmetric RSA encryption	Python	Public dataset	Medium	Traffic flow prediction	IoV networks, V2X communication
Ali et al. [57]	Proposing a vehicular trajectory prediction model using FL and Blockchain.	High efficiency and enhanced traffic management	High overhead and high latency	FL + Blockchain	Python	Local dataset	Medium	Traffic prediction, emergency response	IoV, VANETs, V2X communication
Liang et al. [58]	Proposing a secure and efficient hierarchical decentralized learning.	High accuracy	Increased computational complexity	FL + distributed consensus, and network-level masking	Python	Two benchmark datasets	Large	IoV decentralized learning	IoV, fog computing, FL networks

rate to increase model stability across runs. On three public datasets, their evaluation achieved F1-scores of 0.9751, 0.9851, and 0.9789, showing great accuracy and strong privacy guarantees. Their approach improved the resilience of intrusion detection models against privacy leaks and adversarial attacks, lowering communication costs and effectively spotting security risks in IoV. Also, leveraging FL with Local DP (LDP), Kong et al. [60] proposed a privacy-preserving charging session prediction model that enabled safe data sharing among connected EVs while preserving privacy. Their model used FL to let vehicles cooperatively train a global regression network through parameter exchange under cloud coordination while maintaining raw data that remained dispersed. Their method used LDP to perturb transmitted model parameters, preventing the inference of private user information and reducing privacy risks presented by semi-honest third-party servers. Their method also brought elastic synchronization, which combined local and global models to increase accuracy despite LDP's noise introduction. Real-world charging data experiments confirmed that their model improved charging session prediction accuracy above conventional noisy FL methods. Their approach helped to improve travel convenience for EV drivers, secure energy system optimization, and green transportation options, preserving privacy. Moreover, to minimize inference and gradient leakage attacks, Batool, et al. [61] proposed a privacy-preserving data-sharing system for VANETs integrated with FL LDP. Their approach presented a two-layer security system whereby vehicles applied LDP before sending data to RSUs, ensuring that raw data remained obfuscated and safe. After that, their model trained the FL model parameters at the RSU level, and aggregated trained weights were updated using a central training server. Along with client-side accuracy measures over several training runs, their performance evaluation evaluated accuracy and simulation time for both local and global parameter sharing. Their experimental results showed improved security, resilience against inference attacks, and greater efficiency than conventional FL methods. Using LDP, Ju and Li [62] proposed a privacy-preserving data range query (PPQ) system that sought to improve privacy protection for owners of EVs in Vehicle-to-Grid (V2G) systems. Their method tackled important issues in EV charging services' balancing of privacy, accuracy, and efficiency, often dependent on outside platforms, compromising user privacy. To secure range searches in V2G environments, their approach presented a data encryption optimization model, a data storage optimization (DSO) model, and a Secure Hierarchical Decomposition (SHD) algorithm. Employing efficient data storage and faster data aggregation, they used Optimized Unary Encoding (OUE) to improve the trade-off between privacy and query accuracy, enhancing efficiency. Their simulation results showed that the PPQ scheme resisted distributed Denial-of-Service (DDoS), background knowledge, membership inference, and collusion attacks, among other attacks. Their results confirmed that PPQ preserved great privacy in EV data processing and maintained query accuracy and efficiency. In addition, the AI-based Trust-aware and Privacy-preserving System (ATPS) was designed by Li, et al. [63] to enhance data quality while safeguarding the privacy of vehicular data providers in VANETs. The system introduced two key components: Partial Ordering-based Trust Management (POTM) and Trajectory Privacy Preserving (TPP). The POTM scheme worked by evaluating and managing trust levels among vehicles using a structured ranking system. Fully trusted drones played a crucial role in this process, identifying the most reliable vehicles for data collection while filtering out potential malicious participants. On the other hand, the

TPP scheme leveraged Wasserstein Generative Adversarial Networks (WGAN) along with DP to protect sensitive trajectory data. By introducing carefully calibrated noise, the system ensured strong privacy protection while preserving the usability of the data. Through real-world experiments, the ATPS system significantly improved data quality by 45.76% to 52.57%, while reducing the number of malicious vehicle participants by 15.48% to 16.95%. These results demonstrated that ATPS effectively secured vehicular data, maintained trust in data sharing, and ensured a balance between privacy and utility within VANET environments.

Strong privacy guarantees and important functionalities, including intrusion detection, secure data aggregation, and trajectory anonymization, are given by the integration of DP in IoV applications. Using DP, FL-based models leveraging DP enable distributed training without exposing raw data, thereby lowering the risks associated with centralized data processing. Furthermore, helping to reduce inference and gradient leakage attacks is a DP-enhanced encryption method, preserves data confidentiality while preserving predictive accuracy. However, because of noise perturbation, which can affect real-time performance in large-scale IoV installations, these methods frequently bring computational overhead and accuracy trade-offs. Moreover, reaching an ideal balance between privacy protection and data utility remains difficult and calls for both effective parameter tuning and flexible privacy mechanisms. Notwithstanding these constraints, DP keeps showing promise for privacy-preserving IoV applications, security enhancement, scalability, and resilience against new cyber threats. Table 5 shows DP-based applications in the IoV privacy preservation settings.

Homomorphic encryption-based applications

Rising as a vital privacy-preserving method in the IoV, HE lets computations be done straight on encrypted data without calling for decryption. In vehicle networks where sensitive data such as vehicle trajectories, congestion patterns, and toll transactions must be handled securely while guaranteeing data confidentiality this capability is especially important. While efficient for protecting data at rest or in transit, traditional encryption techniques suffer from preserving privacy during real-time data analysis and model training. By allowing privacy-preserving computations across many applications, including vehicle congestion detection, deep learning-based anomaly detection, and secure FL frameworks, HE overcomes this constraint. Furthermore, integrated with blockchain, HE offers distributed trust models that guarantee safe data sharing in intelligent transportation systems and VANETs. HE combined federated learning, machine learning, and distributed architectures to improve privacy protection while keeping computational efficiency and model accuracy. Using Multilayer Perceptron (MLP) models to process horizontally and vertically partitioned vehicular sensing data, Li et al. [64] proposed a privacy-preserving vehicular congestion detection scheme that maintains data privacy and computational efficiency while their approach solved the heterogeneous encryption problem resulting from various automakers choosing different encryption systems, thereby impeding cooperative model training. Their method combined additive HE schemes and proxy re-encryption to overcome this obstacle, guaranteeing the confidentiality of both learned model parameters and raw sensing data and allowing safe computation. Their formal security analysis verified that their method guarded the MLP learning process against privacy leaks. Conducted on real-world datasets, their

Table 5 DP-based privacy preservation methods in the IoV settings

Authors	Main idea	Advantage	Disadvantage	Method	Simulation platform	Dataset	Size of dataset	Application domain	Environments?
Chen et al. [59]	Proposing FDL-IDM, a privacy-preserving IDS model using FL and DP in IoV.	High accuracy, minimized privacy risks, and reduced overhead.	High overhead	FL + DP + an attention mechanism	Python	Three public datasets	Large	Intrusion detection, IoV security, and privacy-aware threat mitigation	IoV, V2X networks, autonomous vehicle security
Kong et al. [60]	Proposing a model for charging session prediction in connected EVs.	High security	Low accuracy	FL + LDP + elastic synchronization for accuracy optimization.	Python	Real-world charging dataset	Large	EV charging session prediction, green transportation	IoV, EV networks, smart charging infrastructure
Batool et al. [61]	Proposing an FL-based secure data-sharing framework using LDP in VANETs.	Improved model efficiency	Computational overhead in RSU-based FL aggregation	LDP + FL	Python	Private	Small	Vehicular data sharing, FL-based VANET security	VANETs, IoV, V2X networks
Ju and Li [62]	Proposing an LDP-based privacy-PPQ scheme for EV charging services.	Improved privacy protection while maintaining accuracy	High computational overhead	OUE and SHD	Simulated V2G scenarios	Local	Small	EV charging privacy, V2G data security	IoV, smart grid, V2G networks
Li et al. [63]	Proposing a driven trust-aware and privacy-preserving system for VANETs.	Improved data quality, reduced malicious vehicle participation	Low scalability	WGAN-based trajectory privacy preservation with DP and trust evaluation via POTM.	Python	Public	Small	Vehicular data privacy	VANETs

experimental evaluation showed great efficacy and feasibility in training MLP models under several encryption systems for congestion detection.

Aiming to improve privacy protection and data security in VANETs by combining deep learning, blockchain, and HE Chen et al. [65] proposed a Decentralized Privacy-Preserving Deep Learning (DPDL) model. Their model presented a Decentralized VANETs (DVANETs) architecture whereby computing chores were offloaded from centralized cloud services to edge computing nodes to lower network congestion and transmission overhead. Using blockchain for trusted communication among vehicles, RSUs, and EC nodes, their approach established a safe data-sharing mechanism. Moreover, they encrypted transportation data on every EC node using FHE, guaranteeing privacy in deep learning model training employing confidentiality. Their model also made use of blockchain for a distributed ledger-based decentralized model updates, whereby local deep learning models kept their parameters on a distributed ledger for asynchronous sharing, preventing data corruption and illegal access. By proving that DVANETs and DPDL models enhanced network security while lowering computational and communication overhead in VANETs, their simulations validated the efficacy, scalability, and robustness of the proposed system. Moreover, Mohammed et al. [66] proposed a homomorphic, secure, FL-Enabled Pedestrian Detection System (HMFLS) that sought to improve security, lower resource leakage, and speed processing in intelligent transportation systems. As data-generating and processing sources, their model used Base Stations (BS), homogeneous FL servers, surveillance sensors, and traffic light components. Using VGG19 feature extraction from fog nodes and surveillance sensors, their method trained pedestrian and vehicle images using Generative Adversarial Networks (GANs). HE guarantees that computations can be done on encrypted data without decryption, maintaining privacy and confidentiality during vehicle and pedestrian detection. By attaining 98% security accuracy, reducing resource leakage by 50%, and lowering processing time by 52%, their extensive experiments, trained on 28,000 pedestrian and vehicle images, showed that HMFLS outperformed existing detection schemes. With an Android-based interface that makes simple integration into vehicles and mobile devices, their results confirmed HMFLS's efficiency in real-time intelligent transportation systems. In addition, Karim and Rawat [67] suggested TollsOnly architecture sought to improve privacy protection in mobile edge networks and IoV electronic toll transponder data. Their approach addressed privacy issues in urban vehicle tracking systems, in which toll transponders (e.g., E-ZPass) without consistent privacy protections have been placed by cities on sensors. Their approach guarantees that toll data stays encrypted through the computational process by introducing TollsOnly, a fully HE (FHE)-based privacy risk-reducing model. In line with the General Data Protection Regulation (GDPR) and California Consumer Privacy Act (CCPA) compliance criteria, their framework also included blockchain technologies to enable safe and user-owned data sharing. Their results underlined that TollsOnly could be a post-quantum privacy preservation mechanism, safeguarding driver identities and supporting congestion automation in smart transportation systems. Their scalable encryption model for IoV applications helped to close the gap between regulatory privacy criteria and actual toll data protection. Moreover, Ameer and Bouzeffrane [68] proposed a privacy-preserving framework for VANETs that sought to improve privacy and security in Intelligent Transport Systems (ITS). Their approach addressed privacy vulnerabilities brought about by the

dynamic and heterogeneous character of VANETs, thus exposing networks to attacks, including location tracking and Denial-of-Service (DoS). While maintaining data confidentiality using FHE, their method used K-Nearest Neighbors (KNN) as a lightweight yet effective machine learning algorithm for secure vehicle data analysis. Secure third-party computations and privacy-preserving network intelligence were made possible by this encryption system, which lets computations be performed on encrypted data without needing decryption. They underlined that the framework was flexible enough to enable integration with other ML models, enabling adaptation in different VANET settings. To maximize computation times while preserving strong privacy protections, their work also suggested future improvements, including HE combined with DP and secure multi-party computation. Their results confirmed that FHE-enabled VANET security systems could efficiently reduce privacy concerns, safeguard in-vehicle communication and infrastructure data, and secure vehicle interactions from outside hazards.

Employing HE adopted in IoV applications, strong privacy guarantees are given, enabling safe computations on encrypted data without violating confidentiality. While enabling cooperative AI model training in distributed environments, this method improves privacy in many spheres, including traffic congestion detection, pedestrian tracking, and toll transaction security. Though they have benefits, HE-based solutions sometimes bring notable computational overhead, which results in higher processing latency and resource consumption, especially in large-scale vehicle networks. Moreover, including HE with real-time AI-driven programs calls for optimization plans to minimize performance constraints and maintain security. Notwithstanding these difficulties, HE remains a potential solution for improving data privacy in IoV, supporting scalable transportation systems, safe vehicle intelligence, and effective data-sharing mechanisms. Future studies should concentrate on enhancing HE efficiency, lowering computational complexity, and integrating it with other privacy-preserving methods, including DP and secure multi-party computation, to improve security in practical vehicle applications. Table 6 shows homomorphic encryption-based applications in the IoV privacy preservation settings.

Blockchain AI-based applications

Blockchain and AI combined have become a potent strategy for improving IoV security, privacy, and trust enhancement. The very dynamic, distributed, and resource-constrained character of vehicle networks presents major difficulties for conventional centralized security systems. Blockchain is a perfect complement to AI-driven solutions because of its distributed and tamper-proof characteristics, which guarantee safe data exchange, authentication, and model aggregation. Blockchain is being used to improve FL, so allowing distributed AI model training and guaranteed privacy-preserving data exchange. Blockchain-based FL offers safe model updates without revealing actual vehicle data by combining cryptographic methods, including HE and consensus mechanisms. Furthermore, applied for secure knowledge sharing, autonomous decision-making, and anomaly detection in vehicle environments are blockchain-backed AI systems. By enforcing security policies, encouraging honest participation, and stopping malicious data manipulation, smart contracts help reinforce these systems even more. Particularly in fields including federated learning, object detection, and trust management, this section investigates how blockchain-integrated AI is changing IoV security, privacy,

Table 6 Homomorphic encryption-based privacy preservation methods in the IoV settings

Authors	Main idea	Advantage	Disadvantage	Method	Simulation platform	Dataset	Size of dataset	Application domain	Environments?
Li et al. [64]	Proposing a vehicular congestion detection scheme using MLP with HE.	High data confidentiality	High overhead	MLP + Proxy Re-Encryption, and Additive HE	Python	Real-world datasets	Large	Vehicular congestion detection,	IoV, smart transportation
Chen et al. [65]	Proposing a DPDL model for secure VANETs using blockchain and FHE.	Decentralized trust establishment and reduced network congestion.	High overhead, High latency	FHE + deep learning, blockchain, edge computing	Simulation	-	-	VANET security, privacy-preserving	IoV, VANETs, smart transportation networks
Mohammed et al. [66]	Proposing a system for privacy-preserving pedestrian and vehicle detection.	High accuracy, reduced resource leakage	High overhead, Low scalability	HE + FL, and GAN-based feature extraction using VGG19	Python	28,000 pedestrian and vehicle images	Large	Pedestrian and vehicle detection,	IoV, smart traffic systems
Karim and Rawat [67]	Proposing a framework for electronic toll transponder data.	High robustness	High complexity, High latency	FHE and blockchain for secure data sharing.	Python	Public	Small	Vehicle tracking	IoV, smart cities
Ameur and Bouzeffrane [68]	Integrating KNN for secure vehicular data analysis.	High security	High overhead	KNN, Fully FHE	Python	Public	Large	VANET security	VANETs, IoV, Intelligent Transport Systems

and efficiency. Using blockchain as a distributed framework for FL model aggregation, Wang et al. [69] proposed a Blockchain-based Privacy-preserving Federated Learning (BPFL) system that addressed security and privacy concerns in FL for IoV. By combining Multi-Krum technology with HE, their approach improved model aggregation security by ensuring that model updates were aggregated in ciphertext form, thus enabling privacy preservation and verifiability, and hence security. Their model distributed FL updates using blockchain, preventing dominant aggregation control and so reducing server-side threats. They used a reputation-based incentive system to offset negative client-side behavior, punishing bad data contributions and motivating IoV participants to provide honest, high-quality model updates. While their performance evaluation confirmed that their model enhanced FL training efficiency, improved security, and guaranteed privacy protection in vehicular networks, their security analysis showed that BPFL met fundamental security and privacy requirements. Also, Ayaz et al. [70] proposed a blockchain-based FL framework that sought to improve. By combining blockchain's immutability and consensus mechanisms with FL's distributed learning paradigm, their method addressed the mounting difficulties of high mobility, low latency, and data traffic surges in next-generation vehicles. Their approach guaranteed distributed incentives based on blockchain to guarantee dispersed FL model training and security. Their approach also made use of smart contracts and safe transactions to improve the credibility of FL procedures and so stop unwanted intervention in model aggregation. Their method proved to have at least 5% less failure rate than current solutions, even with 30% malicious nodes in the network, verifying its resilience in protecting IoV networks under adversarial circumstances. Their results underlined the interesting possibilities of blockchain-enhanced FL in 6G IoV as well as the difficulties in application, including computational overhead, consensus latency, and scalability issues. Their work suggested future directions of research to maximize blockchain integration with FL for adaptive, privacy-preserving vehicle communication systems. Moreover, Jiang et al. [71] proposed an edge AI framework for object detection in the Visual-IoV. Their approach solved the problems of object similarity and background interference, which historically hampered real-time object recognition in V-IoV settings. Their method combined AI with MEC, allowing effective edge-based processing. Combining symbolic and connectionist AI, they presented an abductive learning algorithm to improve interpretability and robustness in object detection systems. Their framework applied the You Only Look Once (YOLO) classifier using abductive reasoning to enhance model adaptability in challenging V-IoV environments. Using blockchain technology, they guaranteed distributed, decentralized and tamper-proof AI models among IoV edge nodes, improving security and model sharing. Their simulation findings exceeded current AI-based object detection models by showing great robustness and detection accuracy. Their results confirmed that edge AI enhanced by blockchain technologies could be feasible for interpretable V-IoV object detection with security. In addition, Chai et al. [72] proposed a hierarchical blockchain and FL framework for knowledge sharing in the IoV sought to Their approach solves the difficulties of safe AI-based knowledge sharing, in which conventional centralized AI models failed to operate efficiently in distributed IoV networks on a large scale. While a hierarchical FL algorithm was meant to fit the distributed character of IoVs, their method presented a hierarchical blockchain structure to guarantee tamper-proof and decentralized knowledge sharing. Employing machine learning approaches

and securely sharing knowledge with other vehicles, their framework allowed vehicles to learn from environmental data and preserve privacy and decentralization. Using a multi-leader, multi-player game-theoretic model to incentivize data exchange among vehicles, their approach also modeled knowledge sharing as a trading market process. While the blockchain architecture essentially reduced some malicious attacks, guaranteeing safe and dependable vehicle cooperation, its simulation results showed that the hierarchical FL model improved sharing efficiency and learning quality. As well as, Hai et al. [73] suggested a game-theoretic Confidentiality, Integrity, and Authenticity (CIA) security system for IoV, seeking to improve vehicle network security through Blockchain and AI-driven authentication methods. Their approach solved the difficulties presented by IoV's transparency, self-organizing constraints, and vulnerability to hostile attacks. Their architecture presented a three-layer security model comprising vehicle authentication using Physical Unclonable Functions (PUFs), a flexible Proof-of-Work (dPOW) consensus mechanism, and AI-enhanced duel gaming for trust evaluation. Their method guaranteed safe and tamper-proof authentication, reducing side-channel and physical cloning attacks, many of which current blockchain-based IoV verification systems failed to handle. In terms of security robustness, computational efficiency, and transaction overhead reduction, security analysis confirmed the superiority of their mechanism. Their results showed that their blockchain-backed, AI-enhanced CIA model offered IoV systems a very effective and highly secure authentication method.

Blockchain and AI working together in IoV is a quite successful way to solve security flaws, enable distributed model training, and enhance data integrity. These techniques guarantee privacy-preserving computations, reduce single points of failure, and build vehicle network trust utilizing Blockchain-enhanced FL models to stop illegal data access and let cars cooperate in securely based intelligent models. Furthermore, greatly lowering risks connected with adversarial attacks and data manipulation are blockchain-based authentication and anomaly detection systems. Notwithstanding these benefits, blockchain-AI integration creates problems, including computational overhead, higher latency resulting from consensus systems, and scalability issues in large-scale IoV installations. Future studies must focus especially on ensuring real-time adaptability and optimizing blockchain transactions. Blockchain-based AI applications in IoV can keep developing safe, privacy-preserving, and scalable intelligent transportation systems through improved AI model efficiency, resource economy, and blockchain consensus model refining. Table 7 shows blockchain-based applications in the IoV privacy preservation settings.

Adversarial machine learning-based applications

A major tool for improving privacy and security in the IoV and associated intelligent transportation systems is AML, which has attracted much interest. These systems are vulnerable to adversarial attacks aiming to extract sensitive data, alter model outcomes, or interfere with system operations, as increasing reliance on AI-driven models for real-time decision-making exposes them. By including methods that improve model robustness, data privacy, and safe communications, AML-based applications concentrate on resisting such hazards. In IoV, AML is essential for trajectory anonymizing, privacy-preserving data sharing, and security against inference and poisoning attacks driven by AI-driven caches. Differential privacy, GANs, and reinforcement learning help these

Table 7 Blockchain-based privacy preservation methods in the IoV settings

Authors	Main idea	Advantage	Disadvantage	Method	Simula- tion platform	Dataset	Size of dataset	Application domain	Environ- ments?
Wang et al. [69]	Integrating HE and multi-Krum for secure IoV model aggregation.	Ensured verifiable model aggregation	Potential computational overhead due to blockchain integration and HE operations	FL + Blockchain, HE	Python	Private	Small	IoV security, privacy-preserving federated learning,	IoV, V2X net-works, smart transportation
Ayaz et al. [70]	Proposing a blockchain-based FL frame-work to enhance privacy.	Low failure rate, and ensured secure incentive distribution,	High overhead	FL + blockchain, and smart contracts	Python	Private	Large	6G-enabled IoV security	IoV, 6G, block-chain-based FL networks
Jiang et al. [71]	Integrating abductive learning and blockchain for security and interpretability.	High accuracy and robustness, High interpretability	High overhead	YOLO-based object detection, abduc-tive learning for AI interpretability	Python	Public	Small	V-IoV object detection	IoV, Edge AI, Intelligent Transportation Systems
Chai et al. [72]	Proposing a framework for secure and efficient knowledge sharing in IoV.	Improved sharing efficiency and learning quality.	High computa-tional complexity	FL -preserving AI training	Python	Public	Large	IoV knowledge sharing	IoV, intelligent transportation systems,
Hai et al. [73]	Proposing a blockchain, AI, and dual gaming for authentication and trust management.	Improved tamper-proof authentication and security	High complexity	PUFs, dPOW-based blockchain consensus	Python	Public	Large	IoV cybersecurity	IoV, intelligent transportation systems

methods create resilient AI models that preserve data utility while protecting user privacy. Furthermore used to reduce location inference attacks in autonomous cars are adversarial training and privacy-preserving deep learning methods, guaranteeing that real-time decision-making stays safe and trustworthy. In this regard, Wang et al. [74] proposed a differential privacy-preserving deep learning caching framework (DP-DLCF) sought employs a multivariate randomized response mechanism. Their approach perturbed mobile client-sensitive information, so guaranteeing DP protection and minimizing privacy-utility trade-offs via adaptive privacy budget allocation based on the least square method. Their method included selective integrated learning approaches for local model enhancement, optimizing neural network-based caching performance while maintaining privacy, thus enhancing the cache prediction accuracy. Their architecture maximized content distribution in heterogeneous communication networks at the same time as effectively securing user data in telemedicine diagnosis and disease monitoring systems. Their simulation results showed lower privacy leakage, better caching prediction accuracy, and improved performance of privacy-aware deep learning models. Also, Shi et al. [75] proposed a DRL-based TNC route scheduling approach that sought to Their approach dynamically change vehicle paths by considering vehicle locations, time of day, and competition among TNC drivers using a deep Q-network (DQN) algorithm. Their method adopted a distributed architecture, bringing the processing closer to terminal users, lowering computational latency, and accelerating the training pace. Their approach guaranteed that uploaded location information stayed hidden by using a geo-indistinguishability scheme based on differential privacy, safeguarding passenger-sensitive location data. Using open datasets from Didi Chuxing, their simulation studies showed notable decreases in cruise time and increases in driver earnings, thus validating the efficiency and privacy-preserving character of their approach. Moreover, Fan et al. [76] proposed a Two-Layer Privacy-Preserving (TLPP) mechanism that sought through structured data. Their model-derived regional and density features of trajectory points using density-based clustering, while preserving trajectory privacy. Their method learned the distribution of clustered trajectories and produced synthetic trajectory data by using a GAN with a Long Short-Term Memory (LSTM) network. The first layer of privacy protection is formed from these synthetic trajectories, substituting for the original vehicle trajectory data. Their approach presented a customized DP mechanism, which applied region-specific perturbations to the data, forming the second layer of privacy protection, further preserving user privacy. To guarantee data integrity and reduce privacy risks, they also developed a new generator loss function using Wasserstein distance to measure spatial similarity between real and synthetic trajectories. Their model outperformed other privacy-preserving strategies in IoV settings by clearly safeguarding user privacy while maintaining trajectory data usefulness. In addition, with high data utility for perception systems, their proposed DeepClean framework sought to reduce location inference attacks on Autonomous Vehicle (AV)-generated camera data. To create privacy-preserving synthetic images, Adeboye et al. [77] model combined GANs and Variational Autoencoders (VAE), using a hybrid deep-learning approach. Their approach presented a private clustering method to identify private and non-private object regions inside an image, ensuring that non-sensitive areas (e.g., roads) remained visually interpretable, while private objects (e.g., recognizable landmarks, signs) were obfuscated using a Gaussian Mixture Model (GMM). Their method distorted the private object

areas while preserving general scene integrity; hence, location inference attacks were much less successful. DeepClean significantly increased privacy protection by lowering geo-localization attack accuracy to below 4%, preserving synthetic image quality equivalent to current benchmarks. Their results confirmed DeepClean's efficiency against robust attack models and its possibility for privacy-preserving AV vision data processing. Focusing on heterogeneous Radio Access Networks (RANs), beyond-RAN networks, and end-to-end network slices, Wu et al. [78] examined the function of deep learning in privacy preservation inside 5G heterogeneous networks. Their study investigated how deep learning models might reduce privacy risks connected with different cyber threats and inference attacks in IoT-connected autonomous systems while extracting useful insights from vast-scale network data. Their method assessed the viability of adversarial DL models to guard private user data against advanced attacks and investigated privacy vulnerabilities across several network levels. They underlined countermeasures driven by deep learning to improve privacy in cross-domain network communications, autonomous car data exchanges, and user mobility tracking. Their work also highlighted important research issues, including scalability constraints, real-time processing overhead, and adversarial robustness against changing cyber threats in deploying privacy-preserving deep learning models in 5G networks. Emphasizing the integration of deep learning with privacy-aware network architectures to strengthen 5G IoT systems, their results offered a disciplined road map for the next research.

Strengthening data privacy, reducing adversarial risks, and optimizing AI-based vehicle applications through AML integration into IoV security systems have shown notable benefits. Sensitive vehicle information has been effectively protected while preserving computational efficiency using techniques including different private deep learning, reinforcement learning for safe scheduling, and GAN-based data anonymizing. These techniques minimize privacy leaks, stop location tracking, and improve model dependability against changing hazards. AML-based apps do, however, also bring difficulties, including more computational complexity, trade-offs between privacy protection and model accuracy, and real-time adaptation in dynamic vehicle environments. Future developments in AML for IoV should concentrate on maximizing computational efficiency, lowering training overhead, and building more resilient defenses against adversarial AI hazards. AML will remain essential in ensuring intelligent transportation networks using improved strategies, balancing privacy, security, and efficiency in large-scale IoV installations. Table 8 shows AML-based applications in the IoV privacy preservation settings.

Anomaly detection-based applications

Intelligent vehicle networks now mostly rely on anomaly detection to guarantee safe, effective, and privacy-preserving communication between cloud-based services, roadside infrastructure, and autonomous cars. Real-time monitoring systems in the IoV must identify cyber threats, data breaches, and road hazards while preserving computational efficiency, given the growing volume of data transmission in this system. DL, DP, and multi-access edge computing are used by AI-driven anomaly detection methods to enhance intrusion detection and road safety. Whereas machine learning-based intrusion detection systems improve cybersecurity in vehicle ad hoc networks, context-aware privacy-preserving models dynamically change security measures depending on

Table 8 AML-based privacy preservation methods in the IoV settings

Authors	Main idea	Advantage	Disadvantage	Method	Simulation platform	Dataset	Size of dataset	Application domain	Environments?
Wang et al. [74]	Proposing a DP-DLCF for secure data sharing in heterogeneous networks.	High accuracy, ensured DP with adaptive budget allocation,	High computational complexity	DP using multivariate randomized response.	Python	Public	Large	Telemedicine diagnosis	Heterogeneous communication networks
Shi et al. [75]	Proposing a route scheduling approach using DP for location protection.	Improved driver earnings and preserved passenger location privacy	High computational complexity in DQN training	DQN	Python	Didi Chuxing dataset	Large	TNC services, ride-sharing optimization, privacy-aware route scheduling	IoV
Fan et al. [76]	Proposing a mechanism using GANs and DP for vehicle trajectory anonymization.	High privacy protection, improved synthetic data accuracy.	High overhead	GANs with LSTM	Python	Local	Small	Vehicle trajectory privacy	IoV, GPS-based transportation networks, smart mobility
Adeboye et al. [77]	Proposing an adversarial learning-based privacy-preserving model.	Reduced attack accuracy to below 4%, ensured high synthetic image quality	High overhead	VAE, GANs, and private GMM	Python	Private	Large	Autonomous vehicle vision	IoV, AV perception systems, smart transportation
Wu et al. [78]	Investigating DL-driven privacy preservation in 5G heterogeneous networks.	High security	Lacked specific adversarial learning attack evaluations.	DL techniques	Not mentioned	Not mentioned	Not mentioned	5G privacy protection,	5G heterogeneous networks, IoT

environmental conditions in autonomous driving. Advanced CNNs and autoencoders also examine sensor data to identify anomalies in road conditions, fog-assisted IoV systems, and next-generation cars such as 6G-enabled IoV. These methods guarantee that anomaly detection stays strong, flexible, and scalable in many vehicle environments, preserving real-time communication and transportation security. In this regard, Gheisari et al. [79] proposed a Context-Aware Privacy-Preserving method for Autonomous Driving (CAPPAD) guaranteed service efficiency. Their approach introduced dynamic privacy management using the Software-Defined Networking (SDN) paradigm, enabling AVs to flexibly choose privacy-preserving strategies depending on data sensitivity and emergency conditions. Their model used DP or Data Aggregation (DA) based on sensitive and non-sensitive data, respectively. Whereas in an emergency, AVs aggregated and transmitted vital data without significant disturbance, in non-emergency cases, AVs used DP-based obfuscation to minimize privacy leakage and ensure timely responses from RSUs and other cars. CAPPAD outperformed current methods in privacy protection and transmission efficiency while maintaining low-latency emergency response, utilizing extensive simulations using MININET-WIFI, evaluating privacy-preserving effectiveness, computational cost overhead, latency, and computational complexity. Also, Alladi et al. [80] proposed an AI-driven intrusion detection system for IoV that sought to improve cybersecurity by using Deep Learning Engines (DLEs) for attack classification and detection. Their approach solved the constraints of conventional intrusion detection systems, which battled the explosion of vehicle data and the complexity of cyberattacks in major IoV systems. To lower latency and support real-time intrusion detection in dynamic vehicle networks, their method applied DLEs on MEC servers instead of cloud-based platforms. Their architecture categorized vehicle network traffic into possible cyberattack forms, facilitating faster threat detection and mitigation. Their approach attained high accuracy in attack detection, lowered average prediction times, and enhanced the adaptability of intrusion detection in real-time vehicular scenarios according to thorough experimental evaluations on an MEC testbed. Their results confirmed that anomaly detection models driven by AI could solve IoV cybersecurity problems. Moreover, Bibi et al. [81] proposed an AI-based road anomaly detection system for autonomous cars, seeking to improve traffic safety. Their approach addressed growing road surface flaws brought on by poor building materials, climate change, and heavy traffic loads. Using onboard cameras and deep learning models to classify surface flaws, their framework combined Edge AI and VANET-based real-time road anomaly detection, whereby autonomous vehicles recorded road images. For automated road anomaly detection and classification including potholes, bumps, and cracks their model used Visual Geometry Group (VGG-11) and Residual Convolutional Neural Network (ResNet-18). Higher accuracy in spotting road surface problems, and simulation results confirmed the excellence of their deep learning models over current road defect detection methods. Their method allowed autonomous cars to communicate emergency information to approaching ones, so lowering accident risk and guaranteeing better traffic flow. As well as, Yaqoob et al. [82] proposed the CAaDet (Convolutional Autoencoder Aided Anomaly Detection) model sought to detect anomalies during information distribution to improve security in Fog-assisted IoV (Fa-IoV) networks. Their approach addressed important security and privacy issues in Fa-IoVs, which endanger smooth vehicle communication and data integrity. Their method used convolutional layers

and tailored autoencoders for feature extraction and threat detection, integrating deep learning-based anomaly detection. Their model dynamically adjusted hidden neurons to reduce false alarms and increase detection accuracy by analyzing fog node behavior. Employing the NSL-KDD benchmark dataset, the experimental evaluation revealed that CAaDet exceeded current anomaly detection approaches. Their results underlined security flaws in Fa-IoVs and underlined the need for more study on dynamic, AI-driven anomaly detection systems. Finally, Wang et al. [83] proposed an AI-native Vertical Heterogeneous Networks (VHetNets) framework aimed to integrate AI-driven anomaly detection with VHetNets to enhance security and network management in 6G and beyond networks. Their approach leveraged VHetNets as a distributed infrastructure for AI model training, enabling real-time intrusion detection, state monitoring, and security supervision in IoT environments. Their model incorporated distributed AI approaches for anomaly detection and intelligent network management, optimizing resource allocation and network performance. Their framework facilitated the efficient detection of network anomalies while enhancing autonomous decision-making in VHetNets. A case study demonstrated the effectiveness of the AI-native VHetNets-enabled anomaly detection framework, validating its efficiency in detecting security threats across heterogeneous IoT environments. Their findings highlighted the potential of AI-driven VHetNets to provide scalable, adaptive, and intelligent network security solutions.

In autonomous and linked vehicle networks, AI-driven anomaly detection systems offer major benefits in enhancing security, privacy, and efficiency. These models maximize response mechanisms for network anomalies, enable privacy-aware data transmission, and improve real-time threat detection. Still important challenges, though, are high computational overhead, scalability issues, and possible false positives in anomaly detection. Although deep learning models enhance security monitoring and classification accuracy, their deployment in vast-scale vehicle networks calls for additional optimization to balance performance and computational efficiency. Future developments should concentrate on perfecting AI models to lower latency, increase adaptability to dynamic IoV conditions, and strengthen the anomaly detection system's resistance against new cyber threats. Integrating these advances will help AI-based anomaly detection systems remain essential in maintaining safe, private-preserving vehicle communication and securing smart transportation systems. Table 9 shows anomaly detection-based applications in the IoV privacy preservation settings.

Results and analysis

In this part, we investigate several main parts, like analysis of data, the implications of the work, datasets, XAI methods, and their importance in the field.

Analysis of data

Regarding vehicle networks and AI-driven privacy-preserving strategies, the way approaches are distributed reveals important new perspectives on the priority given to several security systems. Reflecting the growing relevance of distributed machine learning techniques, allowing vehicles to cooperatively train models while preserving data privacy, Fig. 3 shows the highest proportion (20%) allocated to FL and privacy-preserving data sharing. Dependency on FL is a reaction to growing worries about centralized data collection, which poses security risks and illegal access. Moreover, including FL

Table 9 Anomaly detection-based privacy preservation methods in the IoV settings

Authors	Main idea	Advantage	Disadvantage	Method	Simulation platform	Dataset	Size of dataset	Application domain	Environments?
Gheisari et al. [79]	Proposing a context-aware privacy-preserving method using DP and data aggregation in AVs.	Low latency in emergency transmissions.	High overhead	SDN-based privacy management, DP for non-emergency cases	MININET-WIFI	Public	Large	secure IoV communication, and emergency response optimization	IoV, V2X, smart transportation
Alladi et al. [80]	Using DLEs deployed on MEC servers for real-time cyberattack detection.	High accuracy, reduced latency	High computational overhead	DLEs for anomaly detection	MEC testbed	Local	Medium	IoV cybersecurity	IoV, MEC, intelligent transportation systems
Bibi et al. [81]	Proposing an road anomaly detection system, using DL models for real-time traffic safety.	High accuracy	High computational overhead	ResNet-18 and VGG-11	Python	Public	Large	Road anomaly detection,	IoV, VANET
Yaqoob et al. [82]	Proposing an DL-based anomaly detection model, using a convolutional autoencoder.	High accuracy, minimized false alarms.	Low scalability	Convolutional autoencoders	Python	NSL-KDD	Large	Anomaly detection	IoV
Wang et al. [83]	Proposing an architecture for anomaly detection in 6G and IoT.	High accuracy	Low scalability	VHetNets	Python	Large	Public	AI-driven security	6G, IoT, VHetNets, Smart Cities

with privacy-enhancing technologies like HE and DP has raised the efficiency of safe data exchange in mobile environments. Comprising 18%, blockchain-enhanced privacy & security is the next important category that emphasizes the rising acceptance of distributed ledger technologies for safe vehicle communication and authentication. Blockchain's transparency and immutability help to improve security by reducing illegal data access and preventing malicious activity. Researchers have been able to enhance privacy without sacrificing data integrity by combining blockchain with AI-driven FL and cryptographic systems. The emergence of blockchain-based federated models shows even more of the possibilities of distributed trust systems in automotive networks. Not far behind, 15% of the research is on HE for secure computation, stressing the need for encryption methods that let computations be done straight on encrypted data without revealing raw data. Applications needing real-time privacy preservation, vehicular congestion detection, secure tolling systems, and privacy-aware autonomous decision-making have especially benefited from this approach. Although the computational overhead is still a difficulty, developments in efficiency-oriented encryption techniques keep improving practical relevance. DP and adversarial learning account for another significant share, 14%, which emphasizes the continuous investigation into noise-based privacy mechanisms and strong AI models able to resist adversarial threats. Especially in areas like location privacy, trajectory obfuscation, and secure data aggregation, DP has been extensively included in vehicle networks to prevent data inference attacks. Conversely, adversarial learning-based defenses have been used to fortify AI models against poisoning attempts aiming at influencing vehicle decision-making procedures. Ensuring the durability of models against adversarial threats remains a top research issue as AI-powered transportation networks grow. Focused on intelligent security monitoring mechanisms using deep learning to detect and mitigate cyber threats in-vehicle networks, AI-driven IDS, and anomaly detection accounts for 13% of the research terrain. By distributing anomaly detection across several communication layers, integrating AI-native VHetNets has further optimized security, improving both real-time responsiveness and attack-mitigating capacity. The adoption of multi-layered deep learning models, including convolutional autoencoders and Recurrent Neural Networks (RNNs), has helped to create highly flexible intrusion detection systems that dynamically react to newly developing threats. 10% or less is dedicated to privacy-preserving data aggregation and object detection, indicating the growing need for safe, quick aggregation methods that guarantee privacy while allowing important features like traffic monitoring and intelligent decision-making. GANs have become popular for privacy-preserving data synthesis since they let synthetic datasets replace sensitive vehicle data while keeping analytical accuracy. Furthermore, privacy-aware approaches improve object detection methods so that AI-driven perception models in autonomous cars do not violate user confidence. Finally, Game-theoretic security models reflect 10% of the growing curiosity in using game theory to model adversarial interactions in vehicle networks. Game-theoretic models improve authentication mechanisms, intrusion response systems, and resource allocation strategies by using strategic decision-making ideas, thus guaranteeing strong cybersecurity defenses in intelligent transportation systems. These models have shown especially great success in reducing real-time security breaches in distributed vehicle systems.

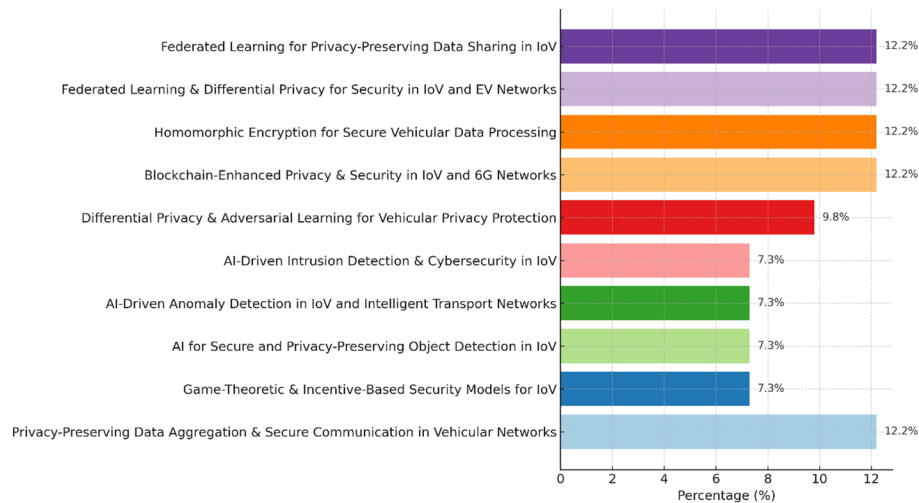


Fig. 3 Distribution of AI-driven privacy-preserving techniques in IoVs

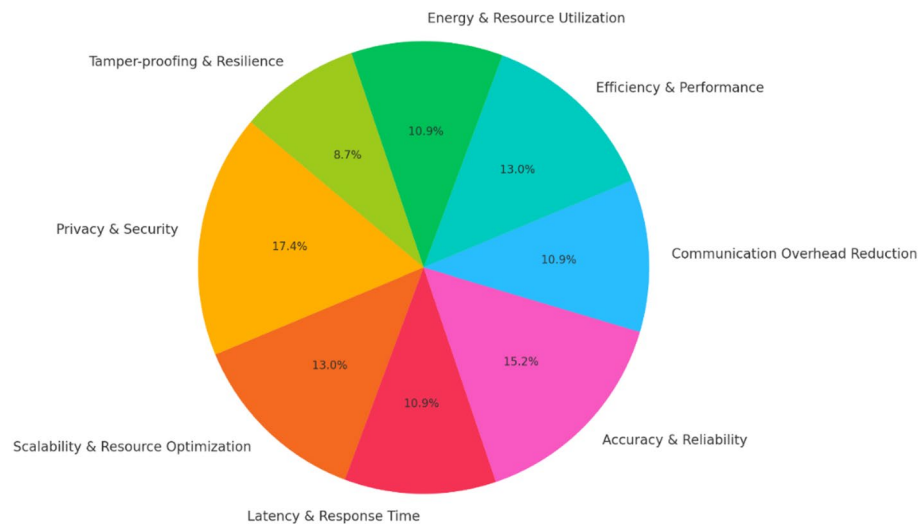


Fig. 4 Distribution of QoS and QoE parameters in privacy-preserving AI for IoV networks

The analysis of privacy-preserving AI systems inside vehicle networks clearly shows that optimal system performance depends critically on several QoS and QoE criteria. Privacy and security are the main issues, as shown in Fig. 4; they account for a major share of the improvements brought about in current research. This trend emphasizes how urgently car networks must be strengthened against adversarial attacks, inference threats, and illegal data access. Among the main approaches used to preserve privacy standards while keeping data utility are HE, blockchain, FL, and DP. The general acceptance of these approaches emphasizes the important harmony in real-world applications between security guarantees and computational feasibility. As seen by the significant allocation of developments in this field, scalability and resource optimization are also basic features of in-vehicle AI installations. Techniques, including hierarchical federated learning, distributed ledger technologies, and decentralized consensus mechanisms, have been suggested to guarantee seamless scalability without overwhelming network resources, considering the dynamic character of IoV environments. These techniques

reduce dependency on centralized authorities by allowing adaptive learning and real-time updates, so mitigating congestion in high-mobility vehicle environments. These scalability-driven solutions guarantee that performance stays constant and efficient even as vehicle AI grows over more extensive systems. Another crucial consideration is latency and response time, especially in mission-critical vehicle applications when real-time decision-making is vital. Reducing communication delays has been much aided by the deployment of lightweight machine learning frameworks and MEC-driven AI architectures. Response times in IoV networks have been further reduced by dynamic data aggregation and privacy-aware caching systems, so improving the general driving experience and safety. Likewise, with AI-driven anomaly detection and deep reinforcement learning methods honing predictive analytics and threat detection in-vehicle networks, improvements in accuracy and dependability remain the priority. Accurate identification of road defects, cyber threats, and privacy violations guarantees the strength of vehicle AI systems. The integration of FL-DP and LDP shows how significant efforts to lower communication overhead have also been. These methods prevent model inversion and inference-based attacks as well as simplify the encrypted data exchange. These methods help to maximize network resources by reducing duplicated transmissions and improving data aggregation methods. Another major area of research has been the energy efficiency of vehicle AI applications; HE and blockchain-based systems include adaptive encryption techniques to reduce too heavy computational loads. Employing optimal cryptographic operations and cutting processing time has resulted in appreciable power consumption reduction throughout intelligent transportation systems. Tamper-proofing and resilience, where game-theoretic models and adversarial training have enhanced the resilience of vehicle AI systems, also have great importance [84].

Figure 5 shows how important issues related to QoS and QoE in privacy-preserving AI models for vehicle networks are distributed. Comprising 25% of all the problems, computational overhead turns out as the most important obstacle. This is mostly the result of the strong reliance on homomorphic encryption, FL, and blockchain, all of which create major processing requirements. These computationally demanding operations

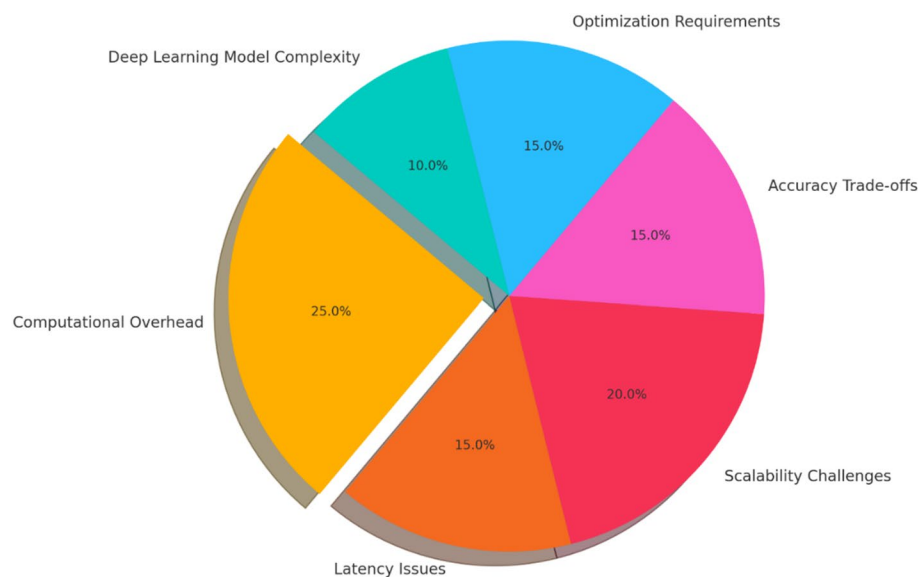


Fig. 5 QoS and QoE challenges in privacy-preserving AI for vehicular networks

sometimes result in resource limitations in real-time systems, which influences the general efficiency of automotive AI models. Representing 15%, latency problems highlight the delays brought about by distributed learning systems, encryption techniques, and blockchain consensus mechanisms. These techniques bring network latency, especially in large-scale IoV environments where real-time communication is essential for safety and decision-making, even while they improve privacy and security. Comprising 20%, scalability issues highlight the challenges of applying privacy-preserving AI models over large-scale automotive networks. To run effectively in real-world large-scale deployments, federated learning, homomorphic encryption, and adversarial machine learning all depend on major optimization. Ensuring perfect scalability gets more challenging as IoV networks grow with increasing numbers of connected cars and smart infrastructure elements. Accuracy trade-offs, which account for 15% of all the difficulties, are another important concern. Many privacy-preserving methods, including perturbation-based approaches and DP, add noise to mask private information. Although these methods help to protect privacy, they frequently compromise the accuracy of AI models. Applications needing great accuracy, such as trajectory prediction and intrusion detection in-vehicle networks, especially find this challenging. Representing 15%, optimization requirements show the need for additional fine-tuning and validation of AI models. Many of the suggested fixes have not been widely used at scale, and their performance in dynamic vehicle environments is yet unknown. Techniques, including privacy-aware trajectory prediction, hierarchical FL frameworks, and blockchain-based authentication, call for further improvements to guarantee they satisfy the high criteria of IoV applications. Finally, the complexity of deep learning models explains 10% of the difficulties. Although AI-driven models greatly advance vehicle intelligence, their complexity sometimes introduces processing overhead requiring large computational resources. Resource-limited vehicle environments can computationally restrict training GANs, CNNs, and reinforcement learning models. All things considered, this study emphasizes the trade-offs included in privacy-preserving AI solutions for vehicle networks. These techniques bring computational, scalability, and accuracy issues that must be addressed for practical deployment, even if they offer strong security and privacy improvements [85].

DL, ML, mathematical modeling, metaheuristic optimization, heuristic approaches, game theory, and reinforcement learning are among the several approaches used in proportionate measures in Fig. 6. In IoV systems, these methods are essential for guarantees of security, efficiency, and privacy preservation. Particularly in real-time traffic analysis, intrusion detection, and privacy-aware trajectory modeling, the predominance of DL (25%) and ML (20%) underlines the increasing dependence on data-driven techniques for intelligent decision-making. These techniques use enormous volumes of vehicle data to identify anomalies, maximize resource allocation, and enhance security in dynamic vehicles. Widely used for organized problem-solving, including optimizing FL aggregation, minimizing computational overhead, and improving network efficiency, are mathematical modeling (15%) and metaheuristic optimization (10%). These models develop exact mathematical representations of privacy and security constraints, guaranteeing strong security solutions. Game theory (10%) and heuristic approaches (10%) offer flexible means to handle adversarial threats and uncertainty in IoV systems. Resilient vehicle AI systems are made possible by these techniques, which also design safe incentive

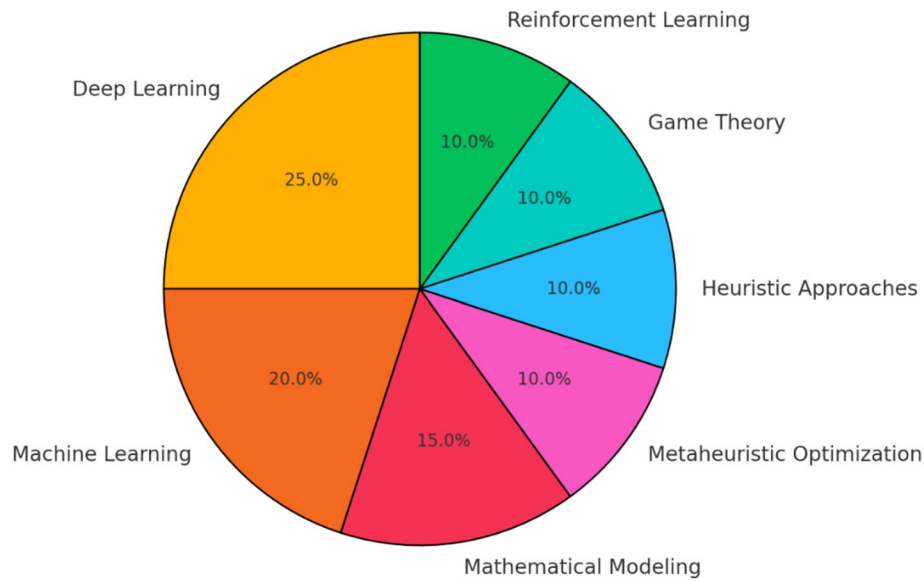


Fig. 6 Distribution of AI-driven privacy-preserving methods in IoV networks

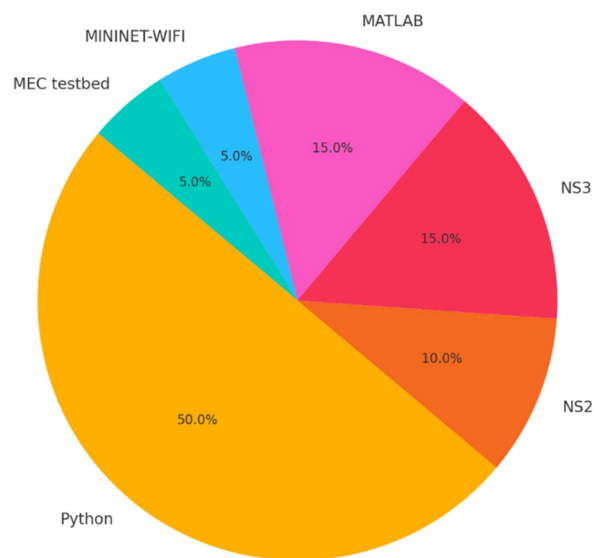


Fig. 7 Distribution of simulation platforms used in privacy-preserving AI for vehicular networks

mechanisms, trust-aware FL systems, and privacy-preserving data-sharing protocols. At last, IoV applications for adaptive decision-making in dynamic environments are progressively including Reinforcement Learning (10%). Its uses span traffic flow optimization, anomaly detection, and secure blockchain integration, transforming real-time vehicle security and privacy mechanisms [27].

Examining simulation platforms used in privacy-preserving AI research for vehicle networks shows a strong inclination for Python-based simulations, which explains most implementations. Python rules as the main simulation environment, shown in Fig. 7, thanks to its vast libraries for machine learning, deep learning, and data analysis, which are fit for creating AI-driven models for vehicle networks. Particularly in MEC and SDN systems, other simulation platforms such as MININET-WIFI and MEC testbed are

especially used for evaluating network-level security and performance. These systems let researchers replicate real-world vehicle communication conditions, so that they can test the adaptability and efficiency of AI models in managing heterogeneous and dynamic network traffic. Furthermore investigated are network protocols, latency, and security concerns in in-vehicle communication systems using well-known network simulators NS2 and NS3 are investigated. These tools let scientists test several security systems, simulate network activities, and assess how privacy-preserving strategies affect data flow efficiency. Another important simulation tool, MATLAB, finds extensive application in mathematical modeling, optimization, and algorithmic validation. Particularly in cryptographic techniques such as HE, it offers a disciplined setting for creating control systems and evaluating vehicle security models. Presumably, because they concentrate on theoretical evaluations, conceptual frameworks, or real-world data analyses rather than simulated environments, a small fraction of research studies specifically mention their simulation platforms [86].

The distribution of several Python libraries used in privacy-preserving AI research inside vehicle networks is shown in Fig. 8. With their great deep learning capacity, TensorFlow and PyTorch, which account for 20% and 18% respectively, rule and help to create advanced AI models for privacy and security uses. For conventional machine learning chores, including anomaly detection and intrusion detection, Scikit-learn (15%) is extensively applied. Accessible deep learning API Keras (12%) supports fast prototyping and experimentation by employing its simplicity. Essential for processing vehicle communication data, sensor logs, and CAN bus messages, NumPy (10%) and Pandas (8%) are vital in data manipulation. Tools for visualizing and network analysis, including Matplotlib (7%) and NetworkX (5%), help to organize vehicle network topologies and interpret AI-driven results. Computer vision work, including road anomaly detection and privacy-preserving image obfuscation, makes use of OpenCV (3%). Finally, Crypto Libraries (2%) offer security-oriented implementations including homomorphic

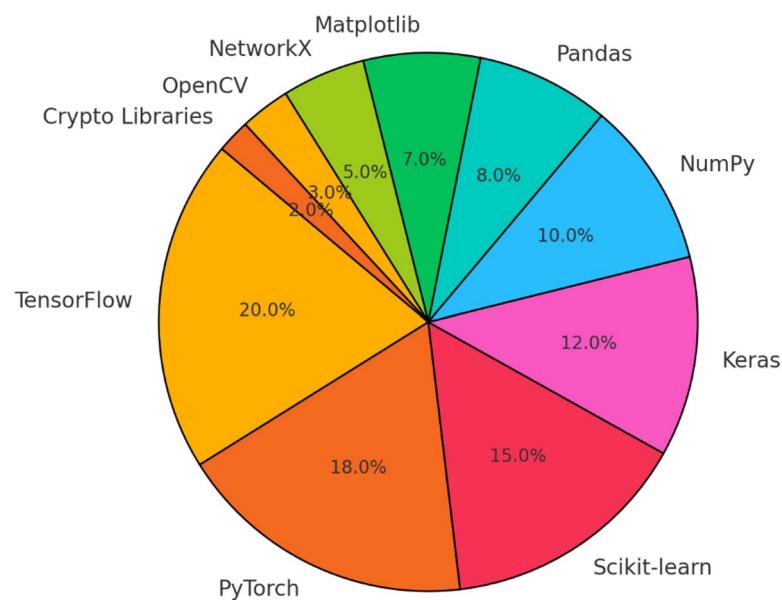


Fig. 8 Distribution of Python libraries used in privacy-preserving AI for IoV networks

encryption, DP, and safe multiparty computation, guaranteeing strong data protection in federated and blockchain-based AI systems [87].

Datasets are absolutely important for the training, testing, and validation of models in the field of privacy-preserving AI and intelligent vehicle networks. The scalability, generalizability, and efficacy of many AI-driven security and privacy solutions depend on public, private, and real-world data availability. An in-depth study of dataset distributions over several research applications in in-vehicle networks, intelligent transportation, and cyber-physical security is presented in Fig. 9. From sites like NSL-KDD, Didi Chuxing, and real-world traffic sensor data, a good amount of the datasets used in research projects come from publicly available repositories. Because of their consistent benchmarks, public datasets are increasingly used so that researchers may evaluate results across several methods. For instance, traffic flow analysis and trajectory prediction depend mostly on datasets like Didi Chuxing, while IDSs frequently use NSL-KDD. Especially in anomaly detection, traffic flow modeling, and cyber-attack simulations, the real-world datasets taken from vehicle sensor data offer a more relevant setting. Local datasets, which researchers gather from particular studies, road networks, or simulated environments, also fall under another interesting classification. Local datasets are particularly relevant for privacy-preserving AI methods, where researchers aim to design solutions that comply with data protection regulations. These datasets comprise synthetic datasets produced by GAN-based models, road anomalies detected, and real-world vehicle trajectory logs. Many studies depend on locally obtained data since privacy issues limit data sharing and hence limit the generalizability, but increase the practical applicability in limited fields. Furthermore, a small number of studies make use of private datasets, usually owned or institutionally controlled. Sensitive uses for such datasets abound in FL systems, blockchain-based privacy, and V2X communication security. Important for assessing privacy-preserving AI frameworks are high-resolution vehicle sensor data, encrypted communications, and adversarial testing scenarios found in private datasets. One main disadvantage of private datasets, however, is their limited accessibility, which makes research result replication and comparison difficult.

Furthermore, changing the terrain of the dataset is a new source for intelligent transportation, IoV, and privacy-aware AI applications. Synthetic datasets created via

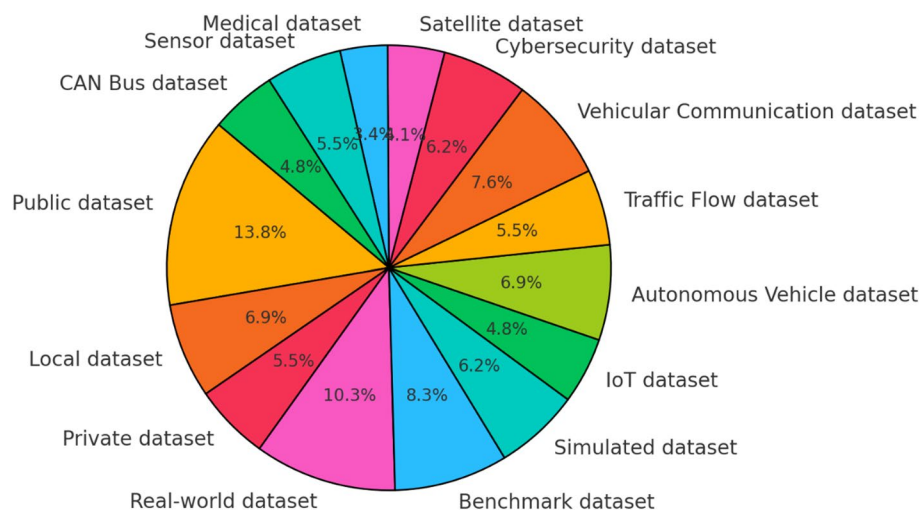


Fig. 9 Distribution of different datasets used in IoV network research

adversarial learning methods and simulation-based datasets from tools like SUMO, Veins, and OMNeT++ are becoming popular, for example. These datasets create controlled environments for benchmarking AI models against privacy concerns and mimic actual vehicle interactions [88].

The distribution of dataset sizes used in research studies on privacy-preserving AI within vehicle networks is shown in Fig. 10. Every dataset is categorized as small, medium, or large; different colors indicate each category. A key understanding of how dataset sizes affect model development, privacy protection mechanisms, and computational efficiency in several automotive AI applications is revealed by the distribution. Large datasets predominate in in-vehicle AI research, indicating the increasing dependence on vast real-world data for training deep learning models, FL systems, blockchain-backed privacy mechanisms, and anomaly detection systems. These datasets offer a high degree of generalizability and robustness for AI models running in dynamic vehicle environments: NSL-KDD for intrusion detection, Didi Chuxing for trajectory prediction, and real-world traffic surveillance datasets. To guarantee both security and scalability, their use presents difficulties related to computational overhead, storage needs, and real-time processing efficiency, which must be resolved through proper data aggregation, distributed learning approaches, and HE techniques. Research involving privacy-aware vehicle communications, federated model training, DP applications, and blockchain-enhanced data-sharing mechanisms makes frequent use of medium-sized datasets in the work. These datasets are appropriate for uses needing moderate-scale real-time analytics, safe AI-driven decision-making, and edge computing optimizations since they balance computational feasibility with model accuracy. Among these are semi-real-time pedestrian detection images, moderate-scale vehicle trajectory datasets, and datasets utilized for distributed data anonymizing methods. Although medium-sized datasets offer a reasonable trade-off between performance and efficiency, they sometimes lack the great data diversity needed for large-scale IoV deployments; hence, synthetic data augmentation and hybrid dataset integration techniques become necessary. On the

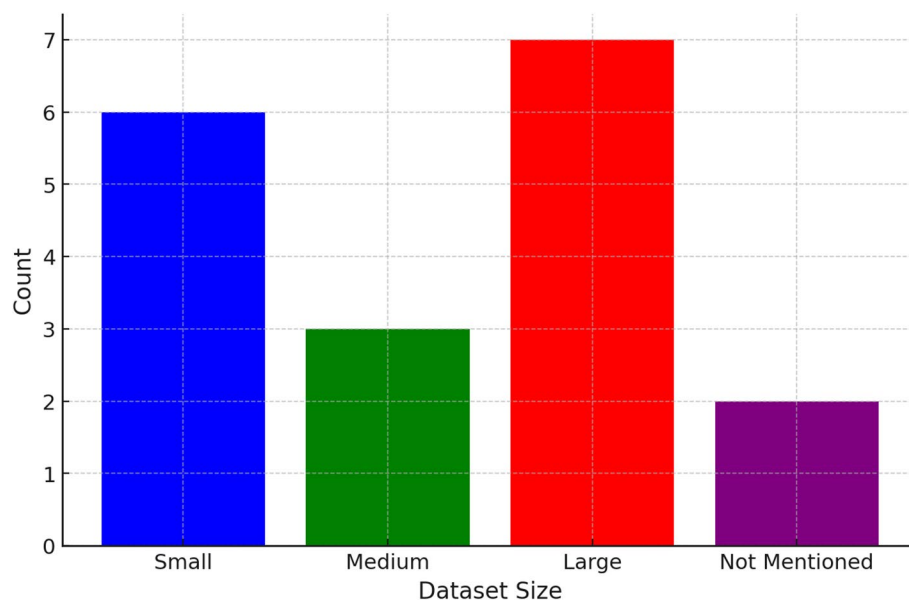


Fig. 10 Distribution of dataset sizes in research

other hand, small datasets are both niche and necessary for algorithm benchmarking, privacy-enhancing cryptographic model validation, and lightweight AI system evaluations. Often small in scope, these datasets are used for privacy-preserving authentication research, early-stage AI testing, and experimental blockchain-backed federated learning. Although small datasets are useful for computationally efficient simulations and edge-AI model validation, their limited real-world applicability and generalizability remain major restrictions. Researchers are looking to synthetic data generation methods employing GANs, federated learning-enhanced data sharing, and real-time vehicular network simulations more and more to overcome these constraints. The distribution of dataset sizes in-vehicle AI research emphasizes the development of data-driven approaches in autonomous networks, maintaining privacy-preserving capability. Large datasets are becoming more and more important as the field moves toward big data-driven AI, large-scale federated analytics, and privacy-aware distributed vehicle intelligence. In AI prototyping, cybersecurity research, and experimental privacy-enhancing mechanism validation, meanwhile, medium-sized and small datasets remain indispensable. Scalable hybrid datasets, privacy-preserving data augmentation methods, and adaptive AI frameworks that elegantly combine edge computing and FL paradigms with big-scale, high-dimensional vehicle data will probably be the main subjects of future work [89].

Figure 11 shows an overview of the research landscape in privacy-preserving AI for the IoV. It shows five main application areas and how much research is being done in each of them. Intelligent transportation and vision privacy (20 studies) and core IoV security and privacy (19 studies) are the two most active fields. Together, they make up the majority of the field. These works concentrate on augmenting vehicular intelligence, safe data sharing, and privacy-conscious perception systems through methodologies including FL, DP, and HE. They meet the need for both decentralized intelligence and strong privacy protections in ecosystems for connected and self-driving cars. The next big area is blockchain-enhanced federated and knowledge-sharing systems (17 research). These systems focus on decentralized trust, auditability, and collaborative model training; however, they are often hampered by latency and energy costs. IoT and 6G-enabled applications (14 studies) expand IoV privacy mechanisms into cross-domain scenarios,

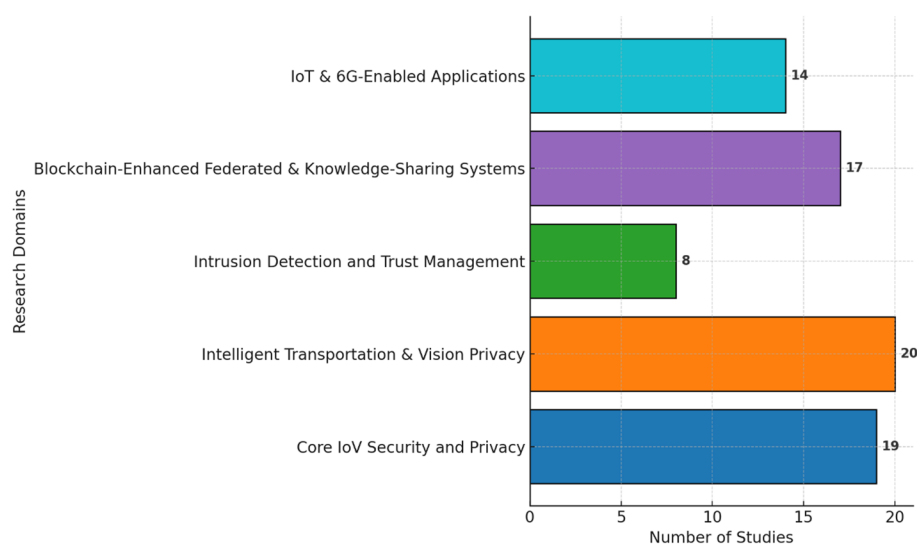


Fig. 11 Categorization of application domains in privacy-preserving AI for IoV

including smart cities, energy-efficient IoT systems, and privacy-preserving V2G communications, illustrating the integration of IoV technologies with next-generation connectivity frameworks. At the same time, IDS and trust management (8 studies) is an important area of security that uses deep learning to defend vehicle networks against threats, including spoofing, poisoning, and inference. Overall, the figure shows a balanced but changing research ecosystem where privacy, scalability, and real-time performance all meet. It emphasizes a distinct progression towards adaptive, hybrid, and reliable AI models capable of functioning effortlessly across vehicle, edge, and cloud layers, signifying a transition from standalone privacy mechanisms to fully integrated, regulation-compliant frameworks for intelligent and secure transportation systems [90].

Figure 12 gives a more detailed look at the distributed settings used in privacy-preserving AI experiments in the IoV. It is divided into 10 primary groups. The investigation indicates that IoV core networks and V2X communication systems are the most commonly examined environments, constituting the structural underpinning of secure and scalable vehicular intelligence. Research in IoV core networks emphasizes privacy-conscious data sharing, decentralized communication, and safe learning frameworks that incorporate automobiles, RSUs, and cloud nodes. These systems use a lot of privacy-preserving methods, like FL, DP, and HE, to let people train models together without giving away their raw data. In the same way, v2x communication, which includes v2v, v2i, and v2p exchanges, is very important for making real-time decisions and coordinating safety-critical tasks. Studies in this field usually focus on secure communications protocols, lowering latency, and strong encryption systems to keep dynamic vehicle interactions safe. Vanet environments and fog/edge computing architectures are also very important parts of contemporary research. Vanets help moving cars talk to one another over a network, which makes traffic flow better, helps avoid accidents, and lets vehicles choose their own way. This category encompasses studies that utilize AI-driven intrusion detection, blockchain-based trust models, and differential privacy-enabled data dissemination to safeguard security and privacy in high-mobility networks. Fog and edge computing systems, on the other hand, let analytics happen with minimal latency by placing compute closer to vehicles. This means that there is less need for centralized

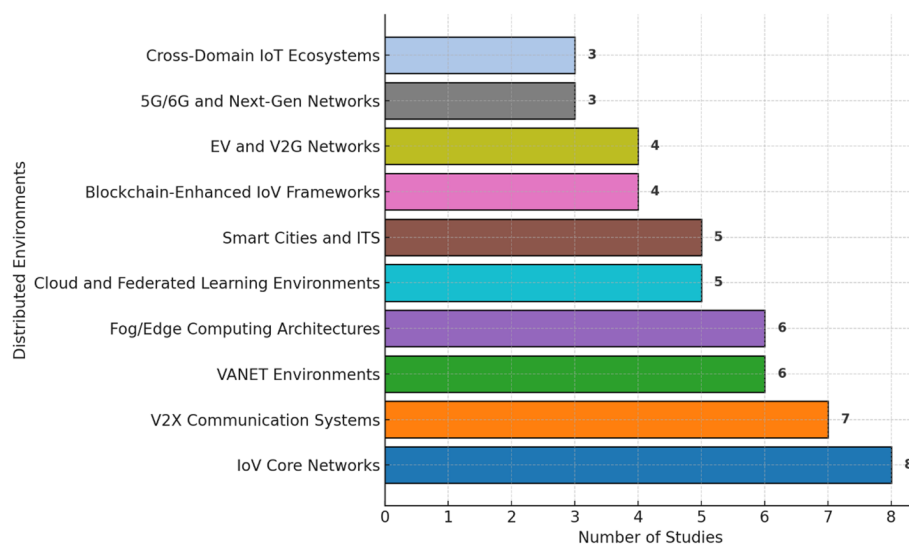


Fig. 12 Distribution of distributed environments used in the studies

servers. These systems are necessary for detecting events in real time, making inferences that protect privacy, and doing computations that save energy in vehicles that are sensitive to latency. The middle part of the picture shows cloud and federated learning environments, smart cities, and ITS. Together, these show how privacy-preserving AI may be used in big cities and on a global scale. Cloud and federated learning research often examines secure model aggregation, noise-based privacy assurances, and cross-silo collaboration, promoting global model enhancement while safeguarding local data privacy. At the same time, its smart city studies take the iov idea and apply it to city infrastructure by using privacy-preserving analytics to improve traffic flow, predict mobility, and keep an eye on public safety. They often use blockchain and AI coordination frameworks to make sure that decisions are made reliably across city-scale systems. Emerging yet potential avenues encompass blockchain-enhanced IoV frameworks and EV/V2G networks, all of which underscore the increasing convergence of security, energy, and trust management. Blockchain-assisted solutions make it possible to check and verify vehicle communications and group training activities in a way that can not be changed. In contrast, EV and V2G research focuses on privacy-aware energy trading, secure charging transactions, and grid coordination, where timing and confidentiality are equally important. Lastly, 5G/6G future-generation networks and cross-domain IoT ecosystems are the next step in the evolution of dispersed IoV settings [91].

The distribution of several XAI approaches applied in the field of vehicle AI and privacy-preserving strategies is highlighted in Fig. 13. These techniques are very important for improving the interpretability and transparency of machine learning models so that stakeholders may know how decisions driven by AI are taken inside intelligent transportation systems. Emphasizing the frequency and relevance of many XAI approaches, the dataset examined spans a broad spectrum of research. Comprising 25% of the studies, SHAP is the most often used technique. Particularly useful for privacy-conscious decision-making in IoV security applications, SHAP offers a game-theoretic method of understanding machine learning models by assigning importance values to individual features. This approach enhances end-user understanding and trust for AI-driven

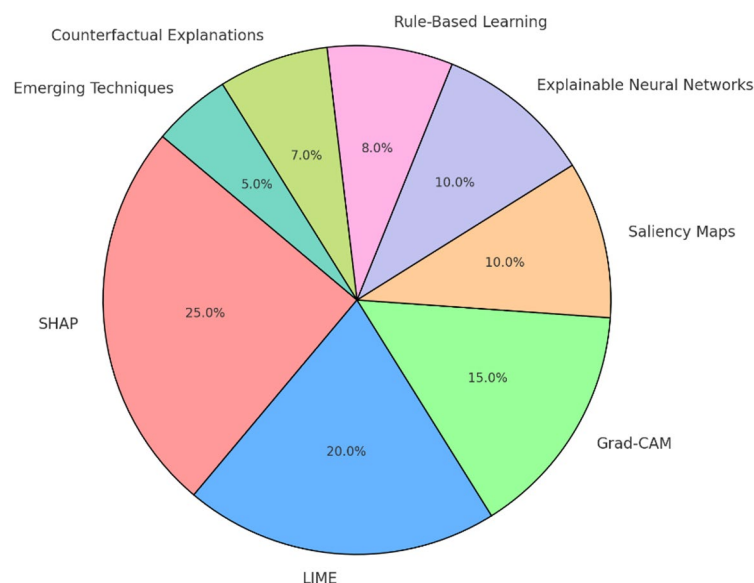


Fig. 13 Distribution of XAI methods used in IoV AI

decisions, thus lowering the risks associated with black-box AI models in-vehicle networks. Appearing in 20% of the studies, LIME is the second most often used method. LIME is preferred since it approximates complicated model individual predictions with simpler, interpretable models, enabling their explanation. In systems of anomaly detection and intrusion prevention, where real-time security depends on knowledge of model behavior, this is especially helpful. 15% of studies are Grad-CAM, mostly used in computer vision applications, including scene recognition in autonomous cars and object detection. This approach helps debug DL models used in self-driving cars by visualizing which areas of an image affect the prediction of a model. Also much embraced are Explainable Neural Networks (10%) and Saliency Maps (10%). Saliency maps improve CNN interpretability by highlighting important aspects of input data that support model decisions. Conversely, explainable neural networks entail designing models with natural interpretability, usually employing attention mechanisms or symbolic reasoning approaches, so ensuring that AI decisions complement domain knowledge. Mostly used in expert-driven models, such as fuzzy logic systems and decision trees, where predefined rules guide the decision-making process. Rule-based learning (8%) is Applications, including autonomous traffic management and vehicle routing, where clear guidelines improve openness and make frequent use of this approach. In privacy-conscious AI systems, counterfactual explanations (7%) have become popular since they provide insights into how model decisions might change should particular input conditions change. This is especially helpful in guaranteeing fairness in driver behavior analysis and risk prediction and in addressing prejudices in AI models. Last but not least, Emerging Techniques (5%) are fresh or hybrid solutions combining several XAI techniques to increase explainability among different AI-driven automotive applications. Among these are hybrid visualization methods, self-explanatory AI models, and neuro-symbolic AI [92].

Table 10 offers an ordered approach for choosing suitable privacy-preserving strategies in the IoV depending on particular privacy issues. Privacy remains a basic issue given the growing dependence on connected vehicles, since real-time exchange of sensitive data across heterogeneous networks causes disturbance. The table arranges privacy issues into ten different spheres and links them to suitable privacy-enhancing strategies and decision criteria. Data privacy in FL, where training AI models cooperatively without sharing raw data is a main goal, is one of the most frequently occurring privacy issues in IoV. While aggregated updates offer significant learning without disclosing raw data, FL combined with DP ensures that individual vehicle data stays on local devices. This approach is generally relevant in autonomous car networks since privacy has to be preserved while raising model accuracy. Location privacy in IoV navigation systems presents another important difficulty since illegal tracking of vehicle positions might compromise privacy. By introducing controlled noise, Geo-Indistinguishability combined with LDP helps anonymize location data, preserving privacy without greatly compromising navigation accuracy. Ride-sharing apps and real-time navigation systems especially benefit from this method. Ensuring that individual sensor readings remain private while yet allowing significant insights is crucial for safe data aggregation in IoV services, including traffic monitoring and vehicle health analytics. While Secure Multi-Party Computation (SMPC) and HE enable useful analytics, they also allow encrypted computations, that is, without a single entity's access to raw vehicle data. This enhances

Table 10 Decision tree for selecting privacy methods in IoV

Privacy concern	Decision criteria	Recommended privacy method	Justification
Data Privacy in FL	Need to train models without sharing raw data?	FL + DP	Prevents raw data exposure while ensuring privacy-preserving updates in vehicular networks.
Location Privacy in IoV Navigation Systems	Need to protect real-time location of vehicles?	Geo-Indistinguishability + LDP	Adds controlled noise to location data, preserving privacy without compromising route accuracy.
Secure Data Aggregation for IoV Services	Need to aggregate vehicle sensor data without revealing individual records?	HE + SMPC	Enables computations on encrypted data, ensuring end-to-end privacy during vehicle data aggregation.
Blockchain-based Privacy in IoV	Need to ensure privacy in a decentralized vehicular network?	Blockchain + Zero-ZKP	Provides decentralized security with privacy-preserving authentication for vehicle identity verification.
Intrusion Detection in Vehicular Networks	Need to detect cyber threats while maintaining data confidentiality?	FL-based Intrusion Detection + DP	Secures anomaly detection systems by preserving privacy during model training.
Vehicular Communication Privacy (V2X, V2V, V2I)	Need to secure communication between vehicles and infrastructure?	E2EE + Group-based Anonymization	Encrypts V2X messages while anonymizing vehicle identities to prevent tracking.
Real-Time Data Privacy for IoV Applications	Need to balance privacy, security, and low-latency processing?	Edge AI + Secure Aggregation Techniques	Enables real-time privacy-preserving AI processing directly on edge nodes with minimal latency.
AI Model Explainability with Privacy	Need to ensure privacy while making AI decisions explainable?	XAI + DP	Provides privacy-preserving model insights while protecting user-sensitive data.
Privacy-Preserving Traffic Analytics	Need to analyze traffic without exposing sensitive driving behaviors?	Secure Aggregation + HE	Allows secure traffic analytics while maintaining the confidentiality of vehicle movement patterns.
Identity Management and Authentication in IoV	Need to verify vehicle identities without compromising privacy?	Blockchain-based Identity Verification + Ring Signatures	Ensures privacy-preserving authentication for vehicle identity in IoV networks.

privacy in-vehicle networks where several stakeholders, including government agencies and insurance companies, need access to combined knowledge. Another interesting strategy is using Blockchain for privacy in IoV, especially in cases of decentralization and security top priority. ZKP blockchain enhances that authentication methods and transactions may be carried out safely without disclosing private data. Where privacy and trust are so important, this approach is vital in IoV identity verification systems, toll collecting, and vehicle ownership tracking. As cyberattacks become more likely, intrusion detection in-vehicle systems calls for AI-based security systems free of compromise of privacy. Employing FL-based intrusion detection combined with DP, anomaly detection across connected vehicles is enabled without disclosing particular attack patterns, ensuring that cybersecurity solutions remain effective while respecting data privacy. Particularly in V2X, V2V, and V2I systems, vehicle communication also needs strong privacy protection. Group-based anonymization combined with End-to-End Encryption (E2EE) ensures that only authorized parties can access vehicle messages, stopping illegal tracking or malicious data interception. For intelligent traffic control systems and emergency response coordination, this is quite helpful. Edge AI combined with safe aggregation techniques offers a privacy-preserving substitute for cloud-based models for real-time AI-based IoV applications, where low-latency processing is critical. This method reduces latency and privacy concerns by letting cars process data locally while sharing just the necessary encrypted information. AI explainability with privacy is another developing

issue whereby privacy-preserving AI decisions have to be interpretable. DP, together with XAI, ensures that model decisions remain understandable without disclosing private training data. Particularly pertinent for autonomous car decisions and regulatory standard compliance is this. Secure Aggregation combined with HE provides a strong solution in privacy-preserving traffic analytics, where large-scale vehicle movement patterns must be analyzed without identifying individual drivers. Using this approach lets governments and researchers examine traffic patterns without violating driver privacy. Finally, a key issue for identity management and authentication in IoV is making sure vehicles can be checked without revealing driver information. For uses like safe toll transactions and vehicle access control, blockchain-based identity verification coupled with Ring Signatures ensures that vehicle identities remain anonymous while preventing fraudulent access, ensuring ideal conditions [93].

Table 11 offers a methodical approach to choosing the most appropriate DL techniques to handle diverse security issues in the IoV. Strong AI-driven security mechanisms are much more important as vehicle networks get more linked and intelligent. Based on particular decision criteria, this decision tree classifies security issues and suggests deep learning models accordingly, guaranteeing the best performance in intrusion detection, privacy preservation, authentication, anomaly detection, and secure communication. Among the most urgent issues in IoV networks is intrusion detection, since cyberattacks can compromise data integrity and vehicle safety. Because CNNs can examine spatial patterns in network traffic data, they are quite good at spotting network anomalies. Likewise, privacy-preserving model training is crucial in federated IoV systems, where FL enhances distributed AI model training without revealing raw data, providing a perfect solution for safe vehicle applications. RNNs shine in processing sequential attack patterns, so offering a potent tool for identifying malware and cyber threats in-vehicle networks when it comes to threat classification. Another important component of IoV security is vehicle anomaly detection, in which case Autoencoders are quite important in learning normal vehicle behavior and spotting deviations suggestive of system failures or malicious activity. GNNs are especially useful for authentication and identity management in-vehicle communications since they help to securely authenticate within V2X networks and model complex interactions between vehicles. Another important issue is privacy-preserving data sharing; GANs offer a quick method to create synthetic data with utility while avoiding sensitive information exposure. Adversarial attacks seriously threaten AI models in IoV security; adversarial training improves model robustness by training against adversarial examples, thus strengthening system resilience. Advanced AI-driven methods are needed for IoV cybersecurity in edge networks; transformer networks have great potential to securely and effectively process massive vehicle data. Likewise, LSTM networks, which are ideal for managing time-series data and forecasting vehicle behavior while maintaining privacy and security, are used in safe AI-driven traffic analysis. Identification of odd trends in vehicle sensor data depends on anomaly detection in IoV data streams; Deep Belief Networks (DBNs) shine in feature extraction and high-dimensional anomaly detection. Self-Supervised Learning (SSL) is necessary to guarantee safe V2V communication, so as to improve secure data transmission and authentication with minimal labeled data. By combining CNNs for effective feature extraction and RNNs for sequential analysis, hybrid CNN-RNN models provide an ideal solution for IoV edge computing, guaranteeing strong security at the edge. RL with

Table 11 Decision tree for selecting deep learning methods in IoV security

Security concern	Decision criteria	Recommended DL method	Justification
Intrusion Detection in IoV Networks	Need to detect network anomalies and cyber threats?	CNNs	CNNs excel at analyzing spatial patterns in network traffic and detecting anomalies caused by malicious intrusions.
Privacy-Preserving Model Training	Need to train AI models without sharing raw data?	FL	FL allows distributed AI model training without transferring raw data, ensuring privacy preservation in IoV.
Malware and Threat Classification	Need to classify different types of cyber threats?	RNNs	RNNs process sequential attack patterns in IoV networks, making them suitable for malware detection.
Vehicle Anomaly Detection	Need to detect abnormal driving behaviors?	Autoencoders	Unsupervised autoencoders learn normal vehicle behavior and flag deviations, helping detect abnormal driving patterns.
Authentication & Identity Management	Need to authenticate vehicles securely?	GNNs	GNNs model relationships between vehicles in V2X networks, providing secure authentication and identity management.
Privacy-Preserving Data Sharing	Need to share vehicular data securely?	GANs	GANs generate realistic synthetic data for secure data sharing without revealing sensitive information.
Adversarial Attack Defense	Need to defend against adversarial ML attacks?	Adversarial Training	Training models with adversarial examples makes them more robust against AI manipulation and attacks.
IoV Cybersecurity in Edge Networks	Need to secure edge-based AI processing?	Transformer Networks	Transformers process large-scale vehicular data efficiently, ensuring security at the edge in IoV systems.
Secure AI-driven Traffic Analysis	Need to analyze vehicle movement securely?	LSTM	LSTMs handle time-series traffic data efficiently, predicting vehicle behavior while ensuring secure processing.
Anomaly Detection in IoV Data Streams	Need to identify unusual data patterns in IoV?	DBNs	DBNs excel at feature extraction and anomaly detection in high-dimensional IoV sensor data.
V2V Secure Communication	Need to ensure secure and authenticated communication between vehicles?	Self-supervised learning	SSL models learn from limited labeled data, enhancing secure V2V communication with minimal overhead.
Securing IoV Edge Computing	Need to ensure secure computations at the edge?	Hybrid CNN-RNN Models	Combining CNNs and RNNs allows efficient feature extraction and sequential analysis, improving edge computing security.
Smart Traffic Management Security	Need to optimize traffic control without compromising security?	RL with DQN	RL-based methods dynamically optimize traffic flow while ensuring secure decision-making in intelligent transport systems.
Trust and Reputation Management in IoV	Need to assess the trustworthiness of vehicles?	Bayesian Deep Learning (BDL)	BDL provides probabilistic reasoning for assessing trust and reliability in vehicular networks.

DQN allows dynamic traffic optimization while preserving security in the framework of smart traffic management security, thus it is perfect for intelligent transportation systems. Finally, dependable communication and the prevention of hostile behavior in IoV networks depend on trust and reputation management. Utilizing a probabilistic method, Bayesian Deep Learning (BDL) evaluates vehicle trustworthiness in real-time, enhancing security and dependability in automotive environments [94].

Figure 14 shows a methodical, multi-tiered security model integrating AI techniques to improve privacy, authentication, and anomaly detection in the IoV. Using AI-driven and cryptographic techniques, the figure shows four main layers of security that each solve particular issues in automotive networks. By using distributed consensus systems, encryption methods, and DL models to reduce cyberattacks and enhance safe vehicle

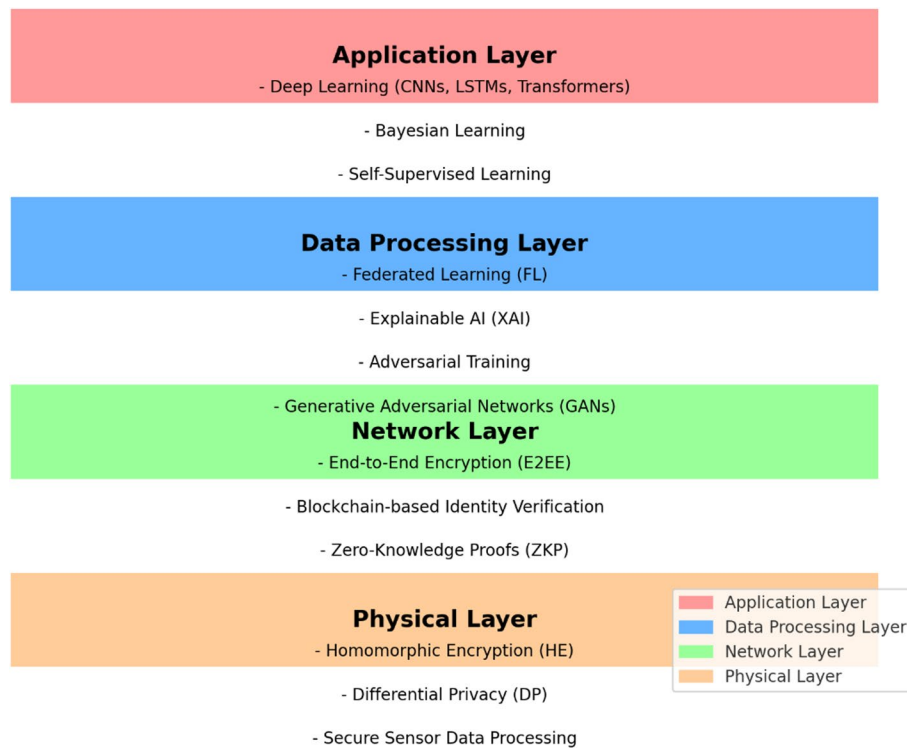


Fig. 14 Multi-layered AI-powered security architecture for IoV

communication, these layers guarantee a complete security architecture. The topmost level, the application layer, emphasizes intelligent decision-making and explainability in AI-driven automotive security systems. Important AI techniques in this layer are Bayesian learning for probabilistic threat assessment and trust-based vehicular communication, DL models for IDS, anomaly detection, and autonomous decision-making, and SSL for improving cybersecurity in V2X networks with limited labeled data. These methods improve security and guarantee great flexibility in the IoV ecosystem. Managing vehicle data while guaranteeing privacy, security, and federated collaboration among infrastructure and vehicles depends critically on the data processing layer. FL, XAI to improve transparency and interpretability in AI-based vehicle authentication and anomaly detection, adversarial training to defend against AI model manipulation and adversarial attacks, and GANs to create synthetic data for privacy-aware vehicular data sharing, constitute core mechanisms in this layer. These methods together protect data privacy and increase the resilience of AI models against cyberattacks. Employing cryptographic and distributed authentication systems, the network layer guarantees safe inter-vehicular and V2I and V2V communications. Blockchain-based Identity Verification provides distributed, tamper-proof authentication for vehicles communicating within the IoV network; ZKP for authentication without revealing sensitive identity-related information, reducing the risks of tracking and illegal access; E2EE secures V2X messages, preventing eavesdropping and unauthorized access. These security systems guarantee that automotive communication stays resilient against adversarial attacks, authenticated, and private. The basis of the architecture, the physical layer, protects encryption, privacy, and sensor data integrity across transmission. HE, which lets computations be carried out straight on encrypted data without needing decryption, enhances continuous data

confidentiality, DP to include statistical noise into datasets to prevent data leakage and unauthorized inference, and secure sensor data processing techniques to ensure tamper-proof collecting and transmission of vehicular sensor data. These systems guarantee that IoV security reaches the hardware and data level, so as to augment privacy and data protection in real-time vehicle applications, going beyond communication and model training [95].

Evaluated over important performance criteria including privacy, scalability, attack robustness, latency, computational efficiency, and real-time adaptability, Fig. 15 offers a comparative analysis of several security mechanisms used in automotive networks. Every method is scored between 1 and 10; higher scores in each category indicate better performance of that particular technique. By allowing distributed AI training without exposing raw data, FL shows strong privacy (9) and scalability (8); however, its complexity of aggregating models across distributed nodes results in challenges with computational efficiency (6) and latency (5). Conversely, HE enhance the highest attack robustness (9) and privacy (10), enabling the safe execution of encrypted computations. Notwithstanding these advantages, it suffers from major latency (3) and computational inefficiencies (4), which make real-time applications challenging to execute successfully. Blockchain offers a well-balanced security framework; it excels in attack robustness (9) and privacy (8), but its latency (4) remains a problem due to the overhead related to consensus mechanisms needed for transaction validation. DP enhance strong privacy (9) and achieves moderate latency (6); thus, it is a useful method for privacy-preserving analytics; nevertheless, its attack robustness (6) and computational efficiency (7) imply that additional optimization could be required to improve its performance. Although its privacy (7) and attack robustness (5) remain moderate and need complementary security layers to reduce possible risks, XAI stands out in real-time adaptability (10) and computational efficiency (9), enabling transparent and interpretable AI-driven security decisions. With only moderate privacy (6) and scalability (5), adversarial training turns out as the most successful countermeasure against adversarial attacks (10). Furthermore, its rather low computational efficiency (4) makes large-scale installations resource-intensive [96].

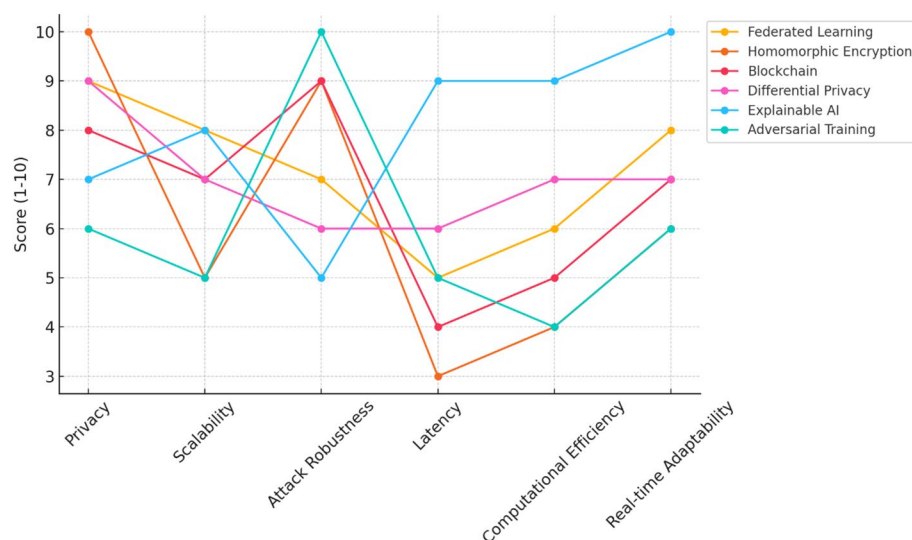


Fig. 15 Comparative analysis of security mechanisms based on key performance metrics

Figure 16 shows the trade-offs between privacy protection level and computational overhead for several AI-based privacy-preserving methods applied inside IoV systems. While the Y-axis shows the degree of privacy protection, similarly rated from 1 (low overhead) to 10 (high overhead), the computational overhead is shown on the X-axis, scored on a scale from 1 (low overhead) to 10 (high overhead). The size of each bubble reflects the real-time adaptability of the technique, whereby larger bubbles match a better fit for time-sensitive vehicle environments, including autonomous driving or V2X communications. The Fig comparison emphasizes the advantages and drawbacks of every method in turn. Positioned at the top-right of the graph, techniques such as HE achieve the maximum privacy score of 10 using their capacity to execute computations over encrypted data without disclosing private information. HE is less perfect for situations where real-time responsiveness is crucial, though, as this comes at the cost of a computational overhead of 10, indicating notable resource consumption and latency. While showing good real-time adaptability, FL sits in a more balanced position and offers strong privacy enhance with a score of 9 and a moderate overhead of 6. It is therefore appealing for distributed vehicular AI applications where privacy and efficiency must coexist. DP then shows a good trade-off, obtaining a high privacy level of 9 while maintaining computational overhead at a reasonable 5 and providing good real-time adaptability. Although blockchain technologies have great privacy (8), their overhead (8) is higher because of the need for consensus validation and block generation delays, which could influence scalability and latency. While Adversarial Training, although great in resilience against adversarial attacks, scores lower on scalability and computational efficiency, represented by privacy 6 and overhead 9, Methods such as GANs reflect different priorities. GANs hold moderate privacy (7) and overhead (7) [97].

Designed to offer a thorough comparison of the cumulative effects of several AI-based security techniques on important QoS parameters within IoV environments, Fig. 17 shows five well-known security techniques embraced in protecting automotive networks. Measuring on a standardized range of 1 to 10 for every parameter, the Y-axis shows cumulative QoS impact scores, providing a consistent evaluation framework. Every bar is split into stacked sections matching five key QoS criteria: latency, scalability,

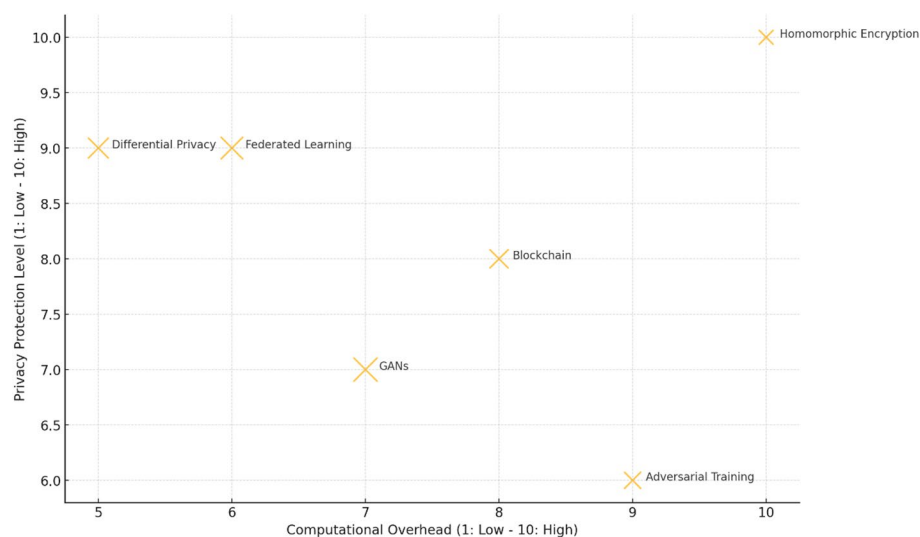


Fig. 16 Trade-off analysis between privacy level and computational overhead in IoV security techniques

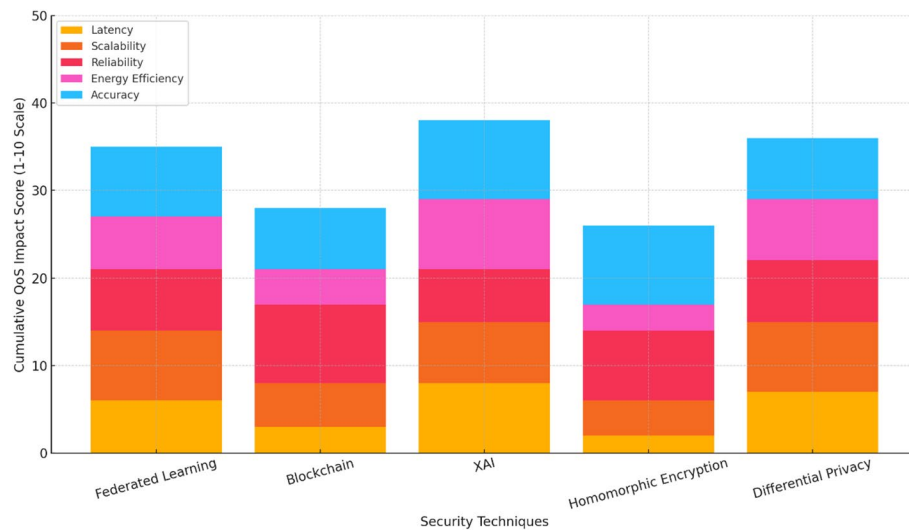


Fig. 17 Comparison of the cumulative effects of various AI-based security techniques on essential QoS parameters within IoV

reliability, energy efficiency, and accuracy, so providing a layered perspective of how every technique influences system performance generally. The study stresses the strategy's clear trade-offs and synergies. Blockchain, for instance, has the highest dependability score (9) because of its inherent immutability and tamper-proof character, but its computationally demanding consensus algorithms cause significant latency penalties (score of 3) and lowered energy efficiency (score of 4). Reflecting its great data protection capabilities, HE is rated high in reliability (8) and accuracy (9), but it is penalized in latency (2) and energy consumption (3) due to the overhead related to encrypted computations. While maintaining reasonable levels of latency, energy consumption, and accuracy, FL and DP show a more balanced performance across all QoS measures, particularly excelling in scalability (8 for both). They are therefore appropriate for large-scale and dynamic IoV environments. Though somewhat lower dependability (6) and scalability (7), XAI performs remarkably in accuracy (9) and energy efficiency (8), offering transparency and lightweight operation. This thorough analysis emphasizes how no one method consistently beats others' overall QoS levels. Rather, the particular QoS priorities and operational restrictions of the IoV application will guide the security technique choice. Applications that give real-time communication and low-latency processing top priority, for example, emergency vehicle coordination, might favor DP or XAI because of their favorable latency and efficiency ratings. Conversely, applications needing high-security guarantees, such as sensitive data aggregation or vehicle identity verification, may choose HE or blockchain even if they have computational disadvantages [98].

Figure 18 shows the relative increase of XAI integration in IoV security studies over the period 2018 to 2025. Three distinct lines, Explainable FL, XAI-enhanced IDS, and XAI for traffic prediction, formulate the chart. Reflecting the increasing awareness of interpretability and trustworthiness in AI-driven vehicle networks, the trend unequivocally shows a continuous upward trajectory in the acceptance of XAI across all sub-domains. From 2 publications in 2018 to 21 in 2025, explainable FL shows a consistent rise, highlighting researchers' focus on privacy-preserving yet interpretable collaborative learning models. Analogous to this, XAI-enhanced IDS shows a notable increase

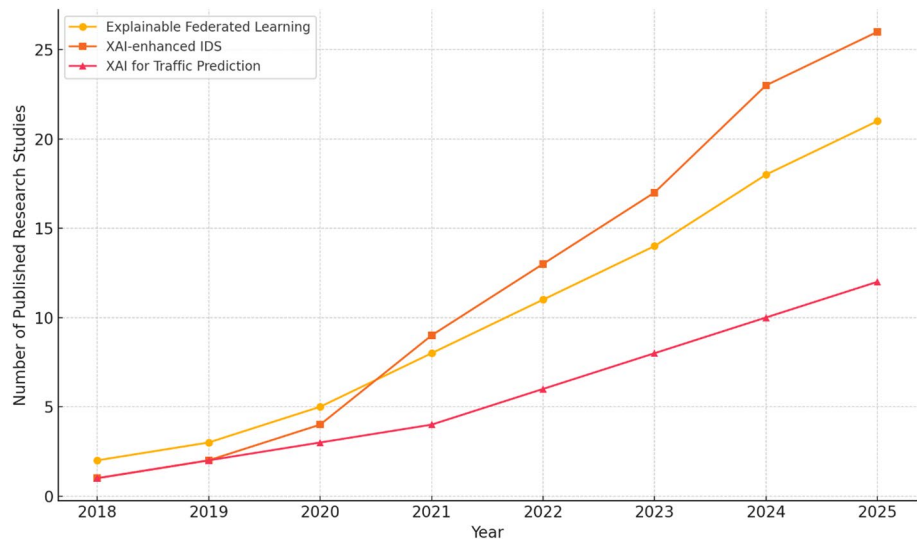


Fig. 18 Comparative growth of XAI integration in IoV security studies over the period 2018 to 2025

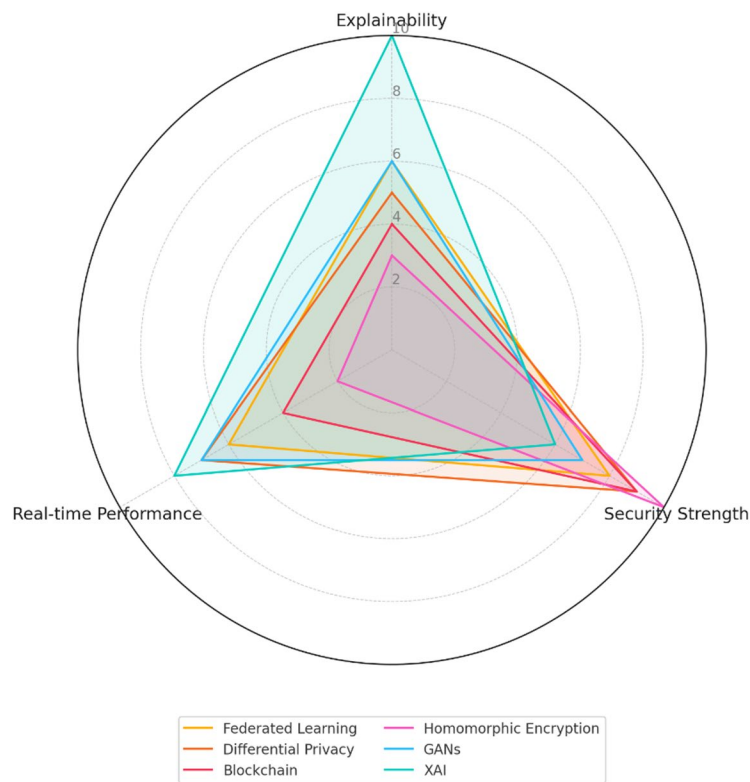


Fig. 19 A comparative analysis of key AI-based security techniques deployed in IoV systems

from 1 in 2018 to 26 by 2025, suggesting a strong emphasis on open intrusion detection for security-critical vehicle environments. As XAI for Traffic Prediction grows steadily but rather modestly, reaching 12 studies in 2025, it suggests a possible research gap and opportunity for further study.

Figure 19 presents a comparative study of important AI-based security methods applied in IoV systems. Based on three key criteria: explainability, security strength, and

real-time performance. Higher scores indicate better performance in that dimension; the scores run from 1 to 10. It emphasizes the inherent triangular trade-off these techniques create. For example, XAI techniques have moderate security strength (6) but score the highest in explainability (10) and retain strong real-time performance (8). Conversely, HE suffers in both explainability (3) and real-time performance (2) due to its computational overhead, while excels in security strength (10). Due mostly to their complexity, blockchain technologies similarly offer great security (9) but suffer from real-time adaptability (4) and explainability (4). Though neither rule in all three dimensions, FL and DP combine to create a more balanced profile across all three. This radar chart emphasizes the need for hybrid or adaptive frameworks able to balance explainability, privacy strength, and real-time efficiency in IoV systems, supporting the case that there is no generally optimal solution. For researchers and business practitioners trying to choose the most suitable AI-driven security technique depending on their particular application needs, the figure provides a decision-making guide [99].

Figure 20 offers a thorough study of the distribution and application of privacy-preserving AI methods among several layers of the IoV architecture. The edge, cloud, fog, vehicle layer, and network layer divide the application of these methods into five major layers. Every segment captures the importance and function of privacy-enhancing devices inside the hierarchical IoV system. Comprising the most 35%, the edge layer highlights its central importance in allowing low-latency, real-time privacy solutions. Edge AI and FL models, which enable distributed data processing directly at the periphery, ensure that sensitive vehicle data, including location, driving behavior, and sensor inputs, is not offloaded to external servers, so reducing exposure risks drives this dominance. The proximity of the edge layer to data sources helps to support privacy-preserving anomaly detection and intrusion detection systems, able to provide instantaneous threat mitigation free from the communication delays connected with centralized processing. Because it can support sophisticated and resource-intensive privacy-preserving models, including HE, blockchain-based consensus, and AI-driven trust management

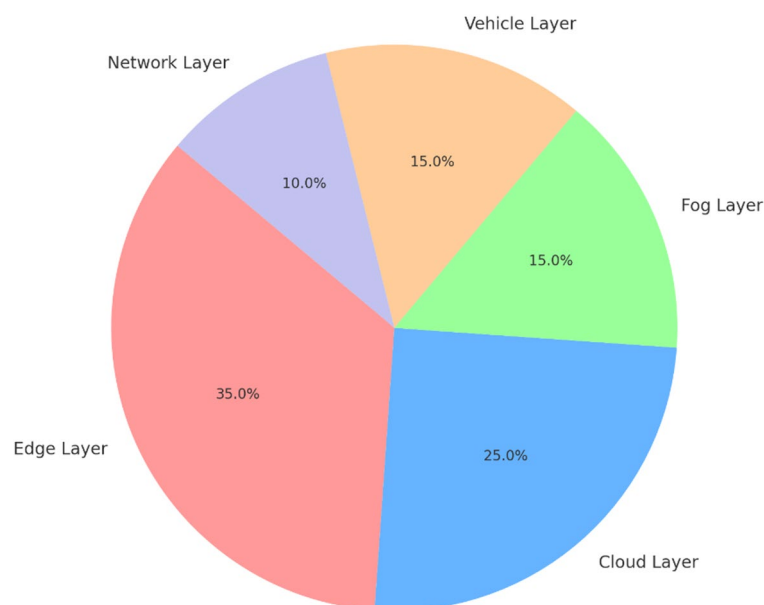


Fig. 20 Distribution of privacy-preserving AI techniques across different layers of the IoV architecture

frameworks, the cloud layer holds a noteworthy 25%. Offering great security at a great cost of latency, these cloud-based models manage tasks including distributed ledger maintenance, encrypted computation, and large-scale model aggregation. Each of the fog layer and vehicle layer helps to distribute 15% overall. Between cloud and edge, the Fog Layer serves as a middle processing tier supporting privacy-preserving computations needing modest latency handling and computational offloading. It guarantees privacy-preserving mechanisms like DP and lightweight encryption schemes, facilitating safe V2I communications and real-time data aggregation. Analogously, the vehicle layer emphasizes onboard AI models buried inside the vehicle, such as XAI-enhanced intrusion detection and privacy-preserving trajectory prediction models. The localized character of this layer ensures that raw data never leaves the car unduly, thus improving user privacy. At last, the network layer makes up 10% of the whole distribution and emphasizes its function in data transmission, security of communication protocols, and vehicle network handshakes. E2EE and group-based anonymizing systems protecting data as it moves between vehicles, RSUs, and cloud/fog nodes define privacy mechanisms at this level. Although the relatively smaller share shows the limited computational capacity of the network layer, it emphasizes its indispensable function in preventing traffic analysis attacks, MITM, and unauthorized interception [34].

Implications of the work in the industry and academia

By including AI-driven privacy-preserving technologies in the IoV, the industry and academia will greatly benefit, and the security, efficiency, and adaptability of intelligent transportation systems will be greatly advanced. From an industrial standpoint, the results of this work enable manufacturers, technology companies, and smart city builders, including FL, DP, HE, and blockchain AI-based security mechanisms, to integrate into practical vehicle applications. These technologies enable producers to improve autonomous vehicle security, protect user-sensitive data, and maximize V2X communications without endangering privacy. Data privacy and cybersecurity issues are the priority in completely autonomous and connected vehicle ecosystems, which the automotive sector is progressively moving toward. Industries can guarantee regulatory compliance with data protection laws, including GDPR and CCPA, by including privacy-preserving AI models, thus building trust among consumers. Ride-sharing apps, EV charging stations, and logistics firms can also use AI-enhanced security systems to stop cyberattacks, lower data breaches, and enable safe data-sharing models across multi-stakeholder platforms. Blockchain and HE in IoV applications also create fresh chances for monetizing vehicle data through safe and transparent mechanisms, opening paths for innovation in traffic management, predictive maintenance, and automated toll collecting. Adoption of these privacy-preserving AI techniques by the whole sector not only improves security but also opens the path for flawless real-time communication, lowering latency, and increasing general transportation effectiveness.

Academically, the suggested approaches and taxonomies form the basis for additional investigation in AI-driven security for vehicle networks. Using the frameworks described in this work, researchers in computer science, cybersecurity, AI, and transportation engineering can create new algorithms that maximize privacy, security, and computational efficiency in major IoV installations. Refining FL models for distributed AI training in-vehicle networks depends critically on academia, which also helps edge

computing platforms to securely process data without centralized exposure. Furthermore, the inclusion of DP into models of vehicle data sharing lets academics investigate fresh approaches to balance data utility with privacy guarantees, so minimizing inference attacks and preserving analytical accuracy. Beyond IoV, HE has ramifications in academia that affect research on encrypted computing in more general sectors, including healthcare, finance, and critical infrastructure security. Particularly in developing strong defenses against data poisoning, model inversion, and adversarial inference attacks, adversarial machine learning-based applications also offer researchers in cybersecurity new challenges and opportunities. Through research on how AI-powered anomaly detection models perform in dynamic vehicle environments, academics can help create next-generation intrusion detection systems with real-time, proactive threat-mitigating capability.

Beyond its direct uses in business and academia, this work promotes multidisciplinary cooperation among government agencies, research facilities, and business companies. The insights gained from this study can be used by policy-makers and regulatory authorities to create uniform frameworks for privacy-preserving AI applications in IoV, ensuring that developing technologies fit legal and ethical standards. The work also helps AI researchers, automotive engineers, cryptographers, and legislators to coordinate across disciplines to produce complete security solutions that handle practical problems. This research can be used by funding organizations and innovation centers to support projects aimed at privacy-enhancing AI for transportation, thus enabling the creation of scalable and effective vehicle AI solutions. Furthermore, the ramifications of this work reach into the classroom, where colleges might include privacy-preserving AI models in their courses so arming the next generation of engineers and data scientists with the tools required to create safe intelligent systems. This work helps to evolve AI-powered vehicle security by bridging the gap between theoretical developments and pragmatic implementations, thus influencing the direction of connected and autonomous transportation in a privacy-centric environment.

Datasets

The fast progress of intelligent transportation systems and vehicle networks has made strong datasets necessary to support research in privacy-preserving AI development. Ensuring data privacy while preserving efficiency is a significant difficulty since linked vehicles produce enormous volumes of sensitive information, including location, driving behavior, and vehicle diagnostics. Several well-chosen datasets have surfaced to support the development, testing, and validation of AI-driven models improving vehicle security, transportation efficiency, and privacy compliance. Among other fields, these datasets cover V2X communication, intrusion detection, trajectory analysis, and privacy preservation grounded on federated learning. The Basic Safety Message (BSM) Dataset is one of the main datasets used in privacy-preserving vehicle AI research. This dataset comprises thorough records of vehicle communication, capturing important criteria including vehicle position, speed, acceleration, and heading. Finding anomalies, spotting harmful behavior, and improving IoV security protocols depend on these records, which are therefore absolutely vital. The dataset is extensively used in the creation of misbehavior detection systems that highlight unusual driving behaviors or malicious data injection attacks. Utilizing BSM data analysis, researchers can apply AI-driven

trust management systems that assess the authenticity of received messages and filter out unreliable sources. This dataset is very important for the development of security algorithms meant to reduce cyber threats aimed at automotive communications. The Didi Chuxing GAIA open dataset is another important collection supporting research in privacy-preserving AI within vehicle networks. This dataset, supplied by top ride-hailing company Didi Chuxing, comprises vast GPS trajectory data gathered from actual ride-hailing activities. It comprises metadata including trip start and end times, distances traveled, and anonymized driver and passenger names. Privacy-preserving trajectory prediction models, route optimization strategies, and dynamic ride-sharing systems are developed in great part using the dataset. Using FL methods allows researchers to train models on this dataset without directly accessing raw user data, preserving privacy while maintaining accuracy. A key resource for intelligent transportation developments, the Didi dataset has been extensively applied for traffic flow prediction, congestion management, and urban mobility research [100].

Another crucial field needing high-quality datasets for training and evaluation is intrusion detection in-vehicle networks. The NSL-KDD dataset is a benchmark dataset for building and evaluating AI-driven IDS. It comprises a varied collection of network traffic records arranged either as normal or in line with particular attack categories. Vehicle security research makes great use of this dataset to create anomaly detection systems for hostile activity, including DoS attacks, packet spoofing, and unauthorized access. Ensuring the resilience of vehicle communication networks depends on being able to discriminate between good and malicious traffic. Using machine learning and deep learning approaches helps researchers reduce false positives and improve IDS detection accuracy [16, 101].

Beyond network security, maintaining driver privacy depends on guarantees of in-vehicle communication integrity. Data gathered from in-vehicle sensors and Controller Area Network (CAN) bus messages form the vehicle sensor and CAN Bus datasets. Real-time vehicle parameter readings for speed, RPM, throttle position, and braking patterns abound in these datasets. Using these datasets, researchers create AI-based intrusion detection models that track CAN bus communications for evidence of cyberattacks or tampering. Finding spoofed or manipulated messages that might allow illegal vehicle control presents a significant obstacle to maintaining in-vehicle networks. Leveraging privacy-preserving AI models helps find anomalies in CAN bus communications to be found without revealing private vehicle diagnostics. Apart from security, privacy issues regarding V2X communications have prompted the creation of specific datasets, including V2X communication datasets. Including the interchange of safety messages, infotainment data, and real-time navigation updates, these datasets record V2V and V2I interactions. Included are features, including message kinds, transmission intervals, and network performance measures, that let academics evaluate the effectiveness of privacy-preserving communication systems. Implementing cryptographic methods, including HE and DP, helps V2X communications to be protected against illegal data access and eavesdropping. Developing distributed trust models and AI-driven authentication systems for vehicle networks depends on this dataset in great part. Apart from conventional vehicle datasets, there exist several domain-specific datasets catered to particular uses, including autonomous driving, intelligent traffic systems, and railway networks. For instance, the BOREAS dataset offers multi-season driving information gathered from

LiDAR, radar, and cameras, among other sensors. This dataset is crucial for improving the perception capacity of autonomous cars since it helps the creation of strong AI models that can adapt to the surroundings. Offering annotated images and point cloud data for railway scene understanding and infrastructure monitoring, the Rail Sem 19 Dataset and WHU-Railway 3D dataset also serve the railway industry. These datasets enable the creation of AI-driven railway safety systems that identify hazards, track conditions, and increase operational efficiency. As vehicle AI develops, the availability of high-quality datasets becomes even more important in advancing privacy-preserving technologies. For researchers and industry professionals striving to improve the security, privacy, and efficiency of intelligent transportation systems, the datasets under discussion provide basic tools. Future AI models can maximize privacy protection while preserving great performance by combining federated learning, homomorphic encryption, and adversarial machine learning approaches. The ongoing growth of these databases, together with developments in AI techniques, will open the path for safer and more privacy-conscious vehicle networks [34, 102].

XAI and its importance

The need for transparency, responsibility, and interpretability in AI-driven decision-making has made XAI integration into privacy-preserving automotive networks a priority for research and development. AI systems are “black boxes,” making it hard for customers, policymakers, and developers to grasp their outcomes as they become more complicated and integrated into IoV applications. Interpretability issues arise in autonomous driving, vehicle anomaly detection, and privacy-preserving AI. XAI provides human-interpretable AI decision explanations while protecting model performance and privacy to close this gap. IoV engineers, lawmakers, and end users need XAI for FL, DP, and HE machine learning models to generate correct forecasts and explain. To avoid false positives and missed security concerns, AI-driven anomaly detection and threat-mitigating systems must justify their alerts and proposals. Developers may improve user confidence, regulatory compliance, and human-AI engagement in intelligent transportation systems by introducing XAI in-vehicle AI systems. XAI is important in FL-based IoV systems that disseminate data to automobiles and edge devices. FL uses dispersed datasets without raw data, making model updates and decision-making harder to explain. LIME and SHAP can show how AI models balance sensor inputs, location data, and vehicle interactions while predicting. When AI systems must detect real-time adversarial attacks, privacy violations, or unusual driving patterns, interpretability is essential. In privacy-preserving DP models, explainability protects AI decisions from dataset noise [103]. By computing on encrypted data without revealing sensitive data, XAI techniques can ensure that encrypted AI models in HE are fair and accurate. XAI provides transparent and verifiable explanations for data access, trust evaluations, and transactions to support Blockchain AI-based distributed decision-making systems. Explainability increases smart contract-based auto transaction automation and decreases disagreements. XAI affects IoV user adoption, industry adoption, and regulation beyond its technological benefits. As governments and regulators tighten AI accountability norms, interpretable AI models become more important. Under the “right to explanation,” EU legislation like the GDPR requires AI systems that affect humans to justify their actions. Autonomous and connected car manufacturers must make their AI-driven technologies

transparent and safe, and private. XAI in-vehicle AI systems can improve regulatory compliance, legal risks, and public trust in AI-powered mobility. XAI helps developers and engineers diagnose and improve models, improving AI security and performance. Consumer adoption of AI-driven decision-making in driver assistance systems, smart mobility apps, and automated traffic management depends on their ability to understand and trust AI recommendations. This work builds more transparent, accountable, and dependable intelligent transportation systems that prioritize security and user confidence using explainability in AI models for IoV [104].

Broader applications beyond automotive use cases

The current study predominantly addresses privacy-preserving mechanisms within the IoV; however, the foundational techniques demonstrate significant applicability to other mission-critical sectors necessitating sensitive data processing, extensive connectivity, and real-time decision-making. FL allows hospitals to work together to train models without sharing raw patient records. This is in line with strict rules like HIPAA and GDPR. FL reduces the hazards of centralization while allowing AI systems from more than one institution to work together to anticipate diseases, create diagnostic images, and provide personalized medicine. DP makes sure that electronic health records and genomic datasets are statistically indistinguishable, while HE lets you do secure cloud-based computations on encrypted radiology images and biomedical signals without having to decrypt them. This keeps the data private all the way through the pipeline. Blockchain-enhanced AI helps handle diverse data streams in smart cities in a way that can not be changed. These streams include smart transportation systems, energy distribution systems, and waste management systems. Blockchain makes ensuring that cross-domain data transactions are open and honest by using consensus-driven ledgers, smart contracts, and storage that can not be changed. At the same time, anomaly detection methods protect surveillance networks, IoT-enabled traffic sensors, and power grids from cyber attacks, insider threats, and advanced inference attacks. AI-based privacy frameworks also protect location monitoring, crowd movement analysis, and digital citizen services. They strike a balance between privacy and openness in real-time governance models. In general IoT ecosystems, privacy-preserving solutions help fix serious security holes in smart homes, industrial automation, and energy systems. FL and DP make it possible for privacy-aware collaborative analytics to happen on edge devices, which means that learning can happen without sharing raw telemetry or user-generated data. In predictive maintenance, robotics coordination, and large-scale sensor networks, AML defenses protect against model poisoning, inference-based attacks, and gradient leakage. Trust frameworks that use blockchain make sure that decentralized authentication, identity verification, and safe data sharing happen across different IoT nodes, removing single points of failure. Our taxonomy shows how flexible it is as a unifying framework for safeguarding data-intensive, safety-critical, and regulation-driven sectors by mapping these privacy-preserving strategies across healthcare, smart cities, and IoT infrastructures. This flexibility shows how important AI-driven privacy-preserving solutions are in a wide range of linked systems [87]. Also, Table 12 shows Cross-domain applications of privacy-preserving AI techniques.

Table 12 Cross-domain applications of privacy-preserving AI techniques

Technique	Healthcare applications	Smart city applications	General IoT applications
FL	Multi-institutional diagnostic models without patient data exchange; personalized medicine	Distributed traffic prediction; collaborative urban planning	Collaborative analytics across IoT nodes (smart homes, industry)
DP	Anonymization of electronic health records and genomic data	Privacy-preserving mobility and citizen service data	Secure aggregation of telemetry and behavioral IoT data
HE	Secure computation on encrypted medical imaging and biosignals	Encrypted processing of surveillance and energy grid data	Confidential analytics on encrypted IoT streams
Blockchain AI	Secure sharing of clinical trial data; decentralized health data ownership	Tamper-proof management of transport, energy, and governance data	Decentralized authentication and identity management in IoT
AML	Defense against poisoning attacks on diagnostic AI systems	Protection of anomaly detection models in surveillance and utility grids	Resilience against inference and model extraction attacks in predictive maintenance
Anomaly Detection	Real-time detection of irregular patient monitoring signals	Intrusion detection in traffic, surveillance, and smart energy grids	Cyberattack detection in large-scale IoT sensor networks

Detailed comparative performance analysis

The use of privacy-preserving technologies in IoV is not only about keeping data safe; it is also about finding the right balance between speed, scalability, latency, and energy use. In this subsection, we conduct a comprehensive performance comparison of key methodologies, DP, standard FL, BFL, HE, AML defenses, and streaming Anomaly Detection, assessing them against various operational metrics. We look at (i) the extra work needed for each training cycle, (ii) the cost of communication, (iii) the time it takes for consensus or cryptographic operations, (iv) the ability to scale horizontally, (v) the effect on accuracy, (vi) the privacy assurances, (vii) the amount of energy used, and (viii) the capacity to work in real time. Table 13 shows the combined performance analysis. When you look at DP-based preservation and Blockchain-augmented FL side by side, you can see that there are some trade-offs. DP simply adds a small amount of extra processing ($O(|\theta|)$) and very little communication cost by adding noise and clipping gradients. This makes it very easy to scale across devices. Its fundamental problem is that accuracy goes worse when the privacy budget (ϵ) is up. BFL, on the other hand, uses immutable ledgers and consensus to assure robust integrity and auditability. However, it has a lot of communication overhead because of block replication and latency caused by consensus, which makes it less suitable for IoV applications that are sensitive to delays. So, even though DP is easier on computers and uses less energy, BFL is better at following rules and being audited, especially in cross-silo or consortium contexts [105].

HE provides strong semantic security and lets you do calculations on ciphertexts, but it also makes computations and communications very expensive, which makes it hard for real-time IoV applications to work. Adversarial ML defenses make systems more resistant to poisoning and inference attacks, but they also make training harder and sometimes less accurate. Streaming anomaly detection is still the best option for real-time use because it does not require a lot of processing power or communication overhead. However, it does not offer any formal privacy assurances unless it is combined with other cryptographic DP approaches. This comparative analysis shows that no one technique is better than all the others; instead, the best way to use them is to mix them together in a way that fits the situation. For example, combining FL with DP for efficiency and

Table 13 Comparative performance matrix of AI privacy-preserving techniques in IoV

Technique	Computation per round	Communication per round	Consensus/ crypto latency	Scalability	Accuracy impact	Privacy guarantee	Energy footprint	Real-time suitability	Typical deployment
Differential Privacy (DP, e.g., DP-SGD)	Low–Moderate (gradient clipping + noise; $O(\theta)$ overhead)	Low (model/gradient payload $\approx \theta $)	None	High ($O(N)$; server aggregation)	Low–Moderate (ϵ -dependent)	Formal ϵ -DP	Low	High (edge-friendly)	Cross-device FL; EHR; mobile sensing
Federated Learning (vanilla FL)	Moderate (local epochs)	Moderate ($ \theta $ per round; optional compression)	Low	High (stragglers aside)	Low (data heterogeneity sensitive)	Medium (privacy-by-locality)	Moderate	Medium–High	IoT edge/fog with coordinator
Blockchain-augmented FL (BFL)	Moderate–High (local train + block verify/sign)	High (block propagation + metadata)	High (consensus latency, TPS-bound)	Medium (block size/TPS limits)	Low (model utility intact)	High integrity/auditability (no ϵ -DP unless combined)	Medium–High	Medium–Low (latency sensitive)	Cross-silo/consortium IoT; audit trails
Homomorphic Encryption (HE)	Very High (cipher ops $\text{poly}(N)$; bootstrapping)	High (cipher-text expansion 5–20 $\times$)	High (crypto eval time)	Low–Medium (compute-bound)	None (exact)/Low (approx HE)	Strong (semantic security)	High	Low (not latency-tolerant)	Cloud computing on encrypted telemetry
Adversarial ML defenses (robust training)	Moderate–High (extra loss terms/adversaries)	Moderate	Moderate	Medium	Low–Moderate (robustness trade-off)	Indirect (resists poisoning/inference)	Moderate–High	Medium	Safety-critical perception/IDS
Anomaly Detection (streaming)	Low–Moderate (light-weight models)	Low (telemetry/alerts only)	Low	High (sharding/edge pipelines)	Variable (use-case dependent)	None (unless combined with DP/HE)	Low	High (real-time friendly)	Edge IDS; RSU monitoring; OT/ICS

privacy that can be proven, or combining BFL with lightweight anomaly detection for smart transportation systems that need to follow the rules.

Adaptability of AI models in dynamic IoV environments

IoV networks are very dynamic, with topologies that change quickly, communication quality that varies, and data sources that are different from one another. This means that AI-driven privacy-preserving techniques that can adapt in real time are needed. FL and AD are two methods that are especially useful in this situation. FL solves the problem of moving cars with distributed and privacy-constrained data by allowing on-device learning with periodic model aggregation. Models in dynamic IoV settings need to be able to handle concept drift, which is when traffic patterns, attack vectors, or driver behaviors change over time. Incremental or ongoing FL techniques update global models more often and allow cars to join in at different times because they are mobile and have spotty connections. Additionally, personalized FL variations enable vehicles or Roadside Units (RSUs) to preserve local modifications of the global model, hence enhancing resistance to non-IID data distributions characteristic of vehicular networks. Federated transfer learning and edge–cloud collaboration are two examples of this. They make sure that new contexts, such as new routes or changes to infrastructure, are quickly included without having to start over from scratch. IoV traffic is very different from one situation to the next, which makes it harder for AD systems to work. In places where vehicle density and communication patterns change quickly, static detection levels can lead to a lot of false positives. To deal with these changes, AD systems are using more and more online learning and adaptive thresholding methods, which keep detection parameters up to date based on the current traffic conditions. Streaming anomaly detection, graph-based temporal embeddings, and context-aware machine learning models are some of the ways to keep an eye on changes in mobility patterns, interactions between RSUs and vehicles, and V2X communication. Additionally, ensemble-based AD methods use both short-term pattern detectors and long-term behavioral baselines to help the system tell the difference between legal abrupt changes (such as lane merging or rerouting) and malicious intrusions. IoV networks get the best of both worlds by combining incremental FL with adaptive AD: they protect privacy and can handle changes in real time. These kinds of hybrid approaches make sure that vehicle systems stay powerful against new threats, speed up decision-making, and stay in line with privacy rules, even while intelligent transportation systems are changing [106].

Interactions among AI-Driven privacy techniques in IoV

In real-world IoV deployments, privacy-preserving approaches are not usually used on their own; instead, they are coupled into multi-layer stacks that cover the vehicle, RSU, and cloud layers. FL with DP is one of the most useful interactions because it allows training on devices with verifiable ϵ -DP guarantees by adding noise and clipping gradients. This makes it possible to scale over huge fleets with little extra latency. But this method comes with a trade-off between accuracy and privacy that needs to be properly balanced. In contrast, FL combined with BFL improves trust, auditability, and resistance against poisoning by recording updates on a distributed ledger, but it suffers from additional communication and consensus latency, making it more suitable for cross-silo scenarios such as collaborations between manufacturers and regulators. FL with Secure

Aggregation employing HE or Secure Multi-Party Computation (SMPC) also hides individual client updates during aggregation. This makes things more private, but it also costs more to compute and communicate at the fog level. Other interactions also have benefits that go well with each other. DP with Blockchain gives you privacy on an individual level and auditability that can not be changed. This is especially valuable in IoV apps that have to keep track of consent and privacy budgets. Blockchain with HE keeps high-stakes data like over-the-air (OTA) software updates and liability records private and authentic, but its high overhead means it can only be used on low-frequency control planes. AML defenses boost resilience when coupled with FL or BFL, minimizing poisoning and inference assaults through strong aggregation rules, albeit at the cost of added computation. Lastly, anomaly detection works with all categories by keeping an eye on real-time V2X traffic, spotting bad behavior or model drift, and starting personalized FL or rollback methods when necessary. In general, these combinations show that privacy, scalability, and integrity should all be improved at the same time, not as distinct aims. FL + DP is better for real-time, large-scale vehicle learning; Blockchain works best as an off-path audit and compliance layer; and HE or SMPC should only be utilized in places where there is enough computing power [107]. Table 14 shows the most common synergies, including the privacy benefits, scalability effects, deployment circumstances, and warnings for each interaction.

An integrative framework for AI-Driven privacy preservation

While numerous studies have proposed frameworks for individual privacy-preserving techniques, these frameworks are frequently limited in scope and do not adequately address the multifaceted characteristics of IoV environments. Based on the taxonomy and comparative analysis from Sects. "Privacy challenges in IoV" – "Results and analysis", we suggest an integrative framework that brings together AI-driven privacy solutions, deployment layers in IoV, and the cross-cutting issues that have come up in this research. This framework solves the problem of fragmentation in current literature by providing a structured model that shows how privacy-preserving methods work together across the vehicle/edge, RSU/fog, and cloud/consortium layers. It also takes into account trade-offs in scalability, latency, and energy efficiency. Data is created quickly at the vehicle or edge layer, and it needs to be protected right away. FL allows multiple people to train a model together without having to share data in one place. DP adds noise to gradients or outputs to keep important driver information from leaking. Anomaly detection models work locally to find unusual activity or outliers in real time. They are the first line of defense against privacy intrusions. However, these methods have limitations in terms of processing power and energy budgets, which means they need lightweight structures and flexible DP budgets. At the RSU or fog layer, where data is aggregated and processed in the middle, it is possible to use methods that use more resources. HE and secure aggregation protocols let you process encrypted updates without anyone else seeing them. AML defenses keep your model safe from poisoning and evasion assaults while it is being aggregated. This layer strikes a balance between the need for computing availability and low latency, although it may still have communication overhead. This can be reduced by utilizing approximate HE, sparsification, or clustering of FL players. At the cloud or consortium level, it is very important to coordinate and manage trust on a large scale. Blockchain is very important since it makes sure that trust is decentralized, that

Table 14 Interaction matrix for AI-driven privacy techniques in IoV

Combination	What is combined	Privacy/Integrity effect	Scalability/Latency effect	Best-fit IoV deployment	Key caveats/mitigations
FL + DP	Local training + DP-SGD/noisy updates	Provable ϵ -DP; mitigates membership/gradient leakage	High scalability; low latency; minimal payload overhead	Cross-device vehicle fleets; city-scale sensing	Utility loss (ϵ -dependent); tune clipping/noise; adaptive DP schedules
FL + Blockchain (BFL)	Signed updates, on-chain audit, incentives	Strong integrity/auditability; deters poisoning	Medium scalability; consensus adds latency; extra replication	Cross-silo (OEMs, authorities); regulated corridors	Use PoA/PBFT, sidechains, layer-2; keep chain off hot path
FL + Secure Aggregation (HE/SMPC)	Aggregation without seeing individual updates	Hides client gradients; combine with DP for end-to-end privacy	Compute heavy at fog; comms $\uparrow$ (HE expansion)	RSU/fog regions with capable hardware	Prefer SMPC or approx HE with packing; compress/sparsify updates
DP + Blockchain	DP privacy + on-chain compliance logging	Formal privacy + verifiable audit of ϵ /consent/events	Low data-path latency if the chain logs metadata only	Compliance-driven ITS, liability records	Avoid logging raw signals; rotate ϵ /keys; off-chain storage pointers
Blockchain + HE	Encrypted data + immutable provenance	Confidentiality + provenance for critical artifacts	Low-rate control-plane only; high crypto/ledger cost	OTA updates, forensics, evidence lockers	Batch/epoch commits; verify off-chain, anchor on-chain
FL + AML (robust agg)	Robust aggregators, poisoning defenses	Resists model poisoning/backdoors	Slight compute $\uparrow$; latency modest	Safety-critical IDS/perception	DP noise can hide anomalies; run aux validators/AD
AD + (FL/BFL/DP)	Streaming AD at edge/RSU with model feedback	Detects drift/attacks; supports rollback and client selection	Very low latency; high sharding scalability	Real-time V2X monitoring, RSU telemetry	No formal privacy; combine with DP/secure agg; calibrate thresholds

vehicle updates are logged in a way that can not be changed, and that all stakeholders' interests are aligned. Blockchain works with FL to provide federated audits and fairness, although care must be taken when choosing consensus protocols to minimize latency problems. The cloud is also where global model aggregation and long-term analytics happen. It sends optimized models back to the edge and fog nodes. This integration of multiple layers is not only additive; it works together. For instance, FL at the edge with DP protects local privacy, and blockchain in the cloud makes sure that audit trails can not be changed. HE at the fog layer lets you do secure calculations on encrypted updates, and AML defenses make everything stronger at all levels. Anomaly detection works as a cross-layer process, always looking for drift or bad behavior. This approach provides a unified design for deploying AI-driven privacy preservation in IoV, addressing fragmented knowledge in past studies and outlining practical routes for scalable, real-time, and regulation-compliant vehicular security systems. The proposed integrative framework is summarized in Table 15, which maps AI-driven privacy-preserving techniques to IoV system layers, highlighting their roles, benefits, and associated trade-offs.

Table 15 Integrative framework linking IoV layers, AI privacy techniques, benefits, and challenges

IoV Layer	Techniques applied	Primary role in privacy	Benefits	Challenges & Trade-offs
Vehicle/Edge	FL, DP, Anomaly Detection	Local model training without data sharing; obfuscation of sensitive data; real-time monitoring	Low latency, user-level privacy, immediate anomaly detection	Limited computation/energy; DP-accuracy trade-off; vulnerability to poisoning if edge devices are compromised
RSU/Fog	HE, Secure Aggregation, AML Defenses	Confidential aggregation of updates; resilience against poisoning and evasion	Strong privacy guarantees, intermediate robustness, moderate scalability	High computation cost (HE); communication overhead; requires efficient clustering/approximation
Cloud/Consortium	Blockchain, FL (global aggregation), Long-term Analytics	Decentralized trust management; tamper-proof logging; incentive mechanisms; large-scale model training	Integrity, accountability, auditability, cross-silo collaboration	Blockchain consensus delays, potential scalability bottlenecks, and energy demands
Cross-cutting (all layers)	Hybrid methods (FL + DP, BFL, HE + DP, AML + AD)	Multi-layer synergy combining privacy, scalability, and robustness	Comprehensive protection, complementary strengths	Integration complexity, heterogeneity of devices, and regulatory compliance

Future work and research challenges

In this section, we are going through future works and research challenges in a detailed manner.

Future works

In this section, we go through future works in a detailed manner.

A. Development of Lightweight Encryption Algorithms for Resource-Constrained IoV Devices.

The growing interconnectivity of vehicles, roadside units, and infrastructure nodes raises important issues about data security and privacy as the IoV ecosystem develops. But a major obstacle is the natural resource limitations found in many vehicle components, especially in On-Board Units (OBUs), embedded sensors, and communication modules. Regarding processing power, memory capacity, battery life, and bandwidth availability, these devices sometimes run under strict restrictions. Consequently, the implementation of conventional cryptographic solutions, such as RSA, full HE, and Elliptic Curve Cryptography (ECC), may result in significant computational overhead, degrading system performance, increasing latency, and impairing the smooth operation of IoV applications, including autonomous driving, V2V communication, and real-time traffic management. To solve this, the design and optimization of lightweight encryption algorithms, especially suited for IoV environments, becomes the main focus of the next investigation. These systems have to find a careful mix between a low computational footprint and strong security assurances. Already showing promise in similar resource-constrained environments like IoT and WSNs, are techniques include stream ciphers (e.g., Salsa20, ChaCha20), block ciphers with reduced rounds (e.g., PRESENT,

LED, SPEEK), and hardware-friendly hash functions. Still, there is much to learn about adapting and rigorously testing such algorithms in the vehicle environment. Moreover, although safe against future quantum hazards, newly developed post-quantum cryptography techniques usually have high computational costs. To future-proof IoV systems without compromising performance, lightweight post-quantum schemes (e.g., NTRU-based cryptosystems, lattice-based cryptography with parameter reductions) should thus be explored. Including hardware-software co-design methods is another vital field of research. One can significantly lower encryption execution times and energy consumption by offloading some of the encryption process to specialized low-power hardware modules such as cryptographic accelerators or Trusted Platform Modules (TPMs). Lightweight encryption should also be flexible for adaptive security models, in which case encryption parameters dynamically scale depending on real-time resource availability, vehicle speed, or network congestion levels, enabling the system to maximize between privacy and performance on demand [108, 109].

B. Advancements in Federated Learning for Diverse Vehicular Data.

The IoV ecosystem generates quite varied data streams by nature: traffic patterns, driver behavior, V2V/V2I communication records, vehicle sensor readings, and multimedia inputs from cameras and LiDAR systems. Because of privacy concerns, bandwidth restrictions, and computational overhead, this heterogeneity presents major difficulties for conventional centralized machine learning methods. Emerging as a potent answer is FL, which allows distributed model training straight on edge vehicles without sending raw data to central servers, preserving data privacy and lowering communication costs. However, using FL in IoV environments with various vehicle data adds several complications. Since vehicle datasets vary greatly depending on geographic location, vehicle type, driving behavior, and environmental conditions, the main research direction is addressing non-IID (Independent and Identically Distributed) data heterogeneity. When trained over such non-IID datasets, current FL systems usually suffer from convergence speed and model accuracy. Future research should concentrate on creating customized FL strategies, such as meta-learning-based approaches, cluster-based federated averaging, or multi-task FL, which let the global model adapt to particular vehicular environments without compromising overall generalization. Still, another significant development is in raising the efficiency of communication. Ensuring stable and low-latency communication between vehicles and FL servers is not easy, given the very dynamic and mobile character of IoV networks. Sparse gradient compression, quantization, and adaptive model pruning help to lower the communication overhead; hence, FL is appropriate for real-time vehicle environments. Using the P2P model, sharing and asynchronous updates where suitable, researchers should investigate new aggregation techniques resistant to intermittent connectivity, packet loss, and topology changes [110, 111].

C. Integration of Differential Privacy Mechanisms in Real-Time Traffic Data Sharing.

Sharing real-time traffic data in the IoV is a big privacy issue because CAVs send sensitive information like location, speed, and routes to RSUs, other vehicles, and cloud systems all the time. This could show how drivers behave and what they do on a daily basis. Differential Privacy (DP) provides robust mathematical assurances by guaranteeing that

the addition or exclusion of individual data has a negligible impact on aggregate computations. However, the integration of DP into real-time vehicular communication is still not well understood. Future research should concentrate on lightweight, rapid differential privacy algorithms tailored for constrained vehicular resources, alongside adaptive privacy budgets that dynamically modify noise levels in response to data sensitivity, network conditions, and quality of service requirements. Because IoV data is always coming in and going out quickly, Streaming Differential Privacy (SDP) methods are needed to keep cumulative privacy without costing a lot of computing power. Federated DP and secure aggregation methods can keep distributed IoV systems safe without sharing raw data. LDP and SMPC are other methods that can protect the privacy of both individuals and groups. Managing long-term privacy budgets by adding to or replenishing them (for example, through randomized participation or selective sampling) will make them more sustainable. Context-aware DP can adjust noise injection to fit the driving conditions, striking a balance between privacy and accuracy. Also, to add DP to the IEEE 802.11p and 5G-V2X protocols, it is important to make sure that the latency is low and the reliability is high. Large-scale simulations and testbeds need to look at how DP affects latency, data usefulness, scalability, and how well it can withstand inference attacks. If we can solve these problems, future IoV systems will be able to share real-time traffic data in a way that is more private, safe, and reliable [112, 113].

D. Blockchain-Based Decentralized Identity Management Systems.

Blockchain-based distributed identity management systems integrated into IoV ecosystems offer a potential answer to handle issues of trust, privacy, and security in-vehicle communications. Often, depending on outside authorities, conventional centralized identity management systems expose private vehicle and driver data to vulnerabilities, including single points of failure, data breaches, and illegal access. By using the inherent qualities of blockchain immutability, transparency, and distributed consensus, decentralized identity systems enable safe and privacy-preserving identity verification without central intermediaries. Unique cryptographic identities recorded on the blockchain can be assigned to infrastructure nodes, RSUs, and vehicles so guaranteeing tamper-proof authentication and traceability of vehicle interactions. By enabling vehicles to prove their identity or credentials without disclosing sensitive personal data, reducing tracking or profiling risks, and further enhancing privacy using advanced cryptographic tools, including ZKPs and Ring Signatures. Smart contracts also automatically enforce access control policies in real-time, allowing conditional, revocable, and fine-grained identity verification procedures catered to different IoV use cases, including toll collecting, fleet management, V2I communication, and cross-border vehicle operations. Future studies should concentrate on creating lightweight blockchain architectures, maybe using sharding, sidechains, or Directed Acyclic Graph (DAG) based ledgers, to reduce computational and communication overhead and maintain integrity and security guarantees. Standardizing initiatives are also essential to guarantee cross-domain interoperability, especially as IoV environments span many countries and communication protocols (V2V, V2X, 5G), and regulatory frameworks are progressively expanding. Blockchain-based decentralized identity management provides a transforming road toward safe, privacy-aware, and trusted vehicle networks, but realizing its full promise calls for holistically addressing scalability, governance, and integration issues [114, 115].

E. Development of Robust Adversarial Machine Learning Defense Mechanisms.

AI-driven models have made IoV systems smarter, more automated, and able to make decisions in real time, but they have also made AML issues worse. Evasion, data poisoning, model inversion, and inference attacks can put driver and vehicle data, safety-critical choices, and vehicle operations at risk. In autonomous driving systems, intrusion detection models, and federated learning-based vehicular networks, tiny adversarial perturbations in sensor data, communication signals, or training datasets can result in misclassification, system instability, or breaches of privacy. So, it is very important to know about strong and scalable IoV adversarial defense tactics. Adversarial training, which makes models stronger by using adversarial examples, takes a lot of computing power and is hard for car hardware which is not very powerful. To make sure that systems are formally resistant to a number of adversarial attacks, researchers have looked into advanced methods including randomized smoothing, ensemble learning, gradient masking, and certified defenses. Also, defensive distillation and autoencoder-based input reconstruction can help lessen real-time traffic adversarial disturbances. These tactics can hurt the performance goals of the IoV system, such as model consistency, computational cost, and real-time responsiveness. Federated adversarial training with DP and blockchain-backed validation makes systems more stable, private, and able to grow. Context-aware defense solutions dynamically change security policies based on the vehicle's status, communication architecture, and threat patterns. This helps the system be more resilient without using too many computing resources. Next, we need to make adversarial defense models that are lightweight, easy to understand, and use little energy to work in the actual world, where cars are very mobile, have different types of hardware, and have spotty internet connections. It is important to test these defenses against a variety of attack vectors utilizing standardized adversarial testing methods, simulation platforms, and benchmark datasets, especially for IoV scenarios. For the intelligent transportation system landscape that is always evolving, AI-powered IoV ecosystem security, dependability, and trustworthiness need good adversarial machine learning defense systems [116, 117].

F. Context-Aware Anomaly Detection Systems.

IoV environments' dynamic and heterogeneous character calls for the development of anomaly detection systems outside of fixed, one-size-fits-all answers. Although they are good at spotting predefined threat patterns and anomalies, traditional AI-driven anomaly detection models are not always flexible enough to fit the several, always-changing environments of vehicle networks, including changing traffic density, different road conditions, network congestion, and environmental factors. By including real-time contextual information ranging from geographical location, time of day, and vehicle speed, to communication topologies in the anomaly detection pipeline, context-aware anomaly detection systems have emerged as a potential direction to address this limitation. More accurate, flexible, and scalable threat detection results from these systems dynamically changing their detection thresholds, feature selection processes, and decision-making strategies depending on the surrounding environment and situational conditions. Anomaly detection thresholds in urban settings with dense V2X communication, for example, can vary from those in highway situations with limited connectivity. Moreover, by including context-aware mechanisms with MEC nodes, low-latency processing, and

real-time anomaly detection are guaranteed without depending just on centralized cloud systems, so lowering response time and network overhead. Advanced implementations factor in contextual variables and use deep learning models such as convolutional auto-encoders, attention mechanisms, and RNNs that can learn temporal and spatial dependencies in vehicle data [118, 119].

G. Secure Multi-Party Computation for Collaborative Data Analysis.

A cryptographic paradigm called SMPC lets several parties jointly compute a function across their private inputs without disclosing those inputs to each other. Within the framework of the IoV, SMPC supports safe and privacy-preserving data analysis among infrastructure, vehicles, and service providers. For example, without revealing individual vehicle information, vehicles can jointly compute traffic statistics or train machine-learning models on shared data. Utilizing this cooperative approach, data utility is improved while preserving privacy and addressing issues of data breaches and illegal access. Using SMPC in IoV means addressing issues including computational overhead, latency, and scalability to guarantee real-time performance and smooth integration into automotive networks. The efficiency and practicality of SMPC are always improving thanks to developments in distributed computing and cryptographic techniques, which make SMPC a reasonable choice for safe collaborative data analysis in IoV systems [120, 121].

H. Privacy-Preserving Data Aggregation Techniques.

Ensuring data privacy and security during cooperative data analysis is critical in the framework of the IoV. Two fundamental methods addressing these issues are SMPC and Privacy-Preserving Data Aggregation (PPDA). SMPC keeps those inputs private yet lets several parties jointly compute a function over their inputs. This enables infrastructure and vehicles in IoV to cooperatively analyze data without disclosing individual datasets, so safeguarding privacy and improving security. For instance, SMPC can help sensor data to be aggregated into spatio-temporal maps without disclosing raw data, so maintain private individual vehicle information. PPDA methods concentrate on data aggregation in a way that stops specific data point disclosure. PPDA guarantees in IoV systems that sensitive data from vehicles is safeguarded during data collection and analysis. For example, a proposed Privacy-Preserving and Verifiable Data Aggregation (PPVDA) system uses homomorphic Message Authentication Codes (MAC) and secret sharing to enable effective aggregation of multi-dimensional data while enabling verification of the correctness and integrity of the aggregation results. Using these methods in IoV brings difficulties, including computational overhead, scalability, and real-time data processing needs [122, 123].

I. Implementation of HE in Vehicular Communications.

Using HE in-vehicle communications offers a possible method to improve data security and privacy inside VANETs. HE guarantees that sensitive data stays private throughout the processing phases by letting computations be done straight on encrypted data instead of decryption. Data privacy and security are raised in VANETs when vehicles routinely exchange data to support uses including traffic control and collision avoidance. Integrating HE allows vehicles to securely share and process data such as location coordinates or speed information, without exposing raw data to possible adversaries. This

method keeps personal privacy as well as the vehicle networks' integrity and dependability. Nonetheless, the practical application of HE in-vehicle communications presents difficulties, including computational overhead, latency, and the requirement for effective key management. Constant research seeks to maximize HE schemes to be more lightweight and fit for the dynamic and resource-limited environments unique to vehicle networks. The security and privacy of intelligent transportation systems should be much enhanced as these developments move forward by the use of HE in-vehicle communications [124].

J. AI-Driven Intrusion Detection Systems.

Future studies in AI-driven IDS should give strong development of resilient, adaptive, and privacy-aware mechanisms top priority to properly protect vehicle networks against ever-advancing cyber threats. Increasing adversarial robustness is a major difficulty since artificial intelligence models are susceptible to adversarial attacks, whereby well-chosen inputs can either evade detection or mislead classification. To support IDS resilience against evasion strategies, research has to concentrate on strong adversarial defense techniques, including adversarial training, defensive distillation, and anomaly-based detection layers. Transparency and trust depend on XAI being embedded into IDS models in concert. Important for real-time vehicle security operations, XAI techniques, including SHAP, LIME, and Grad-CAM, can demystify model decisions, enabling security operators to interpret alerts and respond with more confidence. Furthermore, using FL in IDS presents a viable path for privacy-preserving model training among distributed vehicle nodes. Utilizing FL, vehicles can cooperatively learn strong intrusion patterns without sharing sensitive raw data, so reducing privacy risks and enhancing the detection of localized threats. Future IDS also has to include real-time adaptability utilizing online learning systems that constantly change in reaction to shifting attack paths and traffic patterns. These adaptive models should dynamically update without complete retraining to guarantee timely and context-aware threat mitigation. Improving dataset variation and availability is still another important area of attention. The lack of real-world vehicle scenarios in current datasets sometimes limits IDS efficacy [125].

K. Standardization of Privacy Protocols Across Manufacturers.

Emerging as essential elements in strengthening the cybersecurity of contemporary vehicle networks and tackling the growing risks presented by advanced cyber-attacks are artificial intelligence-driven IDS. By combining cutting-edge artificial intelligence technologies, machine learning, and deep learning, among others, these systems can examine enormous volumes of network traffic data, spotting abnormalities and possible intrusions that conventional signature-based approaches might ignore. For example, the CAN bus, a vital communication backbone in vehicles, lacks natural security mechanisms like encryption and authentication; thus, it is vulnerable to several attack points, including message injection and repeat attacks. By learning normal communication patterns inside the CAN bus and identifying deviations suggestive of malicious activity, AI-driven IDS improves in-vehicle network security. Furthermore, widening the attack surface with the arrival of V2V and V2X communications calls for strong security policies. Proposed to run in edge-computing environments, AI-based IDS systems combine deep learning models with conventional detection methods to manage real-time,

high-throughput traffic, guaranteeing timely threat detection and mitigating effects. Notwithstanding these developments, problems still exist, including the requirement for large labeled datasets for training, the possibility of adversarial attacks against AI models, and the computational limitations of in-vehicle systems. Through the development of lightweight, robust artificial intelligence models able to operate under the resource constraints inherent in in-vehicle environments, ongoing research aims to solve these challenges and open the path for more reliable and safe intelligent transportation systems [126].

L. Development of Privacy-Aware Data Sharing Frameworks.

Developing privacy-aware data-sharing systems is essential to safeguard private information while allowing effective data exchange among infrastructure, vehicles, and service providers in the changing terrain of automotive networks. Usually involving centralized architectures, traditional data-sharing techniques run the danger of single points of failure and higher vulnerability to cyberattacks. Researchers have put out distributed methods using blockchain technology to meet these difficulties. Using ZKPs to safeguard vehicle identities and ensure data integrity and audibility for trusted authorities, a blockchain-based framework has been developed to enable anonymous and auditable data sharing in-vehicle networks. Furthermore, including SMPC methods lets cars cooperatively process data without disclosing personal inputs, so improving privacy. Grounded in game theory, incentive systems have also been investigated to inspire vehicle owners to engage in data sharing, balancing privacy issues with the advantages of shared knowledge. Moreover, the use of LDP techniques helps to minimize the risk of revealing personal vehicle information while allowing the gathering of overall data. Using these cutting-edge models and technologies will help vehicle networks strike a harmonic balance between data utility and privacy, thus building user confidence and supporting the growth of intelligent transportation systems.

XIII. Integration of AI with Secure Hardware Modules.

By improving threat detection, automating key management, and strengthening encryption processes, the junction of AI with secure hardware modules like Hardware Security Modules (HSMs) is transforming cybersecurity. By analyzing enormous volumes of data, AI-driven HSMs can spot abnormalities in real-time, facilitating proactive reactions to possible security breaches. AI systems, for example, can quickly flag possible threats by spotting unusual trends in network traffic or cryptographic operations. Employing predictive times for key rotation based on usage patterns and risk assessments, this integration also helps to automate important management chores, minimizing human error and manual intervention. Furthermore, the creation of safe artificial intelligence accelerators such as GuardNN guarantees hardware-based protection for DL models, safeguarding sensitive computations even in untrusted settings. By creating custom silicon chips meant to improve AI operations and data security inside their data centers, companies like Microsoft are advancing this integration and demonstrating the industry's movement toward combining AI capability with safe hardware solutions. In a time of ever more complex threats, this synergy not only improves the confidentiality and integrity of data but also increases the general resilience of cybersecurity infrastructures.

N. Exploration of Quantum-Resistant Cryptographic Algorithms.

Current cryptographic systems are seriously threatened by the fast development of quantum computing technologies; thus, post-quantum cryptography, also known as quantum-resistant cryptographic algorithms, must be explored and developed. Traditional public-key cryptosystems, including elliptic-curve cryptography and RSA, depend on the computational challenge of tasks, including discrete logarithms and integer factorization. But quantum algorithms, especially Shor's method, can effectively solve these challenges, thus exposing current encryption techniques to vulnerability. Researchers and standardizing organizations have responded by aggressively developing and assessing fresh cryptographic techniques meant to resist attacks from both classical and quantum computers. Recent warnings from cybersecurity authorities highlight the need to switch to these new criteria. Advising companies to get ready for possible quantum computer hacking threats by 2035, the UK's National Cyber Security Centre (NCSC) recommends the use of post-quantum cryptography to protect private information. Reflecting a larger industry trend toward quantum-safe security solutions, industry leaders are also acting proactively; for instance, Cloudflare has included post-quantum cryptography into its offerings to defend against future quantum attacks. Ensuring the confidentiality and integrity of information in the approaching quantum computing era depends mostly on the research and application of quantum-resistant cryptographic algorithms.

O. User-Centric Privacy Control Mechanisms.

By providing complete control over the sharing, storage, and processing of their sensitive driving and location data, user-centric privacy control mechanisms inside the IoV context seek to empower vehicle users. These systems combine sophisticated consent management techniques so that users may dynamically control data permissions depending on contextual awareness, sensitivity levels, and personal privacy preferences. Personalized privacy dashboards and context-aware privacy assistants let drivers see and comprehend data flows between their cars, RSUs, edge nodes, and cloud infrastructure in real-time. Using blockchain-enabled smart contracts also allows users to set precise policies for how their data is accessed, guaranteeing enforceable compliance with privacy laws, including GDPR and CCPA. Future developments should concentrate on using adaptive privacy algorithms able to learn and anticipate user privacy preferences through AI-driven behavioral analytics, enabling automated, context-sensitive changes to privacy policies in response to changing driving conditions. Furthermore, improving privacy guarantees by including DP and secure multiparty computation into user-centric models will enable the use of aggregated vehicle data under an individual anonymity guarantee. In the end, these user-centric privacy control systems will support safe, scalable, and customized experiences in developing intelligent transportation systems as well as raise user confidence in connected vehicle ecosystems.

P. Development of Secure Over-the-Air Update Mechanisms.

Maintaining the integrity, dependability, and cybersecurity of linked vehicles within the IoV ecosystem depends on safe OTA update mechanisms. Future work should focus on creating strong, safe, and efficient OTA architectures, especially for vehicle environments, given the changing threat landscape and the vital part OTA updates play in deploying critical software patches, firmware upgrades, and new security configurations. Design and implementation of cryptographically protected OTA frameworks using

end-to-end encryption, secure boot processes, and Trusted Execution Environments (TEEs) to prevent unauthorized modifications, rollback attacks, and malicious injections during updates is one major area of concentration here. Utilizing immutable logging, distributed validation, and transparent audit trails, which guarantee the traceability of every software transaction, integrating blockchain technology could also improve OTA security. Embedded within OTA systems should be advanced machine learning methods, including anomaly detection and behavior-based analytics, to proactively find and reduce compromised update files or suspicious activity. Moreover, tackling resource constraints and connectivity variability inherent in in-vehicle networks will need lightweight authentication protocols and incremental update approaches that lower bandwidth usage while guaranteeing low latency and minimal downtime. Differential updates, delta patching, and adaptive scheduling techniques will help to maximize resource use and make OTA updates feasible even in conditions of unstable communication typical of mobile IoV settings. Eventually, the evolution of these sophisticated, safe OTA update systems will greatly improve the cybersecurity posture, adaptability, and resilience of the next intelligent transportation systems.

Q. Investigation of Privacy Risks in V2X Communications.

Given the faster acceptance of CAV within the IoV, investigating privacy concerns in V2X communications is ever more important. Future studies should concentrate especially on spotting, simulating, and reducing the wide spectrum of privacy vulnerabilities resulting from constant and large-scale vehicle data exchanges, including real-time location, trajectory data, vehicle diagnostics, and driver behaviors. The thorough investigation of adversarial inference attacks such as location tracking, identity disclosure, and behavioral profiling, which take advantage of freely broadcast safety messages and vehicle metadata within V2X protocols, is one main research direction. With consideration for real-time constraints and high mobility, privacy-preserving mechanisms, including DP, geo-indistinguishability, and context-aware pseudonymizing techniques, should be thoroughly tested and optimized for automotive settings. Investigating cryptographic techniques, including lightweight homomorphic encryption, ZKPs, and SMPC, can also help to preserve privacy without sacrificing the capability and responsiveness needed by safety-critical applications. By giving vehicles dynamic pseudonymous identities verified via cryptographic proofs, integrating blockchain-enabled distributed identity management can also significantly lower identity disclosure risks. Moreover, especially for applications needing ultra-low latency, such as collision warnings or cooperative perception, analyzing the trade-offs between privacy, security, computational overhead, and communication latency becomes imperative. Research has to reach simulation-based privacy-risk assessment models that replicate reasonable adversarial actions, allowing comprehensive validation and benchmarking of new privacy-enhancing technologies under dynamic vehicle network environments. The development of complete, scalable, and practically deployable privacy solutions depends on multidisciplinary cooperation involving cryptography, network security, vehicle communication standards, and regulatory frameworks, ensuring the trustworthiness and acceptance of V2X communications across intelligent transportation systems.

R. Enhancement of Secure Data Storage Solutions in Vehicles.

Since the IoV ecosystem generates and analyzes massive amounts of sensitive data, including location histories, driver behaviors, multimedia sensor information, diagnostic logs, and safety-critical communications, improving safe data storage solutions inside connected vehicles is essential. To protect vehicle-stored data against sophisticated adversarial attacks, including unauthorized access, data tampering, and privacy breaches, future research should extensively investigate advanced cryptographic storage systems, including hardware-accelerated lightweight encryption, homomorphic encryption, and quantum-resistant algorithms. Including reliable hardware platforms such as TPMs or secure enclaves in vehicle architectures will help to guarantee data integrity and confidentiality during the storage, processing, and retrieval phases. Enhanced resilience against single-point failures or malicious changes can come from developing blockchain-based decentralized storage solutions that distribute and safely replicate important vehicle data over several network nodes. Furthermore investigated should be the combination of fine-grained access control systems and Attribute-based Encryption (ABE) to enable dynamic, policy-driven access to stored data, ensuring that only authorized entities, such as maintenance providers, insurers, or trusted infrastructure operators, may access particular subsets of vehicle data based on clearly defined roles or permissions. Further research on privacy-preserving data aggregation techniques, DP mechanisms, and Secure Multi-party Computation (SMPC) can also be very important in enabling vehicles to safely offload data to cloud or edge storage systems without violating user privacy.

S. Assessment of Legal and Ethical Implications of AI in IoV.

The fast integration of artificial intelligence into the IoV raises important ethical and legal questions that demand careful evaluation to guarantee responsible innovation and deployment. Future studies should closely investigate regulatory compliance with new data protection systems, including GDPR, CCPA, and region-specific automotive cybersecurity standards (e.g., UNECE WP.29), to handle the complicated legal environment controlling data privacy, consent management, liability, and accountability inside autonomous and linked vehicles. Clearly defining legal obligations in situations involving automated decision-making, accident liability, data breaches, and malicious cyber events should receive critical attention to guarantee open responsibility mechanisms between manufacturers, service providers, infrastructure operators, and consumers. From an ethical standpoint, especially in areas like insurance pricing, predictive policing within smart transportation, and driver behavior profiling, future studies should carefully examine possible hazards connected with artificial intelligence-driven bias, discrimination, and inequities. Developing XAI approaches to guarantee transparency, audibility, and fairness in algorithmic decision-making will help users and regulatory authorities to comprehend, challenge, and correct choices made by autonomous systems. Moreover, a thorough investigation of privacy ethics has to take place to solve the conflict between people's rights to privacy and informational autonomy and data-driven optimization in-car networks. Researchers should suggest and assess user-centric privacy control mechanisms that enable vehicle users to have transparent, granular control over data collection, sharing, and processing methods.

T. Development of Resilient Network Architectures.

The quick growth and complexity of IoV systems require strong network architectures that can keep communication safe and reliable in changing, dangerous situations. Future research ought to concentrate on the formulation of multi-layered, fault-tolerant architectures that are responsive to congestion, disconnections, and cyber threats such as MITM, jamming, Sybil, and DDoS attacks. Using SDN, NFV, and MEC can move control away from a central point, make better use of resources, and lower latency by processing data in smaller groups. Combining decentralized identity systems with blockchain-based consensus can make authentication and network integrity better. AI-driven methods, especially deep reinforcement learning, can give systems the ability to heal themselves by finding threats and changing paths in real time. To make sure there is redundancy and consistent performance, we should look into hybrid architectures that combine 5G/6G, V2X (IEEE 802.11p), satellite, and mesh networks. To test resilience, latency, reliability, and scalability in real-world mobility and traffic conditions, we need comprehensive simulations and real-world testbeds. This will make IoV communications secure, adaptable, and reliable.

Research challenges

In this section, we go through research challenges in a detailed manner.

A. Ensuring Data Privacy Amidst Extensive Data Sharing.

Protecting data privacy in the IoV ecosystem is a big problem that needs to find a balance between sharing a lot of data and keeping privacy strict. Connected and self-driving cars constantly share private information with each other, infrastructure, cloud servers, and service providers. This makes it easier for hackers to get into, track, and make guesses about the cars' data. Future research must create better ways to protect user data that do not hurt accuracy or real-time performance. Some promising methods are homomorphic encryption for vehicle communication, context-aware differential privacy, secure federated learning, and blockchain-based identity management. Managing dynamic privacy budgets for continuous data streams and making lightweight, resource-efficient mechanisms that work with vehicle hardware are still top priorities. Cross-layer optimization, multi-party computation, and adaptive privacy controls that follow standards like IEEE 802.11p, 5G V2X, and GDPR are also very important. Finally, big tests and simulations should thoroughly test these solutions against modern ML-based inference and reconstruction attacks to make sure that the IoV ecosystem is safe, respects privacy, and works with other systems.

B. Developing Scalable Security Solutions for Expanding IoV Networks.

Creating scalable security solutions for the quickly expanding IoV ecosystem is a big research challenge because there are so many more connected cars, roadside units, and cloud-edge infrastructures. For real-time vehicular networks, traditional centralized security models do not work well because they have high latency, a lot of communication overhead, and are not very flexible. Future research should concentrate on distributed, hierarchical, and decentralized architectures that preserve responsiveness and resilience while effectively managing extensive IoT environments. Some promising areas of research are lightweight cryptography that works with edge computing, intrusion detection that uses federated learning, decentralized authentication that works with

blockchain technology, and adaptive anomaly detection that uses distributed machine learning. Scalable, network-aware security controls should be able to change with the times, such as when topologies, communication standards (DSRC, LTE-V2X, 5G/6G), and environmental conditions change. Also, trust management through consensus algorithms, reputation systems, and decision models that take context into account will be necessary for safe large-scale coordination. Cryptographic primitives that are lightweight and use few resources must also be made for use on board. Finally, testing in real-world situations with different types of mobility and attacks is necessary to confirm performance. To make sure that IoV systems are safe, flexible, and long-lasting, people from different fields, such as networking, cryptography, machine learning, and edge computing, will need to work together to solve the problem of scalability.

C. Mitigating Adversarial Attacks on AI Models in Vehicles.

One of the most important IoV research challenges right now is finding ways to protect AI models in connected and self-driving cars from attacks. Modern cars use deep learning for important tasks like finding things, getting around, and keeping people out. Even small changes to the system can lead to big mistakes or failures. Future work must create strong, understandable, and real-time defense systems that are made for vehicles. This includes adversarial training that puts models in real attack situations, ensemble or hybrid architectures that make it harder for attackers to find single-point weaknesses, and dynamic detection methods like anomaly monitoring, uncertainty estimation, and context-aware fusion. Lightweight adaptive defenses, such as real-time perturbation filtering, input validation, and sensor fusion, should be able to quickly tell the difference between real data and fake data. Adding explainable AI tools like SHAP, LIME, and Grad-CAM will make models more transparent and help people understand and respond to attacks. Standardized benchmarks and realistic testing platforms are also necessary to evaluate robustness across various driving and communication scenarios. To make sure that intelligent vehicular systems are safe and trustworthy, we need to bring together experts from different fields, such as ML security, optimization, sensor fusion, explainable AI, and real-time systems engineering.

D. Balancing Computational Load and Real-Time Processing.

In the IoV, using AI-powered solutions is hard because you have to balance the amount of work that needs to be done with the need for real-time processing. Vehicle networks use advanced technologies like DNNs, blockchain, federated learning, and homomorphic encryption to find problems, protect against cyberattacks, and make decisions on their own. However, the systems on board have limits on CPU, memory, bandwidth, and energy. Future research should concentrate on developing lightweight yet efficient models utilizing techniques such as pruning, quantization, knowledge distillation, and dynamic neural architecture search (NAS) to facilitate real-time intelligence without resource overload. We need adaptive frameworks that split up computation between vehicles and edge/cloud systems based on things like traffic, connectivity, and latency. Researchers can make things more efficient by using hardware-software co-design, low-power edge modules, optimized FPGA or GPU/TPU accelerators, and predictive task offloading through MEC. Setting standard benchmarks for accuracy, latency, and energy use will also be very important. To achieve this balance, people from different fields need

to work together on ML optimization, real-time engineering, and edge computing to make sure that IoV operations are safe, efficient, and responsive.

E. Establishing Interoperability Among Diverse IoV Systems.

Getting different IoV systems, devices, and protocols to work together perfectly is still a big problem. The IoV ecosystem brings together different parts, like V2V, V2I, V2P, and V2C, that use different technologies and standards, such as DSRC, LTE-V2X, and 5G-V2X. Unified communication frameworks, cross-platform middleware, and standardized protocols are all necessary to make sure that automakers, infrastructure providers, telecoms, and regulators can share data safely, reliably, and quickly. Future research ought to concentrate on semantic data integration, standardized ontologies, and unified data formats to facilitate consistent interpretation among stakeholders. Blockchain-based decentralized systems can help make sure that data sharing, identity management, and authentication are all clear. Hybrid gateways and adaptive bridges that can translate between different IoV protocols are also important, as are AI-driven, self-adaptive solutions that can find compatibility problems in real time. Setting up frameworks for interoperability testing and certification will speed up the process of making things more standard and improve the scalability, security, and resilience of future IoV ecosystems.

F. Ensuring Trustworthiness in AI-Driven Decision-Making.

A major challenge to the growth of the IoV is making sure that AI-driven decision-making in vehicular networks is trustworthy and open. AI decisions in navigation, intrusion detection, and anomaly response must be correct, understandable, and accountable because automotive systems are safety-sensitive. Future research should focus on creating explainable AI (XAI) models specifically designed for automotive settings, offering clear, context-sensitive justifications for actions taken in response to unforeseen incidents or cyber threats. Probabilistic and uncertainty-aware techniques, including Bayesian deep learning and deep evidential learning, facilitate the quantification of confidence levels and the improvement of transparency. It is also important to set up standard benchmarks to test AI's trustworthiness, interpretability, and fairness in real-world IoV settings. Combining federated learning, blockchain-based trust frameworks, and secure multi-party computation can also make it possible to make decisions that can be verified while keeping privacy. Making AI models more resistant to attacks will help keep them safe from being compromised. Adding human-in-the-loop strategies and systems for ongoing feedback can help humans and AI understand each other better. This will make intelligent vehicle ecosystems safer, more reliable, and more trusted by the public.

G. Managing Data Integrity in Decentralized Networks.

In the IoV era, where decentralized architectures and distributed networks like federated learning, blockchain-based vehicular systems, and peer-to-peer V2X frameworks rely on secure and consistent data exchange between vehicles, RSUs, and cloud platforms, keeping data integrity is a big research problem. As attacks like data manipulation, replay, and Sybil get better, future studies need to come up with better cryptographic and consensus methods to make sure that data can be sent without being changed, without hurting performance or scalability. This includes lightweight cryptographic hashes, digital signatures that can be used by many people, and ledger protocols that work best

on IoV devices with limited resources. To solve IoV-specific problems with latency and mobility, we should look into hybrid consensus methods like DPoS, PoA, or PBFT. Adding machine learning to edge nodes for real-time anomaly detection, along with secure timestamping, version control, and logging in distributed ledgers, will make things more open and easy to check. Also very important are systems for managing identities that are secure and the ability for different manufacturers and infrastructure providers to work together. To make vehicular networks that are trustworthy, reliable, and smart, we need to deal with these problems.

H. Protecting Against Physical and Cyber Threats.

It is very important and difficult to protect IoV systems from both physical and cyber threats that come together. This requires solutions from many different fields. Vehicles, RSUs, and infrastructure now work in both the physical and digital worlds. This makes them easy targets for hybrid attacks that combine hacking, data interception, and DoS with hardware tampering, sensor spoofing, or sabotage. Future research should formulate comprehensive defense strategies that amalgamate physical security measures, cybersecurity protocols, and advanced anomaly detection systems specifically designed for automotive contexts. This includes making ECUs, Trusted Execution Environments (TEEs), and secure cryptographic co-processors that cannot be hacked from a distance or in person. To find and stop physical-layer attacks like LiDAR interference, GPS spoofing, or camera deception, strong sensor-fusion methods should also be used. Advanced machine learning models that can detect adversaries in real time and adaptive blockchain-based cybersecurity frameworks can make systems more resilient by using decentralized authentication and secure OTA updates. We also need full risk assessment models to find weaknesses at the cyber-physical interface and guess where attacks might happen. Finally, to make sure that communication is safe between mixed legacy and connected vehicle systems, automakers, cybersecurity experts, and policymakers will need to work together closely to create unified standards and regulatory frameworks. This will help create safe and resilient intelligent transportation ecosystems.

I. Addressing Legal and Ethical Implications of AI in IoV.

Solving the difficult legal and ethical issues related to automated decision-making becomes vitally important as the IoV progressively combines advanced artificial intelligence technologies. Especially those involving autonomous driving, accident liability, data privacy violations, and possible discriminatory biases, future studies must carefully investigate the legal accountability and ethical obligations resulting from artificial intelligence-driven vehicle actions. Clarifying legal frameworks that define responsibility, whether it comes from end users, software developers, or automakers, remains a central difficulty needing multi-stakeholder analysis and policy formulation. Moreover, researchers have to look at ethical concerns like algorithmic transparency, making sure that, in line with society's values and legal requirements, AI-driven decisions in autonomous vehicles and traffic management systems are both interpretable and justifiable. Compliance with privacy rules like GDPR, which stress individuals' rights to understand and challenge automated decisions impacting them, depends on developing XAI approaches especially suited for automotive settings. Researchers also have to assess possible ethical hazards, including unintentional bias reinforcement through AI models

and training datasets, which would disproportionately disadvantage some user groups. Such issues can be reduced by methodically investigating ethical frameworks such as fairness-aware machine learning and objective FL algorithms catered to the automotive sector. At last, addressing cross-border legal harmonization and international standards for privacy-preserving AI deployment in IoV scenarios calls for multidisciplinary cooperation involving legal scholars, ethicists, technologists, regulatory authorities, and industry representatives. Such group efforts will enable thorough rules, strong ethical standards, and coherent legal systems necessary for the responsible and fair integration of artificial intelligence into intelligent transportation systems [127, 128].

J. Ensuring Secure Firmware and Software Updates.

The fast development of the IoV calls for regular firmware and software updates to preserve the best performance, improve security, and add fresh features. Delivering these OTA updates securely poses major technical difficulties, though. Future studies have to concentrate on creating strong cryptographic authentication and verification systems able to thwart supply chain exploits, firmware injection, rollback attacks, and malicious or unauthorized update attempts. Investigating advanced digital signature systems, secure bootloaders, blockchain-backed verification processes, and quantum-resistant cryptographic algorithms especially suited for limited vehicle environments helps to ensure end-to-end integrity and authenticity of updates. Designing lightweight but safe OTA update systems able to support seamless, incremental software updates without compromising vehicle network latency or computational resources is another vital area. Furthermore, researchers have to discuss the security consequences of using cloud and edge computing infrastructure to distribute updates at scale, guaranteeing safe data transmission channels, anomaly detection in update distribution patterns, and real-time threat monitoring. Another exciting path to guard OTA update processes against physical interference and advanced cyberattacks is the development of TEEs and safe hardware modules incorporated inside vehicle components. Moreover, a systematic study is needed to identify privacy-preserving mechanisms inside OTA systems, to protect private user information during software transmission and concurrently follow strict regulatory requirements. Last but not least, future research has to give top priority to creating standardized interoperability rules and regulatory frameworks controlling safe OTA operations so enabling industry-wide adoption of best practices to guarantee resilient, trustworthy, and privacy-conscious firmware and software update strategies for next-generation IoV installations [129, 130].

K. Managing Network Latency and Reliability.

Particularly considering the strict demands of real-time communication, safety-critical operations, and autonomous driving capabilities, managing network latency and dependability becomes a key challenge in the fast-developing IoV. Especially when combining heterogeneous network architectures including V2V, V2I, and V2C interactions, ensuring consistently low-latency communications becomes increasingly difficult as IoV systems grow. Future work should focus on creating creative, latency-aware routing systems and dynamic resource allocation techniques able to maximize data flow paths in fast-changing vehicle network topologies. These initiatives should support dynamic prioritizing and effective management of network traffic by using methods including

MEC, SDN, and network slicing inside 5G and beyond 5 G infrastructures. Furthermore, creating AI-driven predictive models and reinforcement learning-based solutions helps to proactively find congestion areas, allowing adaptive real-time network configuration changes and guaranteeing continuous, dependable communications even in highly congested traffic conditions. Investigating advanced data aggregation techniques and context-aware communication systems is another important focus of research since they minimize pointless data exchanges, drastically lowering network load and latency. Moreover, by utilizing distributed identity management, fault-tolerance, and tamper-resistant logging of significant network interactions, blockchain-enabled decentralized mechanisms can improve dependability. To fully validate dependability and latency goals, researchers also have to give strong priority to thorough performance evaluations and simulations under reasonable scenarios, including urban mobility models, highway platooning, and mixed autonomous-human driving conditions. To guarantee not only performance but also resilience against cyber threats, enabling the deployment of efficient, dependable, and ultra-low-latency IoV networks capable of meeting the needs of future intelligent transportation systems, addressing these challenges will ultimately need cross-layer optimization strategies, integration of secure hardware accelerators, and strong cryptographic measures.

L. Enhancing User Privacy in Data-Driven Services.

Particularly considering the exponential increase of data exchange among connected vehicles, roadside infrastructure, and cloud-based analytics platforms, improving user privacy in data-driven services within the IoV ecosystem remains a vital but ever-difficult task. The widespread use of data-intensive apps personalized navigation, usage-based insurance, driver behavior monitoring, predictive maintenance, and cooperative autonomous driving, requires large-scale data collecting, processing, and sharing, which generates natural privacy hazards for vehicle owners and operators. Future research should concentrate on developing privacy-preserving technologies that rigorously protect user data at every level of collecting, transmission, processing, and storage to meet these difficulties. Emerging mechanisms like FL, DP, and HE must be further optimized and easily incorporated into IoV architectures to offer strong protection without appreciably compromising service quality or adding intolerable computational overhead. Furthermore, context-aware and user-centric privacy controls must be created to empower consumers with precise, transparent control over how their data is shared and handled, enabling dynamic changes to privacy preferences depending on real-time settings, situational awareness, and sensitivity levels of data. To guarantee privacy while maintaining analytical accuracy, researchers should investigate advanced anonymizing and pseudonymizing strategies, including k-anonymity and geo-indistinguishability, to prevent the re-identification of people from aggregated or shared data sets. Blockchain technology is another vital path since it uses distributed identity management and immutable data transaction logging to improve traceability, responsibility, and openness of data consumption. Quantifying privacy risks and trade-offs depends on comprehensive evaluation systems, which also provide unambiguous benchmarks for industry adoption, regulatory compliance, and user confidence. Designing complete privacy-preserving frameworks that match technological capabilities with ethical standards and legal requirements will depend on cooperation among cybersecurity experts, artificial intelligence specialists, automotive

engineers, regulatory bodies, and end users, ensuring user trust and enabling the sustainable adoption of intelligent, data-driven IoV services [131, 132].

XIII. Developing Resilient Systems Against Sensor Failures.

It is very important to do research on how to make IoT systems that can handle sensor failures, since autonomous and connected vehicles need accurate sensor data all the time to work safely. Integrated systems that use LiDAR, radar, GPS, cameras, and other sensors are at risk of becoming unreliable if their sensors fail due to hardware problems, environmental interference, cyberattacks, or signal degradation. Future research should concentrate on resilient sensor-fusion algorithms that utilize redundancy, contextual awareness, and adaptive filtering to preserve situational awareness in the face of partial failures. Deep learning-based fusion, anomaly detection, and fault diagnosis can help find and fix problems with sensors in real time. IoV systems will be able to quickly reorganize data pipelines when problems happen if they have fault-tolerant and self-healing features at both the hardware and software levels. Using distributed AI and edge computing can lower latency even more and make it easier to quickly adapt to sensor limitations in a specific area. Authentication backed by blockchain can protect sensor data and stop bad actors from changing it. Lastly, it is important to create full simulation platforms and real-world testbeds to test performance in a wide range of failure and attack scenarios. Making sensors more durable will greatly improve the safety, reliability, and public trust in autonomous IoV systems, which will help with the rollout of next-generation smart mobility.

N. Implementing Secure Vehicle-to-Everything Communications.

Realizing the full possibilities of intelligent transportation systems inside the IoV depends on implementing safe V2X communications, a crucial and technically demanding research area. Securing these interactions presents a difficult task given that V2X comprises V2V, V2I, vehicle-to-pedestrian, and V2C communications. Especially with the proliferation of connected devices and fast-expanding network topologies, this presents a significant challenge. Advanced cryptographic protocols suited especially for the high-mobility, low-latency, and bandwidth-constrained character of vehicle networks must be developed in future work. Without compromising the strict latency needs of safety-critical vehicle communications, lightweight cryptographic schemes, quantum-resistant cryptography, and effective key-management frameworks are needed to prevent eavesdropping, spoofing, and message injection attacks. Furthermore, investigations should be on authentication and identity management systems grounded on distributed platforms such as blockchain to guarantee verifiable and tamper-resistant message exchange among infrastructure parts and vehicles. Moreover, researchers should look at the integration of privacy-preserving methods, including DP, pseudonymization, and secure multi-party computation, to protect private data sent during V2X contacts, thus reducing the risks connected with location tracking, profiling, and illegal data inference. Furthermore, resilience against network-layer and physical-layer attacks, including DoS, jamming, replay attacks, and Sybil attacks, demands strong intrusion detection systems driven by sophisticated machine learning algorithms capable of real-time anomaly detection and threat mitigation within dynamic vehicle networks. Since safe V2X solutions must operate flawlessly across many automotive platforms, communication protocols

(such as DSRC, LTE-V2X, and developing 5G/6G standards), and legal regimes, developing interoperability standards and frameworks is also vital. Lastly, validating the efficacy and resilience of safe V2X implementations depends on thorough simulations and field tests in reasonable environments, guaranteeing their viability for actual deployment. By effectively tackling these several security issues, transportation safety will be much improved, user confidence will be raised, and the implementation of large-scale, safe, intelligent transportation systems will be sped forward [116, 133].

O. Ensuring Energy Efficiency in Security Mechanisms.

Within the fast-developing field of the IoV, controlling the complexity of heterogeneous network environments presents a major research challenge. IoV ecosystems are progressively combining several technologies as they grow: 5G and future 6G cellular networks, Dedicated Short-Range Communications (DSRC), MEC, VANETs, cloud infrastructures, and several legacy communication protocols. Every technology adds unique communication standards, data rates, latency characteristics, dependability, and security mechanisms, complicating robust connectivity and seamless interoperability. Thus, future work has to concentrate on creating adaptive network architectures able to dynamically control protocol adaptation, communication routing, and resource allocation across several heterogeneous environments. Investigating advanced cross-layer management algorithms enhanced by AI, including reinforcement learning, federated learning, and deep neural networks, to maximize data transmission, minimize latency, and guarantee QoS under highly varied network conditions, is part of this. While isolating and reducing cross-domain interference or performance degradation, research must also investigate advanced network slicing techniques, which segment shared physical resources into logical networks suited to particular IoV applications such as real-time autonomous driving control, infotainment streaming, and emergency response. Another important difficulty is ensuring safe interoperability across many communication standards, which calls for the creation of unified, lightweight security frameworks including distributed identity management, quantum-resistant cryptography, and adaptive authentication systems fit for many network architectures [134].

P. Ensuring Energy Efficiency in Security Mechanisms.

Given that complex security mechanisms often carry significant computational and energy costs, energy efficiency presents a critical and technically difficult research challenge within the field of IoV security. Autonomous and electric cars depend mostly on limited onboard computational resources and battery capacities; thus, security strategies that balance strong protection against low energy consumption are quite important. Future studies should thus concentrate on creating lightweight cryptographic algorithms, adaptive authentication systems, and energy-aware artificial intelligence-driven anomaly detection systems tailored especially for automotive settings. Designed to fit the energy limits of vehicle-embedded systems, this includes investigating resource-efficient encryption methods, including symmetric encryption systems, lightweight HE variants, and elliptic-curve cryptography. Context-aware adaptive security systems should also be looked at dynamically changing security parameters, including cryptographic key lengths, encryption intensity, and authentication frequency, depending on real-time risk assessment, driving conditions, battery life, and V2I connectivity. To

greatly reduce computational overhead and improve security analytics and intrusion detection, advances in energy-efficient AI algorithms, especially model compression, quantization, and FL with sparsified gradients, are vital. Furthermore, research should investigate hardware-software co-design strategies using dedicated low-power security modules such as TPM or secure enclaves to effectively offload cryptographic operations, so as to preserve battery life without sacrificing security. Under realistic mobility and network conditions, exact energy costs and effectiveness of security solutions must be quantified using experimental validation, including real-world vehicular datasets and thorough simulations. In the end, tackling this problem will guarantee the broad acceptance of strong yet energy-efficient security mechanisms, supporting secure, sustainable, and scalable IoV implementations in the next smart transportation ecosystems [134, 135].

Q. Addressing Scalability Challenges in Federated Learning.

Given the fast-expanding scale and complexity of linked vehicle networks, scalability presents a major research challenge in implementing FL for IoV applications. Future studies have to look at new approaches and sophisticated architectural designs able to effectively control the significant rise in participating vehicles, data heterogeneity, and communication limitations inherent in IoV systems. Crucially, developing hierarchical or multi-tiered FL models with edge, fog, and cloud layers will help to enable efficient distribution of computational tasks and best model aggregation techniques across several network levels. Research efforts should also concentrate on developing effective communication protocols tailored especially for resource-constrained, high-mobility vehicles, including model compression, sparsification, gradient quantization, and asynchronous aggregation techniques, so lower overhead. Overcoming data heterogeneity, a major obstacle in IoV federated scenarios, calls for creative ideas, including personalized federated learning, adaptive clustering systems, and meta-learning frameworks that let localized model customizing without compromising global performance. Developing lightweight encryption and safe multi-party computation techniques integrated with FL guarantees data privacy and security without taxing network resources in another crucial area. Evaluation of the performance, communication efficiency, convergence speed, privacy-preserving capabilities, and general system resilience against adversarial attacks at scale depends on thorough testing using realistic simulation platforms and experimental vehicles. Finally, addressing scalability issues in FL will enable its practical implementation in large-scale IoV systems, enabling secure, privacy-preserving, and efficient distributed learning across diverse, highly dynamic vehicles [113, 127].

R. Developing Robust Anomaly Detection Systems.

Particularly considering the increasing complexity, size, and dynamism inherent in modern vehicle networks, the development of strong anomaly detection systems for IoV environments remains a major research challenge. Future studies should concentrate on developing sophisticated, real-time, AI-driven anomaly detection systems able to rapidly and precisely identify several cyberthreats, unusual vehicle behavior, data breaches, and network anomalies. By capturing complex spatiotemporal dependencies inside streaming automotive data, developing adaptive deep learning models such as hybrid CNN-LSTM networks, attention-based mechanisms, and transformer architectures can

greatly improve anomaly detection. Moreover, context-aware detection techniques that dynamically change sensitivity depending on environmental, vehicle, or network conditions have to be looked at to lower false alarms and improve detection accuracy. To give interpretability, transparency, and justification for alerts, that is, to build more trust and actionable insights for system managers and vehicle operators, XAI techniques must be included in anomaly detection systems. Research initiatives also have to investigate federated anomaly detection methods using distributed learning technologies while maintaining data privacy, thus facilitating cooperative threat detection between vehicles without centralized data aggregation. Dealing with scalability calls for lightweight anomaly detection models tailored for edge computing systems with limited resources, preserving computational efficiency without compromising real-time responsiveness. Rigorously validate the efficacy, resilience, and scalability of these anomaly detection frameworks under various adversarial conditions and dynamic IoV network scenarios through extensive evaluations employing realistic simulation scenarios, benchmarked datasets, and practical vehicular testbeds. In the end, overcoming these difficulties guarantees the dependability, security, and resilience of anomaly detection systems in heterogeneous, large-scale IoV environments [116, 136].

S. Ensuring Compliance with Global Privacy Regulations.

Ensuring compliance with worldwide privacy rules presents a difficult and important research question within the developing IoV ecosystem. The spread of linked cars creates enormous volumes of sensitive data across different geographical areas, thus intensifying the difficulty of following different legal regimes, including the GDPR, CCPA, and developing international privacy rules. Using federated learning, DP, homomorphic encryption, and secure multiparty computation that allows seamless and compliant cross-border data sharing while safeguarding personal and location-related information, future research must concentrate on the development of advanced privacy-preserving technologies. Investigating privacy-by-design solutions specifically for IoV platforms, that is, privacy-enhanced data anonymization, context-aware consent management, dynamic user-centric privacy controls, and automated compliance audits, is vital. Further research on intelligent compliance monitoring systems using AI and blockchain technologies could present transparent, tamper-proof mechanisms to track data usage, preserve records of user consent, and provide real-time evidence of compliance for regulatory assessments. Furthermore, harmonized frameworks able to adapt to changing privacy environments and actively address possible conflicts between various regulatory requirements depend on interdisciplinary research efforts involving legal scholars, technology experts, legislators, and ethicists. Rigorous testing across simulation platforms, real-world pilot deployments, and thorough privacy impact assessments will be vital in validating that IoV technologies remain not only functionally robust and secure but also aligned with strict worldwide privacy demands. To build user confidence, regulatory acceptance, and sustainable development inside IoV ecosystems all around, these complex compliance issues must ultimately be addressed [137, 138].

T. Fostering Collaboration Between Stakeholders.

Ensuring the long-term success and security of IoV systems depends mostly on encouraging efficient cooperation between several stakeholders, such as automotive

manufacturers, regulatory authorities, cybersecurity experts, researchers, and end users. Dealing with this difficulty calls for creating clever, safe team models that can fit different goals, technical capacities, and legal surroundings. Creating distributed ledger or blockchain-based decentralized platforms should take the front stage in research efforts to safely enable data sharing, trust management, and group decision-making procedures. Utilizing further investigation into privacy-preserving collaborative techniques, such as FL combined with secure multiparty computation, stakeholders can jointly analyze sensitive data without violating intellectual property or privacy. Establishing interoperable standards and open communication protocols, especially meant to link policy, academia, and industry, will also be very important. Investigating standardized data models, open governance structures, and AI-driven negotiation tools is therefore essential to balance competing interests, maximize resource allocation, and build mutual trust utilizing compromise. Real-world testbeds and cross-sector pilot projects can provide useful insights into collaboration dynamics by utilizing practical analysis of possible obstacles, including competitive tensions, proprietary data concerns, and different privacy expectations. Stakeholder incentives and properly mediating conflicts depend on multidisciplinary research, including social sciences, economics, cybersecurity, law, and engineering. Resilience, innovation capacity, and the general integrity of IoV systems will be much improved by strong, open, and privacy-conscious cooperation between stakeholders, benefiting all users engaged in the vehicle ecosystem [139, 140].

Conclusion and limitations

The IoV is growing quickly, and more and more people are using CAVs. This has made data-heavy environments where vehicles, roadside infrastructure, and cloud or edge services are always sharing information. These changes make it easier to get about, safer, and more efficient, but they also make privacy and security problems worse. Conventional security procedures are ill-suited for the real-time, diverse, and distributed nature of the IoV; thus, AI-driven privacy-preserving methodologies are under active investigation. This paper organizes and summarizes AI-based privacy techniques into six groups: FL, DP, HE, blockchain-integrated AI, AML, and AI-driven anomaly detection. It also talks about what each technique is supposed to perform and what trade-offs have been reported. To sum up, blockchain-integrated AI can help with decentralized auditability; AML wants to make systems stronger against inference and poisoning threats; FL lets models learn without sharing raw data; DP gives statistical privacy guarantees at the cost of utility; HE lets you do math on encrypted data; and anomaly detection methods try to find possible privacy or security violations in real time. Nonetheless, the literature reveals ongoing obstacles, such as processing overhead, scalability in the context of high mobility, and adaptability to dynamic settings. Furthermore, the lack of established evaluation standards hinders direct comparisons among methodologies. Our comparative synthesis elucidates the advantages and limitations of different methodologies and delineates potential design configurations (e.g., FL + DP at the vehicle/edge tier with more robust cryptography delegated to fog/RSU infrastructure). Due to the diversity of datasets, measures, and deployment circumstances, the current evidence does not substantiate conclusive performance rankings. Consequently, the proposed taxonomy and design recommendations should be seen as decision-making tools rather than obligatory formulas. (i) using lightweight models and adaptive aggregation to cut down on overhead;

(ii) making it possible to deploy across edge/fog/cloud; and (iii) creating standardized benchmarks and reporting standards for latency, energy, and privacy risk. Work on these areas will help make AI-enabled privacy more compatible with the realistic limits of next-generation vehicle systems.

This review has several limitations. First, it limits sources to English-language publications that are indexed in major scholarly databases. This may leave out work from specific regions and non-English perspectives. This choice is based on indexing coverage and methodological consistency, not neglect. Second, by focusing on peer-reviewed literature, we may have missed out on useful information from industry white papers, standards drafts, technical studies, and policy briefs that are often published before journals. Third, the big differences between datasets, threat models, measurements, and experimental setups make it hard to compare studies and do a meta-analysis. Because of this, our results focus on patterns and trade-offs instead of clear ranks. Fourth, the six-domain taxonomy is based on expert knowledge and research, but it has not been tested with real users or in large-scale settings yet. Fifth, many results come from simulations or small-scale prototypes, which makes them less reliable in real IoV situations (such as latency during congestion, energy budgets on embedded devices, intermittent connectivity, and following the rules). Future work should use multilingual sources, checked grey literature, shared benchmarks, and evaluations of real vehicular traces across edge/fog/cloud tiers.

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All authors contributed equally.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

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